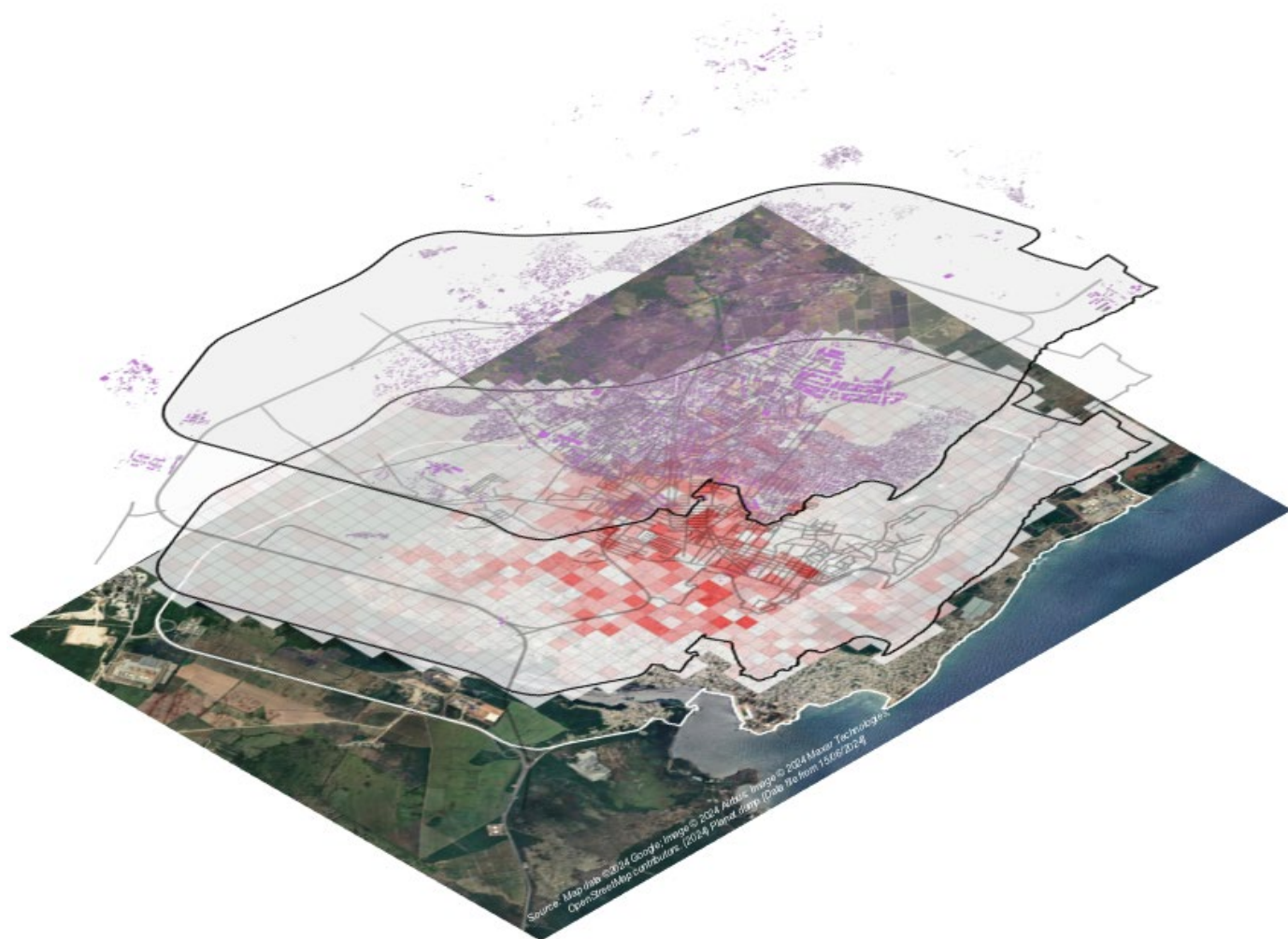


Using Open-Access Spatial Data for Enhanced Sanitation Planning

Leonardo Porto Nazareth

MSc Thesis Identifier WSD.24-1072190

September 2024



Using Open-Access Spatial Data for Enhanced Sanitation Planning

Master of Science Thesis
by
Leonardo Porto Nazareth

Supervisor
Hans van der Kwast

Mentor
Shirish Singh

Examination Committee
Hans van der Kwast, Dr. Bhushan Shrestha and Shirish Singh

This research is done for the partial fulfilment of requirements for the Master of Science degree at the
IHE Delft Institute for Water Education, Delft, the Netherlands.

Delft
20/09/2024

Although the author and IHE Delft Institute for Water Education have made every effort to ensure that the information in this thesis was correct at press time, the author and IHE Delft do not assume and hereby disclaim any liability to any party for any loss, damage, or disruption caused by errors or omissions, whether such errors or omissions result from negligence, accident, or any other cause.

© Leonardo Porto Nazareth September 2024.

This work is licensed under a [Creative Commons Attribution-Non Commercial 4.0 International License](https://creativecommons.org/licenses/by-nc/4.0/)



Abstract

The conventional approach to sanitation often relies on sewerage solutions (offsite systems), which involve extensive networks of pipes and centralized treatment facilities. While effective in well-established urban areas, this method may not be suitable for developing countries with mixed formal and informal settlements. Innovative frameworks and approaches, such as Sanitation21 and Citywide Inclusive Sanitation (CWIS), propose the application of sanitation solutions tailored to the local context in order to implement a sustainable and effective system, taking into account population density, water availability, existing infrastructure, and socioeconomic and cultural aspects. Despite all efforts, global sanitation challenges persist, with 3.5 billion people lacking access to safely managed sanitation.

This research explores the potential of applying open-access spatial data to improve early-stage sanitation planning by adopting open-source Geographic Information System (GIS) software. A significant barrier to effective sanitation planning is the lack of up-to-date and reliable data needed, even considering official sources such as censuses. Although there is an increase in new open-access data, its application in sanitation planning remains limited. To address this gap, this study aims to identify, catalog, and utilize essential open-access spatial datasets available globally, offering a data-driven approach to select the most appropriate sanitation systems according to local contexts.

The proposed methodology integrates gridded population datasets and additional spatial data, such as street, road, and building footprints, into sanitation planning with the support of developed QGIS scripts that automate the planning process steps. Applied in a case study in Boca Chica, Dominican Republic, this approach demonstrated the value of granular spatial data for better understanding the population distribution among the study area and other local characteristics. The findings show that gridded population data provides more detailed insights than conventional census data aggregated at the municipal level, allowing for more accurate characterization of the subarea and more assertiveness in system selection. However, primary information, such as water availability and existing sanitation systems, still depends on local knowledge. Another underscored advantage of using spatial data is the possibility of assessing not only the characteristics of subareas isolated but also considering the context of neighboring areas.

The proposed methodology is flexible and can be replicated in various contexts, including developing countries, democratizing access to valuable data and modern tools, contributing to more informed and consistent sanitation planning.

Acknowledgments

I would first like to express my gratitude for the immeasurable and unconditional support of my family, who, throughout my life, have taught me to value what is most important: the people. A special thanks to my beloved Maria Luiza, who inspires me to be a better person every day. May this long year apart make us a more resilient and united family.

I would also like to express my appreciation to the excellent professionals I have had the opportunity to learn and work alongside throughout my career: engineers José Carlos de Melo, Ivan Paiva, and most recently, Fernando Bernardes—my sincere thanks for sharing some of your vast knowledge with me. In particular, I thank my greatest professional reference, coincidentally, my father, Pery Nazareth, for all your demanding yet fair guidance and, above all, your generosity and constant support.

I extend my gratitude to my colleagues and new friends I had the pleasure of meeting and sharing this year with at IHE. To the outstanding professors and teachers, especially Professor Hans van der Kwast, who taught and inspired me even before I arrived at IHE, and Professor Shirish Singh, for all the knowledge and support.

Lastly, I thank the open-source community, particularly OSGeo and the QGIS community. Through the tireless efforts of these thousands of individuals, knowledge and science continue to evolve freely and democratically, as they should always be: accessible to all.

"No soy pobre, soy sobrio, liviano de equipaje, vivir con lo justo para que las cosas no me roben la libertad."

Pepe Mujica.

Table of Contents

Abstract	6
Acknowledgments	7
Table of Contents	8
List of Figures	10
List of Tables	12
Chapter 1 Introduction	13
1.1 Background	13
1.2 Research problem	14
1.3 Research justification	14
1.4 Research objectives and research questions	14
1.4.1 Research objective	14
1.4.2 Research questions	14
Chapter 2 Literature review	15
2.1 Sanitation	15
2.1.1 Types of sanitation systems	15
2.1.2 Sanitation approaches and frameworks	18
2.1.3 Sanitation planning process	19
2.1.4 Required data and challenges for early-stage sanitation planning	20
2.2 Spatial data	23
2.2.1 Importance of spatial data for sanitation planning	23
2.2.2 Open-access data	25
2.2.3 Open-source GIS software	26
Chapter 3 Methodology	28
3.1 Cataloging open-access spatial data for sanitation planning	28
3.1.1 Criteria for selection of the datasets	28
3.1.2 Primary spatial datasets – Gridded population	29
3.1.3 Secondary dataset – Streets and building footprints	31
3.2 Data analysis and processing workflow	33
Chapter 4 Results	45
4.1 Case study, Boca Chica – Dominican Republic	45
4.1.1 Introduction	45

4.1.2	Climate and Geological Overview	46
4.1.3	Population – Dataset Selection.....	47
4.1.4	Existing information and data	51
4.1.5	Delimitation of the areas – First step	51
4.1.6	Creation of subareas – Second step.....	61
4.1.7	Calculate sanitation system and indicators – Third step	61
Chapter 5	Discussion	70
Chapter 6	Conclusion and Recommendations	74
References		75
Appendices		84
Appendix A. -	Research Ethics Declaration Form.....	84
Appendix B. -	Layout of the sanitation systems suggested for the subarea.....	86
Appendix C. -	Subarea Characteristics, Indicators, and Calculated Results Tables.....	93
Appendix D. -	Authorization Letter for Use of Aggregated Data.....	96
Appendix E. -	Declaration on Use of AI	98

List of Figures

Figure 1: Annual costs per household (1983 USD) for conventional sewerage, simplified sewerage, and on-site sanitation, Natal, northeast Brazil (Sinnatamby, 1986).....	17
Figure 2: Unit cost of sanitation technologies (Türker, 2011)	17
Figure 3: Comparison of population data: Municipality level (left, ONE-RD (Geoportal, n.d.)) vs. 100-meter grid level (right, WorldPop & Center for International Earth Science Information Network (CIESIN), 2018).	24
Figure 4: Neighborhood divisions (white lines) in San Pedro Macoris, Dominican Republic, highlighting varying types of occupation and population density within a specific neighborhood.....	25
Figure 5: Three Types of polygons adopted in the workflow	33
Figure 6: Flowchart of the workflow for the steps of analysis and pre-processing of the spatial data	34
Figure 7: Examples of subarea polygons generated from different combinations of input layers	36
Figure 8: Illustrative examples of clipping primary data to the AOI Polygon.....	39
Figure 9: Illustrative examples of step 6 - Creating and consolidating subarea polygons with primary data.....	41
Figure 10: Suggested sanitation system decision tree.....	42
Figure 11: Additional recommendations decision-tree	43
Figure 12: Illustrative examples of manually dividing a subarea polygon and the automatically recalculated primary data	44
Figure 13: Location of the municipality of Boca Chica and its two districts (La Caleta and Boca Chica)	46
Figure 14: Comparison of 2020 Population datasets with 2022 Census by province, Dominican Republic	48
Figure 15: Comparison of 2020 population datasets with the 2022 Census by municipalities, Santo Domingo Province	49
Figure 16: Spatial distribution of different population datasets in the study area.....	50
Figure 17: Delimitation of the area of interest for the study case	53
Figure 18: Areas with existing sanitation system and location of existing wastewater treatment plant.....	54
Figure 19: Record of connections of water supply users in Boca Chica.....	55
Figure 20: Workflow for delimitation of zones with different average water consumption....	56
Figure 21: Gridded population density for the study area.....	57
Figure 22: Official division of neighborhood of the study area	58
Figure 23: Workflow for delimitation of administrative boundaries layer subdivisions	60
Figure 24: Consolidation of Base layers into a single Subarea layer.....	61
Figure 25: Visualizations of the “Subareas - Suggested Sanitation” Layer. Average water consumption (left) and population density (right) at the end of the planning period.....	64
Figure 26: Identification of subareas with data discrepancies	64

Figure 27: Suggested sanitation system for the start of planning (left) and end of planning (right)	65
Figure 28: QGIS form displaying details of the selected subarea.....	66
Figure 29: Detailed visualization of a densely populated low-income area, highlighting optional datasets of streets (OpenStreetMap) and building footprints (Overture Map).....	67
Figure 30: Final delimitation of the Subareas with the respective suggested sanitation system for the end of planning	68

List of Tables

Table 1: Common steps in the planning frameworks and specific recommended actions within each framework	19
Table 2: Type of sanitation system per population density.....	21
Table 3: Specifications of WorldPop constrained dataset.....	30
Table 4: Specifications of WorldPop unconstrained dataset.....	30
Table 5: Specifications of the HRSL dataset	30
Table 6: Specifications of the GHSL Copernicus dataset	31
Table 7: Specifications of OpenStreetMap dataset	32
Table 8: Specifications of Overture Map dataset	32
Table 9: Population by type of sanitation suggested for the case study.....	69

Chapter 1 Introduction

1.1 Background

Despite significant efforts over recent decades, the global sanitation crisis remains a critical challenge. According to the World Bank Group (2023), 3.5 billion people still lack access to safely managed sanitation. The United Nations Sustainable Development Goal 6 (SDG 6) aims to ensure the availability and sustainable management of water and sanitation for all by 2030.

One of the primary reasons for this is that the conventional approach to urban sanitation, particularly in developed countries, has focused on centralized sewer systems. While these systems can be highly effective in high-density urban areas, they are often unsuitable for the mixed formal and informal settlements characteristic of many developing cities. Informal settlements frequently lack basic infrastructure, including water supply and sanitation services, and face rapid and unplanned demographic growth. This has often resulted in underutilized infrastructure and communities unable to connect to the sewer network due to financial, technical, or social barriers (Gambrill et al., 2020).

To address this challenge, alternative frameworks such as the Sanitation21 framework and the Citywide Inclusive Sanitation (CWIS) approach have been developed, emphasizing a holistic view that integrates technical, social, economic, and environmental factors. The CWIS approach, for instance, advocates for equitable access to improved sanitation services through a combination of sewerred and non-sewerred systems tailored to local contexts. These frameworks, focused on service delivery, advocate that each selection between the two types of sanitation systems is strongly influenced by the characteristics of the area in which they will be implemented (Narayan & Lüthi, 2020; Tilley et al., 2014).

The early-stage planning phase of a sanitation project offers a unique opportunity to explore various arrangements to address current sanitation challenges. However, this phase can be constrained by limited data and time, resulting in inefficient alternative analysis. Open-access spatial data and GIS software offer significant potential to overcome these challenges by allowing the acquisition and adoption of reliable, granular, and up-to-date geographic information. The increasing availability of open-access spatial data and the surge of open-source GIS software have made these methods more accessible, even in developing countries. Spatial data can support the early-stage sanitation planning by allowing the characterization of the subareas of the project, considering crucial aspects such as population density, infrastructure availability, and other critical factors, therefore facilitating the identification of the most appropriate sanitation solutions for each area (Ye, 2018; Hammouri et al., 2022). Despite these advantages, the use of open-access data in sanitation planning remains limited compared to other sectors. Most spatial data applications in sanitation have focused on operation and maintenance rather than the initial planning stages. There is a need for more systematic adoption of GIS and open-access spatial data in the early planning phases to ensure that sanitation system selection is context-appropriate and sustainable (De Moura & Procopiuck, 2020; Kerstens et al., 2016).

This thesis aims to catalog and explore existing primary open-access spatial data to assist planners and other stakeholders—such as sanitation utilities, planners, engineers, and decision-makers—during the early-stage planning of a sanitation project. By adopting open-source GIS software and spotlighting available datasets, the research pretends to fill the data gaps that hinder effective sanitation planning in developing countries and propose methodologies that can be replicated. In addition, although not essential for early-stage planning, secondary datasets were incorporated, which help better characterize the area and are critical for subsequent phases of sanitation planning. The final goal is to enhance early-stage sanitation planning with solutions tailored to the local context, ensuring more consistent sanitation systems.

1.2 Research problem

There is a significant gap in the adoption of open-access spatial data for early-stage sanitation planning. This phase of a sanitation project is crucial, as it offers the opportunity to explore different sanitation solutions tailored to the area. However, it is often restricted by a lack of data and limited time, which leads to an inefficient analysis of alternatives. Most existing studies focus on a specific locality, adopting data and methodologies that are not replicable in other regions. This research proposes a methodology that integrates gridded population datasets to fill the gap in population data that is often outdated or unavailable and lacking the necessary detail. In addition, it considers local information, such as water availability and average consumption, and explores secondary data that can be valuable in later stages of sanitation planning.

1.3 Research justification

This research confronts the global sanitation challenge by focusing on the underutilized potential of open-access spatial data and GIS in early-stage sanitation planning. Standardizing and democratizing spatial data analysis methodologies aims to enhance accessibility, especially in developing countries. Finally, the study contributes to scientific knowledge and social well-being by promoting equitable and adequate sanitation systems.

1.4 Research objectives and research questions

1.4.1 Research objective

The following objective was followed to achieve the aim of this research.

1. To investigate the potential of open-access spatial data to enhance early-stage sanitation planning.

1.4.2 Research questions

- What primary data is required for early-stage sanitation planning is available as open-access spatial data?
- What are the specifications of each of the datasets?
- What methods can be employed to increase the accessibility to open-access spatial data acquisition, analysis, and processing?
- How can the open-access data be used to enhance early-stage sanitation planning?

Chapter 2 Literature review

2.1 Sanitation

Since 2015, the United Nations has the Sustainable Development Goals (SDGs), divided into 17 sets of goals representing global challenges, and indicator 6.2.1 (a) represents the Proportion of the population using safely managed sanitation services. According to the World Bank Annual Report of Global Water Security and Sanitation Partnership, 3.5 billion people still lack access to safely managed sanitation, and the current rhythm of progress will not be enough to achieve access for all until the target year of 2030. Recent reports from 2022 show that to achieve the SDG by 2030, the yearly capital investment should be sixfold, making explicit that currently, global investments are not enough to lead to achieving the goals (World Bank Group, 2023). Still, the lack of financing is not the only reason why sanitation coverage is not increasing as expected worldwide. The focus on infrastructure has led to expensive investments in sewer networks and wastewater treatment plants without considering the local context. This often resulted in underutilized infrastructure because local communities lacked the capacity to operate and maintain the systems or because residents faced barriers to connecting to the sewer network. (Gambrill et al., 2020).

Safely managed sanitation is crucial for enhancing human well-being and environmental quality. Poor sanitation is related to several diseases, including diarrhoeal, neglected tropical diseases (soil-transmitted helminth infections, schistosomiasis, and trachoma), vector-borne diseases, stunting in children, antimicrobial resistance, and environmental enteric dysfunction (World Health Organization, 2018). Many investigations demonstrate the impact of improving sanitation on reducing diarrhoeal diseases and other fecal-oral diseases (Haller et al., 2007; Mara et al., 2010; Minh & Hung, 2011). Considering the non-health-related socioeconomic benefits, one of the positive impacts is productivity at the workspace, as improved sanitation facilities save time that would be spent searching for places to defecate, especially in areas where open defecation is common. It contributes to economic growth by allowing people to focus more on their work. Concerning educational benefits, proper school sanitation facilities can increase attendance, particularly for girls, stimulating long-term educational and economic improvements. Moreover, it helps prevent environmental degradation by reducing contamination of water bodies and soil, supporting agriculture and fisheries, making regions more attractive to tourists, and scaling up the local economy (Minh & Hung, 2011).

2.1.1 Types of sanitation systems

There are two main types of sanitation systems: onsite sanitation and offsite, also known as sewerage sanitation systems. Onsite sanitation includes technologies that treat human waste locally or store it temporarily. Local waste treatment systems, such as septic tanks, separate the wastewater into two parts: a liquid part called effluent, which flows out into a drain field or soak pit, and a thicker mixture of solids and liquids called fecal sludge, which remains in the tank. This fecal sludge must be periodically removed and transported to a treatment plant for safe processing. Those that store waste, known as container-based solutions, keep excrement in

specially designed containers and subsequently transport it to a facility for treatment. Offsite systems, which rely on sewer networks for conveyance, can be categorized into decentralized and centralized types. Decentralized are small-scale systems that rely on a sewer network to collect and transport wastewater from a limited area, such as a few blocks, to a local treatment plant. Centralized systems, in contrast, use extensive networks and pumping stations to convey wastewater to a centralized treatment plant for appropriate disposal (Ghangrekar, 2022). Tilley et al. (2014) further describe these systems briefly:

"Onsite Sanitation: *A sanitation system in which excreta and wastewater are collected and stored or treated on the plot where they are generated.*

Offsite Sanitation: *A sanitation system in which excreta and wastewater are collected and conveyed away from the plot where they are generated. An offsite sanitation system relies on a sewer technology for conveyance.*

Decentralized Wastewater Treatment System (DEWATS): *A small-scale system used to collect, treat, discharge, and/or reclaim wastewater from a small community or service area."*

(Tilley et al., 2014)

Deciding on the most suitable type of system for each area is primarily a matter of technical assessment; it means determining the local conditions and what type of system is feasible given these characteristics. For example, onsite systems depend on the relation between the infiltration capacity of the soil and the amount of effluent generated (commonly measured as surface area per population equivalent) to ensure proper operation, with separation of the liquid portion and the ability to store fecal sludge safely (Diaw et al., 2020; Gao et al., 2019). On the other hand, sewage collection systems depend on a minimum hydraulic flow to carry the wastewater away from the collection point, so areas with low water consumption are technically restricted to this type of system (Mara, 1996).

In many cases, both types of sanitation are technically feasible, so the decision as to which is the best solution must be evaluated, considering different criteria such as costs (of implementation and operation of different technologies) and socio-cultural aspects, which can either favor or make the implementation of certain systems unfeasible. Implementation costs are closely related to population density. For example, low-density areas are better suited to onsite systems, as the relation surface area per population equivalent favors soil infiltration, while a sewerage system would require a disproportionately high length of pipe per connection, driving up costs. In contrast, regions of high population density with a minimum of water consumption favor offsite systems due to the cost-benefit of the shorter connections per user. This gain in scale means that network systems become cost-competitive as population density increases (Mara, 2005; Nelson & Murray, 2008).

Estimating the costs of implementing a sanitation system in a generalized manner is challenging. However, Mara (2005) provides valuable insight into the variation in costs depending on the type of sanitation and technology applied and the significant influence of population density. Figure 1 demonstrates the substantial decrease in the cost of sewerage systems (conventional and simplified sewerage) as population density increases from 0 to 100 inhabitants per hectare.

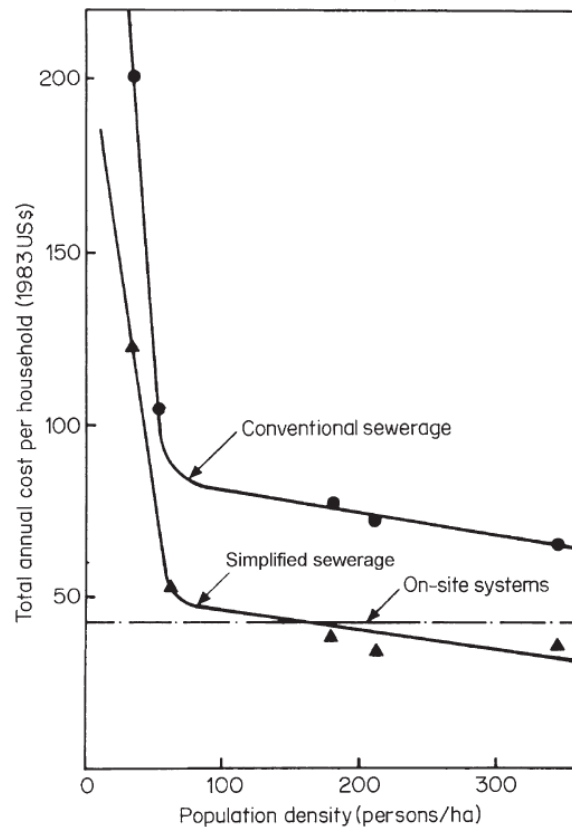


Figure 1 Annual costs per household (1983 USD) for conventional sewerage, simplified sewerage, and on-site sanitation, Natal, northeast Brazil (Sinnatamby, 1986)

Türker (2011) compares the costs of various onsite and offsite sanitation technologies worldwide and demonstrates that prices for the same technology can vary significantly. This variation is influenced by local factors such as soil type, topography, population density, and local materials and labor costs. As shown in Figure 2, the price ranges for onsite and offsite sanitation technologies often overlap.

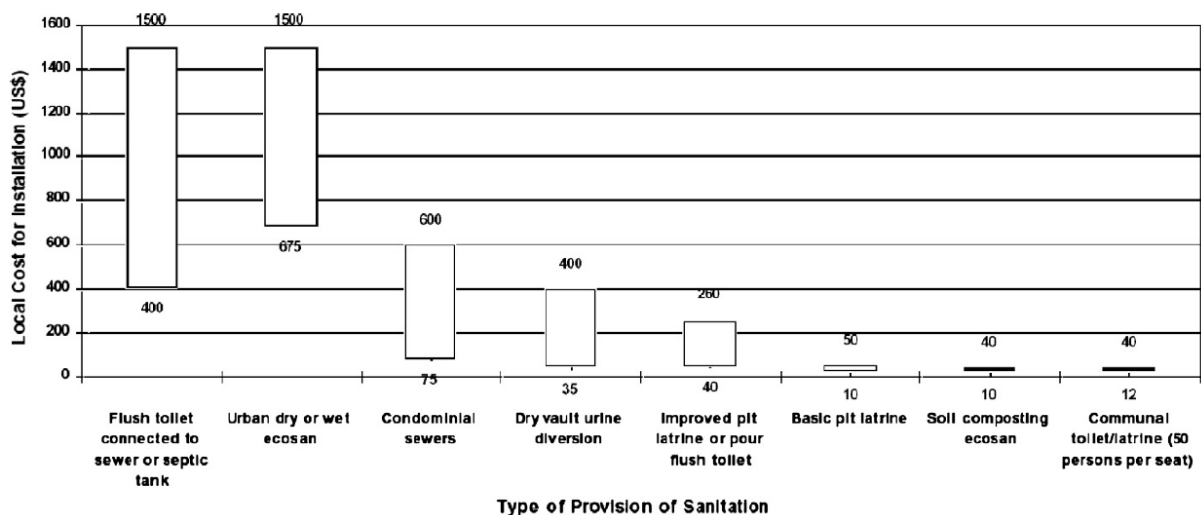


Figure 2 Unit cost of sanitation technologies (Türker, 2011)

2.1.2 Sanitation approaches and frameworks

Sewered sanitation systems are the most commonly adopted solution for urban areas in developed countries, collecting and transporting wastewater from residential areas to centralized wastewater treatment plants. They are efficient systems that guarantee safe management of collection and transportation, avoiding human contact with the wastewater and reducing the costs of managing high-density occupied areas. However, implementing sewerage solutions, without considering local characteristics, resulted in several unsuccessful sanitation projects worldwide, especially in developing countries, mainly caused by the cities' rapid and disorganized demographic growth. Many cities in the global south are characterized by a mix of formal and informal settlements, with the informal ones often marked by low-income households, lack of access to water and other basic infrastructures, disorganized occupation, and a variety of population densities (Gambrill et al., 2020).

The hardware component to bring safely managed sanitation to various settlements is well-known and applied in many developing countries. Technologies such as container-based sanitation, pits, septic tanks, vacuum trucks, conventional and simplified sewerage systems, decentralized and centralized fecal sludge, and wastewater treatment plants can be applied according to each location's demographic and physical characteristics. Even though the technology is available, integrating the different local contexts of the same city into a unique system is still a challenge (Parkinson et al., 2014; Schrecongost et al., 2020). In addition to local demographic, socioeconomic, and physical characteristics, lack of political will and financial and human resources are common barriers to the success of sanitation planning (Gambrill et al., 2020).

In response to the existing challenges, alternative frameworks have been formulated over the past decades, focusing not only on the hardware but also on the other aspects involving sanitation systems. For instance, the Sanitation21 framework was formulated in 2006 by the International Water Association (IWA) Task Group. Later, it was merged with the Community-Led Urban Environmental Sanitation (CLUES) planning guideline developed by the Department of Water and Sanitation in Developing Countries (Sandec) of the Swiss Federal Institute of Aquatic Science and Technology (Eawag) (Parkinson et al., 2014). In 2016, a series of discussion meetings resulted in the emergence of the Citywide Inclusive Sanitation (CWIS), which took advantage of lessons learned with the previous frameworks and project experience and incorporated some aspects of the SDG to create:

"An approach to urban sanitation, where all members of the City have equitable access to adequate and affordable improved sanitation services through appropriate systems of all scales (sewered & non-sewered), without any contamination to the environment along the entire sanitation value chain." (Narayan & Lüthi, 2020).

Although a more holistic view, prioritizing service delivery and considering social, environmental, economic, and organizational aspects potentially increases the feasibility of a sanitation system, it also increases the complexity of the planning process (Schrecongost et al., 2020). Furthermore, this multi-criteria evaluation may be considered for several technologies and different local contexts within the same city, which is common in developing countries, making the decision-making process even more arduous. Besides efforts to develop methods and tools to assist in different phases of sanitation planning coming from different researchers,

there is still no solution that encompasses the various possibilities of existing systems and technologies (Spuhler & Lüthi, 2020).

2.1.3 Sanitation planning process

The planning process of a sanitation system can be divided into different phases, from the early-stage planning phase to the implementation of the designed system. The name and detailed steps of each phase vary according to the region and the framework being adopted. Kvarnström & McConville (2007) evaluate three sanitation planning frameworks called Open Planning, Household-Centred Environmental Sanitation (HCES), and Sanitation21 and show that they share common steps besides being called and organized differently, as seen in Table 1. For instance, for both Open Planning and Sanitation21 frameworks, the identification of the technical options is underneath the second step, whereas for the HCES, it is done along the first and fourth steps.

Table 1 Common steps in the planning frameworks and specific recommended actions within each framework (Kvarnström & McConville, 2007)

Common Step	Open Planning	HCES	IWA Sanitation21
Recognition of Planning Domains	Identification of affected stakeholders	- Differentiate zones within urban environment - Problems should be solved close to their source	Identify key actors in each decision-making domain
Analysis of Objectives/Interests	- Stakeholder participation in problem identification - Terms of Requirement	Stakeholder participation in determination of needs and priorities	Identify the interests of key groups
Analysis of External Drivers	Identification of boundary conditions	- Assessment of current situation - Enabling environment	Understand external factors
Analysis of Technical Options	- Define sanitation system boundaries - Terms of Requirement	- Define system boundaries and current capacities - Identify a wider range of options	Analysis of existing systems and new systems
Assessment of Management Requirements	Terms of Requirement	- Assess current capacities and responsibilities of organizations - Need to ensure support from municipalities	- Identify the capacities within each domain for implementation and long-term management of a system - Identify the management requirements for the technical systems

Common Step	Open Planning	HCES	IWA Sanitation ²¹
Critical Assessment of Feasibility	Choice of most appropriate solution based on Terms of Requirement (Stakeholder participation)	Evaluation of feasibility of service combinations (Stakeholder participation)	- Assess whether systems meet the objectives in each domain - Assess whether systems can be managed in each domain

During the early-stage planning process, the study area must be divided into subareas based on similar characteristics and constraints. This helps to select the priority areas and narrow down the possible alternatives for each, facilitating the following steps of comparing viable solutions (Hammouri et al., 2022). In the context of this thesis, early-stage planning is the initial step in sanitation planning when there is no preliminary information regarding the area. The objective is to gather helpful information to select the most suitable type of sanitation system given each subarea of the city's characteristics, considering onsite and offsite sanitation, as discussed earlier.

Evaluating the alternatives of technologies is done in further steps of sanitation planning after the early-stage planning phase, and it is outside the scope of the current study. Effective early-stage planning successfully identifies the most suitable systems and benefits the subsequent phases by narrowing down the possible technologies for each subarea and generating helpful information and indicators. For example, it is expected to inform not only the number of inhabitants or households per subarea during the technology evaluation process but also socioeconomic indicators of low, middle, and high-income areas, estimation of land availability, and estimation of street and network length, among others. Adopting high-detailed with great data accuracy in early-stage planning is unnecessary because it will be refined in further steps. Still, it is essential to find a balance between the quality of the data and the effort and resources employed in this step.

2.1.4 Required data and challenges for early-stage sanitation planning

In the early-stage planning phase, which includes an initial assessment of the area for a sanitation system, resources are often limited. Acquiring the data required to evaluate the most suitable system, which depends on accurately characterizing the area, is challenging. Population density – emphasized in technical recommendations and guidelines as a primary factor – relies on census data that often lacks the necessary level of detail, is outdated, or is non-existent, particularly in developing countries. Along with demographic data, water availability, and average consumption, as well as the identification of areas with existing sanitation systems, are also critical and rely on local personnel's knowledge or field surveys (Kuffer et al., 2022; Ministry of Urban Development, Government of India, 2008). As discussed in Chapter 2.1.1, the significance of these primary data becomes evident when considering different local contexts. In the case of medium-density areas where no technical constraints limit sanitation options, additional factors like proximity to existing systems, street, and road accessibility, and the socioeconomic and cultural characteristics of the population must be considered to ensure that the solution effectively meets the community's needs (Mara, 2008; Ministry of Urban Development, Government of India, 2008).

Additionally to the population density, existing water supply and sanitation systems are considered critical data; additional information, when available, can further enrich the understanding of the area of interest and its surroundings. For instance, identifying informal settlements and low-income areas suggests exploring cost-effective, resilient, and adaptable sanitation solutions. Other infrastructure elements, such as paths, streets, and buildings, are closely related to the sanitation systems and might be adopted as complementary data (Mara, 2008; Nelson & Murray, 2008). Incorporating additional data in early-stage planning also facilitates later phases by enabling multi-criteria frameworks like Sanitation21 and CWIS; when combined with data collected in subsequent stages, this allows for the effective ranking and selection of the most suitable technology for each project subarea (Furlong et al., 2020; Hammouri et al., 2022; Ketema et al., 2014; Loetscher & Keller, 2002).

Based on existing recommendations and guidelines, the primary data considered in this research includes:

- Demographic data to determine population density.
- Delimitation of areas with existing sanitation systems and the type of system in use.
- Delimitation of areas with water availability and average water consumption per person.

Secondary, or additional, data includes:

- Identification of informal settlements.
- Identification of low-income areas.
- Mapping of roads and paths.
- Building footprints.

Another important consideration is that the classification of low, medium, and high population density areas varies significantly across the world. Demographic and local characteristics, such as the average number of people per family, the number of families per household, average building size, average building height, and the level of urban planning, significantly influence these classifications. For example, urban areas in Asian countries tend to have a higher average population density than other parts of the world, especially in developing countries (Angel et al., 2021). The Indonesian Ministry of Public Works uses the criteria shown in Table 2 to select the appropriate sanitation system based on population density (Kerstens et al., 2016).

Table 2 Type of sanitation system per population density

Type	Residential population density (people/ hectare)	Treatment systems (household/system)
Rural	Low (<100)	Onsite (1 household/system)
	High (>100)	Community Based ¹ (50–100 households/system)
Urban	Low (<100)	Onsite (1 household/system)
	Medium (100–250)	DEWATS (500–5,000 households/system)
	High (>250)	Offsite (10,000–50,000 households/system)

Note. Adapted from (Kerstens et al., 2016).

¹ The Community based is a DEWATS consisted by a simplified sewer system and an Anaerobic Baffled Reactor combined with a Filter (Kerstens et al., 2015).

Kabange and Graham (2017) conducted a study to evaluate how population density affects sanitation costs in the Kotoko community of Suame in Kumasi, Ghana, a low-income and high-density peri-urban area. For the study case area, they concluded that at a population density of 160 inhabitants per hectare, the sewerage solution would be more cost-effective than the onsite solution. The study compared ventilated improved pit (VIP) latrines with simplified sewerage, which are considered low-cost technologies. A similar study was performed by Manga et al. (2020) in the informal settlements of Soweto, Johannesburg, South Africa. This study compared the simplified sewerage with two onsite technologies: urine diversion dry toilets (UDDT) and VIP latrines. It also evaluated the cost-effectiveness of different scenarios for the sewerage systems, including connecting to an existing network and considering the treatment plant costs with and without pumping stations. In this case, the population densities at which sewerage becomes more cost-effective than onsite technologies ranged from 112 to 198 inhabitants per hectare. Both studies included capital and operational costs in their analysis.

The population density values where offsite sanitation is recommended in the abovementioned studies are considerably lower than those presented in Table 2, adopted in Indonesia. Furthermore, no studies have compared the costs of simplified sewerage systems with more expensive but still commonly adopted technologies in Latin America, such as septic tanks. Such a comparison might reveal an even lower population density threshold where low-cost sewerage systems become more cost-effective than medium and high-cost onsite technologies, representing the range where both onsite and offsite sanitation are feasible solutions, requiring further evaluation of other factors, as mentioned in Chapter 2.1.1.

2.2 Spatial data

Open spatial data has become increasingly important in recent years, mainly because of technological advances, the growing number of open data initiatives, and the effort of the Open Geospatial Consortium (OGC) to create the standards. By making geographic information freely available, scientists, professionals, and decision-makers can better work with it and create innovation across diverse sectors, including sanitation, urban planning, and environmental management (Ye, 2018). At the beginning of the 2000s, the rise of open-source Geographic Information System (GIS) software popularized and facilitated spatial data analysis that had already been used decades earlier (GRASS GIS, n.d.).

The OGC is a not-for-profit organization with representatives from government, industry, nonprofits, and academia. It has 30 years of existence and works on standardizing spatial data formats and protocols to guarantee they can be used across different software and platforms. It facilitates the sharing and integration of geographic information, making spatial data utilization more effective. There are several types of spatial data, each encompassing multiple formats. In this thesis, two types of data will be utilized: raster and vector data. Raster data consists of pixels, where the size, known as resolution, varies by data source. Each pixel holds values, representing information such as temperature or elevation. Satellite and aerial images are examples of raster data, where each pixel contains several band values rendered into colors. Vector data, on the other hand, represents geographic features using points, lines, and polygons, which are well-suited for discrete data like locations, boundaries, and infrastructure. Each vector feature can contain attributes with information relevant to its specific context (Minghini et al., 2020).

2.2.1 Importance of spatial data for sanitation planning

Spatial data is widely employed in multiple fields, from environmental management and urban planning to emergency responses like humanitarian crises. Spatialized information brings an extra dimension of analysis that makes it possible to relate each spatial element to the surrounding elements and to analyze, simulate, predict, and visualize relationships that are not possible when using non-spatialized data. The implementation of GIS in the water and sanitation sector has been growing in the past decades, driven by technological advances and the increasing accessibility of tools and data that, lately, were resource-intensive (Hammouri et al., 2022). There are several examples of adopting spatial data and analysis in diverse sanitation scopes; the majority focused on monitoring, assessment, and operation and maintenance (De Moura & Procopiuck, 2020). However, the adoption of GIS tools and spatial data, specifically during the sanitation systems planning process, is still lacking compared to other infrastructure sectors (Kerstens et al., 2016).

A comprehensive review of existing frameworks and their outcomes related to GIS was conducted by Kerstens et al. (2016). Between the evaluated frameworks are the previously presented Sanitation21, and the Community-Led Total Sanitation, among others, that focused on different aspects of sanitation like financial models, risk, and service delivery assessments. It is highlighted that only the “GIS-based sanitation analysis” of the nine evaluated sanitation planning frameworks results in a GIS-based output. However, despite integrating GIS, this framework is not ready for adoption using different criteria. Furthermore, a new framework has been developed and adopted for strategic planning and allocating investment in sanitation at the national level in Indonesia. The developed framework was applied in this case study using proprietary GIS software and spatial data from multiple sources, including the United Nations population dataset, Google, and the Joint Monitoring Program supported by the World Health

Organization in partnership with UNICEF. Still, the framework is not publicly available for further evaluation and reproduction in different locations.

Using spatial data and analysis on sanitation planning offers several advantages that enhance decision-making, as shown. Gambrill et al. (2020) state that the local aspects of a given area strongly influence the selection of the most suitable sanitation system and adoption of geographic information, such as the spatial distribution of the population density and socioeconomic data, the register of available infrastructures, the delimitation of areas with water and sanitation services, among other information, helps visualize the project's area in its integrality, enhancing decision-making. In addition, the manual delimitation of the subareas is facilitated by the possibility of using aerial and satellite images, which are globally available nowadays, to support the more precise demarcation of their boundaries (Kumar et al., 2023).

A crucial aspect of geographic information is the unit or resolution of the data. A standard method to generate global population data is processing several different data sources and types using algorithms and storing the result as the estimated number of people living in a grid of a given unit if using vector format or resolution if using a raster format. The grid size can vary depending on the dataset. However, for example, the Gridded Population of the World v4 provides the number of inhabitants globally distributed in square grids with 1-kilometer sides (Data & Center (SEDAC), n.d.). The boundaries of sanitation projects usually do not coincide with the artificial boundaries, such as cities and districts, or even smaller divisions, such as neighborhoods or census blocks. Official government data, such as population and socioeconomic information, is usually available at the level of census tracts or census blocks. Practically, this can range from data representing a set of neighborhoods to that of an individual neighborhood. In many developing countries, this data is often only available at the municipality level, corresponding to a census county level, and does not accurately reflect the population's distribution within the municipality. In contrast, granular gridded population distribution data provides a much more precise representation of the population's distribution, as illustrated in Figure 3 (Hammouri et al., 2022). The level of aggregation of census-published data can be limited for several reasons: to restrict sensitive data that could be used to identify information of a given household to the lack of resources or capacity to survey and process fine-granulated information (United Nations, 2017).

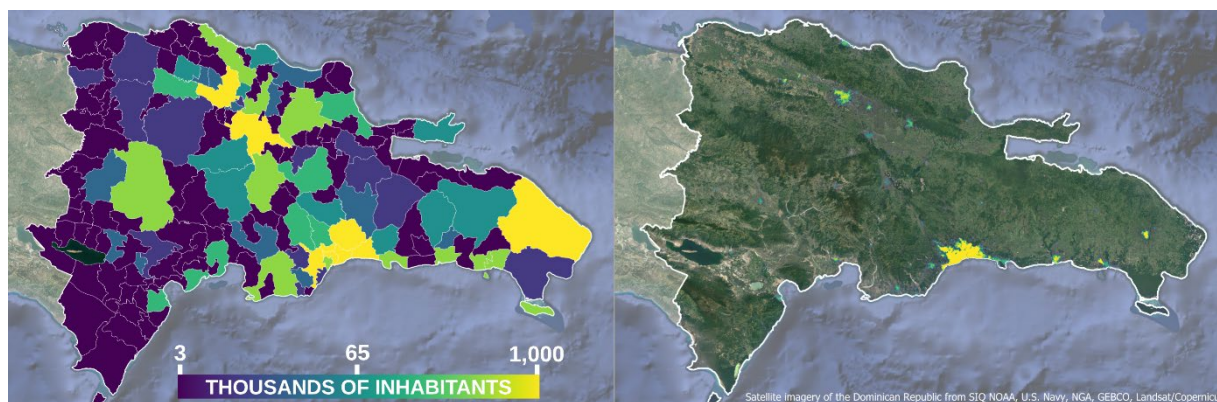


Figure 3 Comparison of population data: Municipality level (left, ONE-RD (Geoportal, n.d.)) vs. 100-meter grid level (right, WorldPop & Center for International Earth Science Information Network (CIESIN), 2018).

Even when population data is available at the neighborhood level, it only provides an average for the entire area, failing to distinguish between sparsely and densely populated sections within that neighborhood, as illustrated in Figure 4. This can hamper sanitation projects because sewered sanitation and decentralized systems, primarily gravity-based networks, have boundaries limited by the watershed in which they are located. Due to the specific nature of the

sanitation sector, it is beneficial to use granular population density data at the unit of a block or group of blocks to ensure that the population of each subarea is accurately represented. This finer resolution allows for more precise planning and better alignment with the actual needs of the area (Pajares et al., 2021).

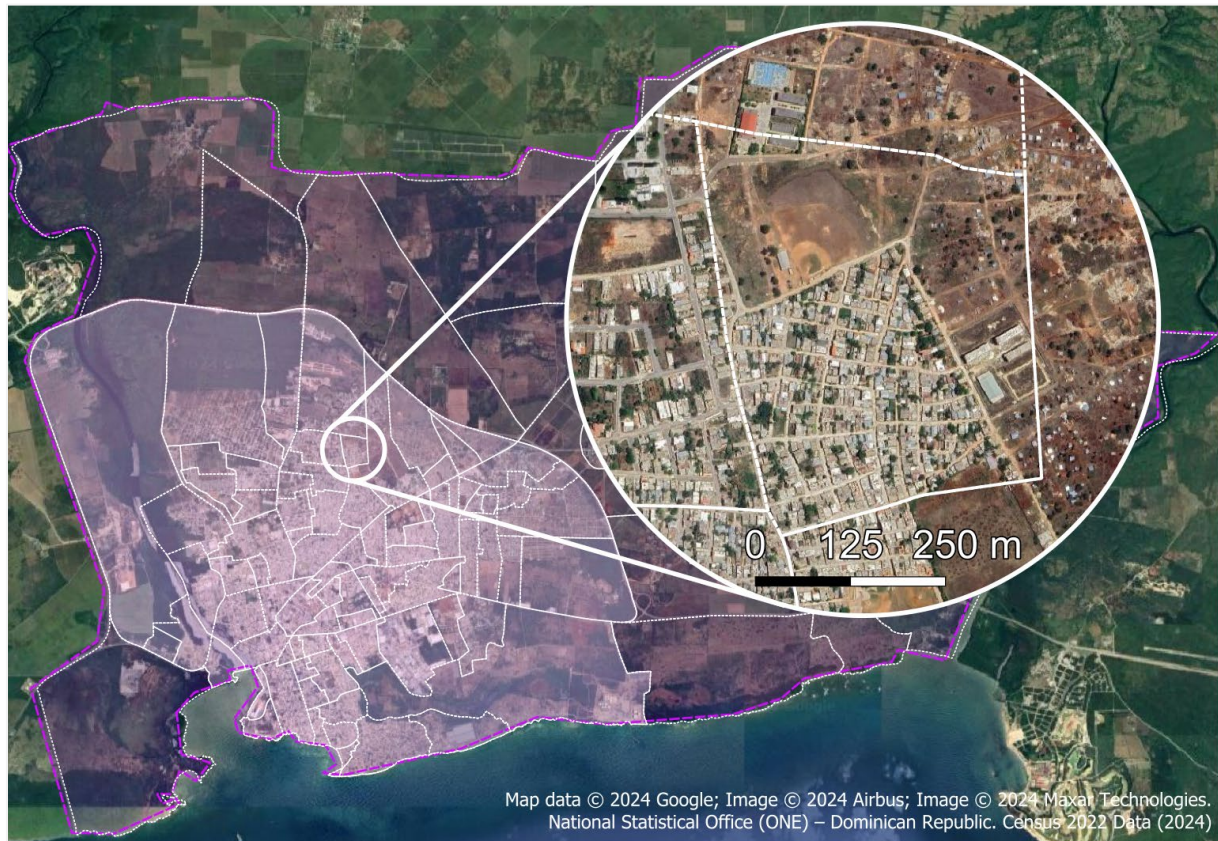


Figure 4 Neighborhood divisions (white lines) in San Pedro Macoris, Dominican Republic, highlighting varying types of occupation and population density within a specific neighborhood.

2.2.2 Open-access data

As supported by previous discussions, the need for more data and information is a well-known problem in developing countries and a barrier to increasing sanitation coverage worldwide. Primary data, such as population density, are usually available only at city or district levels in the global south and are typically outdated. This makes early-stage planning of sanitation systems in these regions even more challenging due to the difficulty of adequately characterizing the project areas. There is a demand for alternative sources of information with sufficient granularity and quality to fill these data gaps and improve the planning and implementation of sanitation systems.

Over the past decade, several initiatives have strived to produce new open-access datasets, driven by the growing availability of data and advancements in technology that enable the processing of significant amounts of information. These datasets are primarily produced by universities that contribute scientific knowledge and innovative models in collaboration with multilateral organizations and large companies that provide modern equipment with advanced processing capabilities. Most of these initiatives are focused on addressing data gaps, particularly in developing countries (Tatem, 2017). Because of the involvement of research

groups in producing these datasets, many studies have also been published as open science, enabling the reproduction and enhancement of the models. For instance, various global gridded population datasets have experienced refinements by incorporating new methodologies or updated data sources, resulting in an improved version of the previous ("High Resolution Settlement Layer (HRSL)," 2016).

Because of the substantial amount of existing data and the rapid arrival of new data, it is important to ensure that it follows specific standards to make it accessible (Blatt, 2016). The Findability, Accessibility, Interoperability, and Reusability (FAIR) principles are recommended for leveraging the effectiveness of open data ("The FAIR Principles," n.d.). The findability aspect applies to data and metadata, where metadata is a detailed explanation of the data and how that assists in using it effectively. Accessibility ensures that everyone can easily find and obtain data. Interoperability means different systems and software can work with the data, especially open-source software. Reusability ensures that data is provided under licenses, such as open or Creative Commons licenses, allowing it to be used in other projects. The OGC Standards discussed previously follow the FAIR principles regarding the data format and transfer standards. Standardized data transfer protocols are essential for facilitating communication between computer programs. Application Programming Interfaces (APIs) provide a set of specifications and protocols that rule how programs interact to exchange information and data. Web Services (WMS) and Web Feature Services (WFS) are specific API implementations designed for geospatial data access and visualization, with the advantage of easing access using a regular website address, not requiring technical skills. The OGC has specific standards for each method (Chester, 2023).

The adoption of spatial data in the sanitation sector is concentrated on operation and maintenance, with few examples of the adoption in early-stage sanitation planning, as discussed previously. Moreover, the few existing early-stage planning using spatial data adopt datasets specific to the area of the project, not open-access spatial data that follows a global standard, which would ease the reproduction of the steps in other parts of the world. In contrast, there are some examples of the energy sector adopting open-access spatial data to overcome the lack of data in developing countries. Cader et al. (2018) conducted a case study in Tanzania, demonstrating that the quality of open spatial data enabled a finer level of granularity in the spatial distribution of the population, highlighting the potential of open-access data for electrification planning. Another interesting project from the energy sector is the Energy Access Explorer (EAE) Platform, which analyzes energy access worldwide using open data. The platform uses population and socioeconomic spatial data to identify areas with high potential demand for electricity expansion in developing countries (Walter et al., 2023).

2.2.3 Open-source GIS software

The term GIS can be defined differently, but a simple form is "a system for capturing, storing, checking, manipulating, analyzing and displaying data which are spatially referenced to the Earth," as identified by Maguire (1991). These are essential tools for working with spatial data and generating the expected outcomes. Local governments frequently rely on external consultants to conduct GIS analysis due to lacking local expertise (Gondhalekar et al., 2013). However, it conflicts with the recommendation to minimize the influence of external consultants, especially during the initial stages of sanitation planning, avoiding the bias for a specific sanitation system and ensuring solutions are tailored to the local context (Spuhler & Lüthi, 2020; Von Dobschütz & Wafler, n.d.).

The history of open-source GIS software comes from the beginning of the 1980s when the U.S. Army Corps of Engineers Construction Engineering Research Laboratory (USA/CERL) started exploring GIS for several purposes and developed the first version of the GRASS GIS. In the beginning, GRASS GIS was developed under the Public Domain license and converted into a General Public License in 1999 (“GRASS GIS History,” n.d.). In the early 2000s, some open-source GIS desktop software emerged. QGIS stands out for its growing and active community, continuous improvements, and the ability to integrate other software, such as GRASS, providing a complete GIS solution (Seibel, 2023).

The emergence of diverse open-source GIS software has significantly lowered the barriers to accessing advanced spatial data tools. With the rise of several initiatives providing reliable open-access spatial data at a global level and with an appropriate level of granularity, as previously presented, the adoption of open-source GIS software could fill the evidenced gap in adopting this data on early-stage sanitation planning. Adopting these novel resources democratizes the implementation of the most advanced methodologies, empowering local communities to develop more effective and context-specific sanitation solutions.

Chapter 3 Methodology

The proposed methodology aims to explore, prepare, and use open-access spatial data to bridge the lack of available data for early-stage sanitation planning, especially in developing countries. To make all steps of the process accessible and reproducible, the project uses QGIS (QGIS Development Team, 2024), an open-source GIS software, with each step comprehensively explained.

3.1 Cataloging open-access spatial data for sanitation planning

The first step of the research is to identify and catalog open-access spatial datasets considered essential for early-stage sanitation planning. This methodology involves specifying the types of datasets needed, outlining the criteria for their selection, and defining the structure for describing the selected data.

3.1.1 Criteria for selection of the datasets

The following criteria were adopted during the dataset selection to ensure they can be effectively adopted in early-stage sanitation planning:

1. **Open-access:** The datasets must be freely accessible to the public.
2. **License:** The datasets should be provided under open licenses, such as Creative Commons licenses, which permit free use, distribution, and modification. This ensures the data can be legally copied, reused, and shared.
3. **OGC standard format:** The data format must comply with the Open Geospatial Consortium (OGC) standards, ensuring compatibility and interoperability across various GIS software and platforms.
4. **Resolution (granularity):** As discussed previously, the granularity of the datasets is essential to ensure a good characterization of the subareas. It does not apply to infrastructure datasets. For demographic data, the resolution should be at a block or group of blocks level, with a maximum of 100 meters for gridded population data.
5. **Global availability:** The datasets must be available globally, particularly in developing countries, to ensure the benefit of areas most impacted by the lack of data.

Besides that, the data were evaluated using FAIR (findability, accessibility, interoperability, and reusability) principles as a guide. Some aspects of the principle considered in the above criteria, such as interoperability and reusability, are ensured by adopting the OGC Standards. However, some aspects, like the quality of metadata and the documentation of the dataset, were not considered an excluded criterion.

3.1.2 Primary spatial datasets – Gridded population

As discussed previously, demographic open-access spatial datasets are essential for enhancing early-stage sanitation planning. These datasets have two significant advantages over traditional census data: they can be updated more frequently, and the information's granularity allows planners to understand better the population's distribution in the area of interest. This avoids the problem known as the modifiable area unit problem (Openshaw, 1984), which occurs when data is aggregated over a large area, a common issue with official government data. Many open-access gridded population datasets have emerged, allowing researchers, planners, and government agencies to implement them in various applications, such as economic assessments, infrastructure planning, combating vulnerability, and humanitarian projects.

There are two main approaches to creating the gridded population dataset: top-down and bottom-up; the present research is focused on the top-down because most of the globally available population datasets adopt it. In this method, population counts from census data are broken down into smaller grid cells across a region. Some simple top-down methods assume that the population is spread evenly across these cells, while more advanced methods combine census data with additional information, like land cover or night-time lights, to estimate better where people are likely to live; these methods are called dasymetric models. Constrained population datasets, a subset of top-down methods, limit population distribution only to grid cells where buildings are detected using building footprints or settlement data from satellite imagery. This leads to more concentrated population distributions in areas with human activity. The final dataset quality depends on both the census data and additional data used, and inaccuracies in any of these inputs will affect the accuracy of the gridded population (Hierink et al., 2022; Kuffer et al., 2022).

Due to the premises of the current research, the selection of population spatial datasets was limited to open-access datasets with global coverage and a resolution of at least 100 meters. The four selected datasets presented below are not an exhaustive list; others might be included in the future. The methodology proposed in this research is flexible enough to work with different data types, resolutions, and standards of information stored.

3.1.2.1 *WorldPop*

The WorldPop dataset, developed by the School of Geography and Environmental Sciences at the University of Southampton in the United Kingdom, provides several different datasets, including two top-down generated models adopted in the current research: the constrained and unconstrained. The methodology developed for the *WorldPop* research group adopts Random Forest to create a semi-automated dasymetric model, as explained in detail by Stevens et al. (2015). In this research, the version of *WorldPop* datasets was adopted and adjusted to United Nations population estimates. Tables 3 and 4 show the specifications for *WorldPop* constrained and unconstrained datasets, respectively.

Table 3 Specifications of WorldPop constrained dataset

WorldPop constrained	
Specifications	Description
Type of the data	Raster
Format of the data	GeoTIFF
Year of dataset data	2020
Resolution of the dataset	100 meters
Data coverage	Available globally
Official data source	https://hub.worldpop.org/geodata/listing?id=79
Official API	https://www.worldpop.org/sdi/introapi/
License	Creative Commons Attribution 4.0 International (CC BY 4.0) license

Table 4 Specifications of WorldPop unconstrained dataset

WorldPop unconstrained	
Specifications	Description
Type of the data	Raster
Format of the data	GeoTIFF
Year of dataset data	2020
Resolution of the dataset	100 meters
Data coverage	Available globally
Official data source	https://hub.worldpop.org/geodata/listing?id=69
Official API	https://www.worldpop.org/sdi/introapi/
License	Creative Commons Attribution 4.0 International (CC BY 4.0) license

3.1.2.2 High-Resolution Settlement Layer (HRSL)

The HRSL dataset was generated using a model that employs machine learning techniques to detect buildings from high-resolution satellite images (sub-meter resolution) and redistribute the population based on these building footprints. The population data is then allocated proportionally to the identified buildings within the census units, resulting in a population map with a resolution of approximately 30x30 meters (Tiecke et al., 2017). Table 5 shows the specifications for the *HRSL* dataset.

Table 5 Specifications of the HRSL dataset

High-Resolution Settlement Layer (HRSL)	
Specifications	Description
Type of the data	Raster
Format of the data	GeoTIFF and Comma-separated values (CSV)
Year of dataset data	2020
Resolution of the dataset	30 meters
Data coverage	Available globally
Official data source	https://dataforgood.facebook.com/dfg/docs/high-resolution-population-density-maps-demographic-estimates-documentation
Alternative data source	https://gee-community-catalog.org/projects/hrsl/
License	Creative Commons Attribution 4.0 International (CC BY 4.0) license

3.1.2.3 Global Human Settlement Layer (GHSL) - Copernicus

The methodology for creating the Global Human Settlement Layer (GHSL) population grids uses Sentinel-2 imagery and Landsat high-resolution satellite data (10m and 30m) to map the global distribution and density of built-up areas. This process resulted in a global 100-meter resolution population grid from 1975 to 2030 in 5-year intervals (European Commission. Joint Research Centre., 2023). Table 6 shows the specifications for the *GHSL* Copernicus dataset.

Table 6 Specifications of the GHSL Copernicus dataset

GHSL Copernicus	
Specifications	Description
Type of the data	Raster
Format of the data	GeoTIFF
Year of dataset used	2020
Resolution of the dataset	100 meters
Data coverage	Available globally
Official data source	https://data.jrc.ec.europa.eu/dataset/2ff68a52-5b5b-4a22-8f40-c41da8332cfe#dataaccess
Alternative data source	https://gee-community-catalog.org/projects/ghsl/
License	Creative Commons Attribution 4.0 International (CC BY 4.0) license

3.1.3 Secondary dataset – Streets and building footprints

Although not essential, infrastructure datasets can offer valuable insights during the early-stage planning phase and become increasingly important in subsequent stages of sanitation planning, particularly in technology selection and cost evaluation. For the proposed methodology of this research, particularly the datasets of street registers, and building footprints might be helpful.

Street and path data can assist in identifying informal settlements, often characterized by irregular street patterns, and assess the feasibility of specific sanitation systems that require access to effluent collection and transport. Building footprint data can be used to evaluate building density, which sometimes serves as a proxy for population density. These datasets are typically available in vector format, and two different datasets were selected, as presented in the following.

3.1.3.1 OpenStreetMap

OpenStreetMap is a famous collaborative mapping initiative launched in 2004 in the United Kingdom. It has grown exponentially since then, with more than two million contributors worldwide and a community that is still growing. Many large companies rely on *OpenStreetMap* data, and the available data covers streets, paths, roads, points of interest (such as commercial, governmental, and administrative buildings), rail trails, powerlines, and building footprints, among others. A recent evaluation of the building footprint data conducted by Zhou et al. (2022) found that cities in Europe and Africa have the most mapped buildings for this dataset. Table 7 shows the specifications for the *OpenStreetMap* dataset.

Table 7 Specifications of OpenStreetMap dataset

OpenStreetMap	
Specifications	Description
Type of the data	Vector
Format of the data	GeoJSON, CSV, shapefile, among others
Year of dataset	Not applied, the data is constantly updated
Data coverage	Available globally
Official data source	https://www.openstreetmap.org/
Alternative data source	https://download.geofabrik.de/
License	Open Data Commons Open Database License (ODbL)

3.1.3.2 Overture Maps Foundation

The *Overture Maps Foundation* is a recent initiative founded in 2022 that claims to join forces to create “reliable, easy-to-use, and interoperable open map data” (Overture Maps Foundation, 2024). The initiative includes companies like Microsoft, Amazon, Meta, and the Humanitarian *OpenStreetMap* Team (HOT). One of *Overture Maps'* goals is to consolidate various datasets into a single, unified dataset, create unique identifiers for better navigability, and adopt *GeoParquet*, a modern data format that optimizes the querying and analysis of large datasets. One of its advantages is filling gaps in mapped infrastructure where *OpenStreetMap* data is lacking by incorporating other available datasets. For example, in areas with limited *OpenStreetMap* building footprint data, it combines data from Microsoft Building Footprints and Google Open Buildings. Table 8 shows the specifications for the *Overture Maps* dataset.

Table 8 Specifications of Overture Map dataset

Overture Maps	
Specifications	Description
Type of the data	Vector
Format of the data	GeoParquet
Year of dataset	Not applied, the data is updated monthly
Data coverage	Available globally
Official data source	https://docs.overturemaps.org/getting-data/
Alternative data source	https://github.com/overturemaps/overturemaps-py
License	Community Database License Agreement – Permissive v2 (CDLA)

3.2 Data analysis and processing workflow

The analysis and pre-processing of each dataset acquired for the area of interest were conducted using QGIS, an open-source GIS software. As recommended, the Long-Term Release (LTR) version was adopted to ensure stability. Several scripts and a user-friendly interface were developed within QGIS to streamline the analysis and pre-processing, utilizing tools like the Model Designer and PyQGIS, the Python API for creating scripts and plugins available in the software. The objective is to create tools that make the entire process more accessible for individuals without advanced GIS software skills. The workflow was divided into sub-processes that align with the steps of a typical sanitation planning process, starting with delimiting the project area, identifying local characteristics (attributes), inputting data and parameters, and generating the desired outcomes. The steps, parameters, and calculations are comprehensively detailed to ensure the process can be replicated using other GIS software if needed.

One central aspect of the workflow is the three types of polygons that help organize the spatial data and ensure that all necessary attributes (data and information) are considered for each subarea. The first type of polygon is the Area of Interest (AOI) Polygon, which is the delimitation of the project area and the boundaries where the spatial data must be clipped in case the spatial dataset is more extensive than the area of interest. The second type is the Auxiliary Polygons, a series of different layers that indicate the boundaries of essential aspects of the planning. The third is the Subarea Polygons, the smallest areas automatically generated during the process. These consolidate the spatial data, calculate indicators, and suggest the most suitable sanitation system for each subarea based on criteria that will be further detailed in subsequent sections. Figure 5 illustrates the three types of polygons.

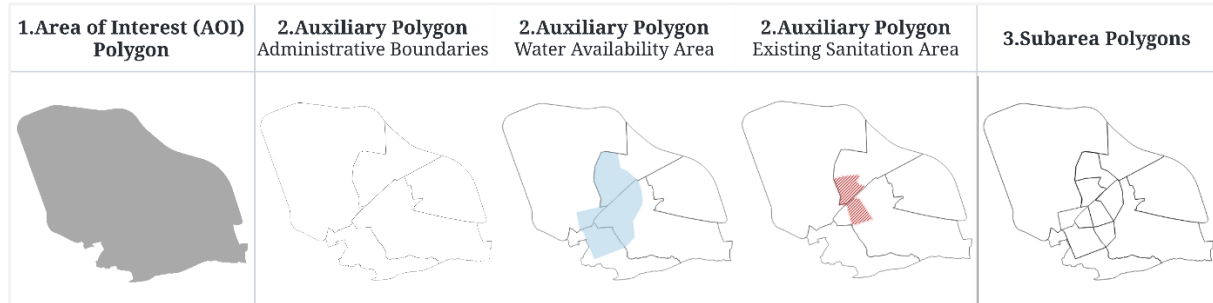


Figure 5 Three Types of polygons adopted in the workflow

Figure 6 illustrates a flowchart of the proposed workflow.

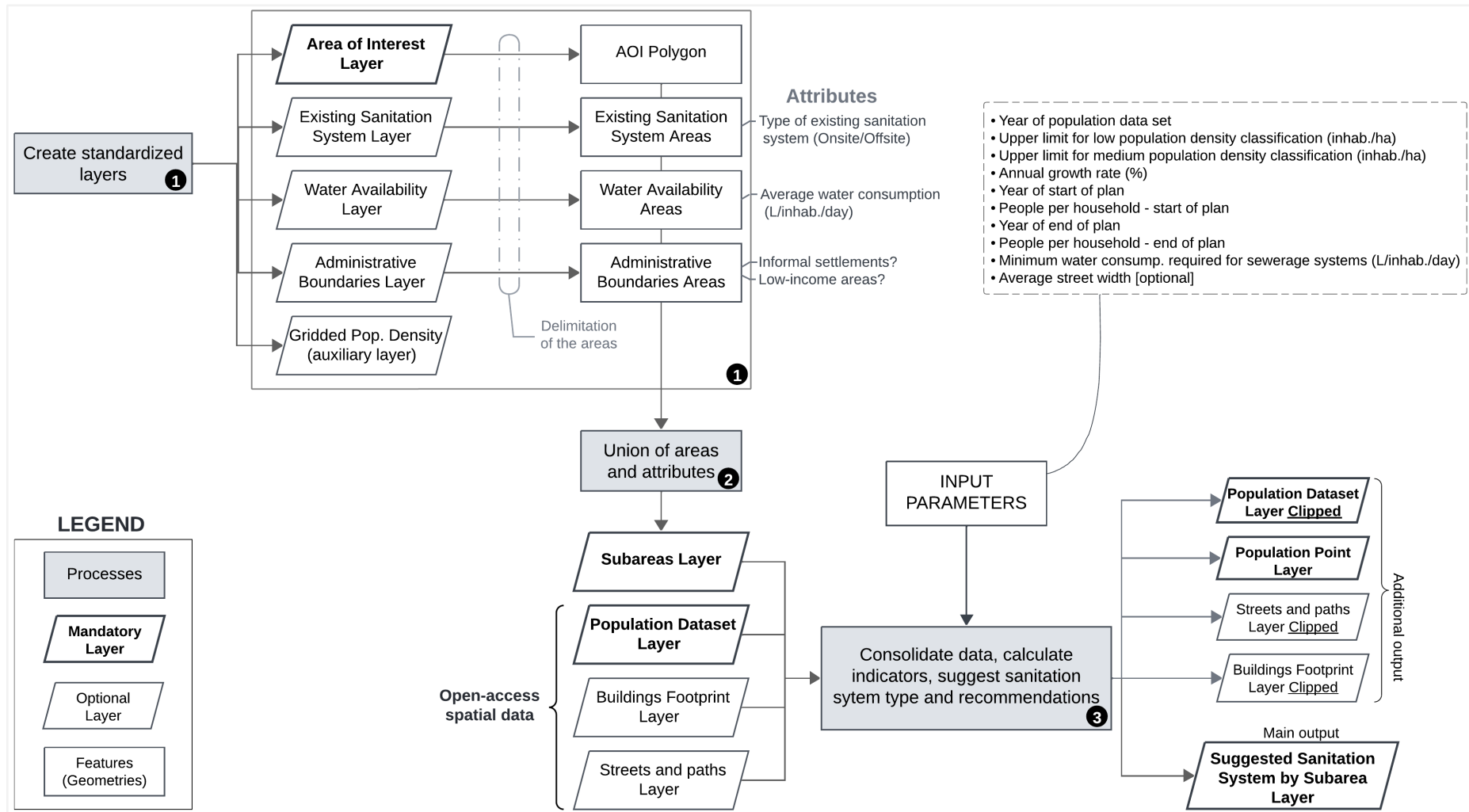


Figure 6 Flowchart of the workflow for the steps of analysis and pre-processing of the spatial data

As shown in the flowchart above, the entire process is divided into three sub-processes to ensure that all mandatory information and data are accurately provided and readily available. Additionally, each sub-process includes multiple exception-handling and error-checking mechanisms to prevent errors during the processes.

1. Create standardized layers (Create Base Layers)

This process generates vector layers representing the Auxiliary Polygons that the planner utilizes to delineate the project areas, input required, and additional information based on local characteristics. The standardized layers incorporate automatizations to facilitate the delimitation of the areas and ensure the planner is informed of the mandatory data. In addition, attributes are configured with the correct data types (Boolean, numerical, string) and appropriate attribute values.

Area of Interest (AOI) Layer – Used to delimit the project area, ensuring all information and data are clipped to this boundary. Only data within the AOI polygon are considered during processing. This layer includes the following attributes:

- **uuid** – A string attribute containing the unique identification of the polygons.

Existing Sanitation Systems Layer – Used to delimit the areas (polygons) with an existing sanitation system. Only areas where the sanitation system is safely managed should be included. This layer includes the following attributes:

- **uuid** – A string attribute containing the unique identification of the polygons.
- **ExistingSanitation** – A Boolean attribute indicating whether a sanitation system exists in the area.
- **SanitationType** – A string attribute specifying the type of sanitation system (onsite or offsite).

Water Availability Layer – This layer is utilized to delimit the areas with an existing water supply system. This layer includes the following attributes:

- **uuid** – A string attribute with the unique identification of the polygons.
- **WaterAvailable** – A Boolean attribute indicating whether water is available in the area.
- **AverageWaterConsumption** – A numerical attribute to inform the average water consumption in the respective area. The average consumption must be informed in liters per inhabitant per day.

Administrative Boundaries Layers – Adopted to delimit official or unofficial boundaries based on administrative divisions, roads, avenues, waterbodies, etc. This layer includes the following attributes:

- **uuid** – A string attribute with the unique identification of the polygons.
- **InformalSettlement** – A Boolean attribute indicating areas where the occupation is informal.
- **LowIncomeArea** – A Boolean attribute indicating low-income areas.

Population Density Grid (optional) – Assist in defining administrative boundaries by generating a grid (either rectangular or hexagonal) representing population density for a specified year (target year) using a linear growth projection. It requires the selection of a population dataset and the entry of parameters:

- **id** – An integer attribute with the unique identification of the grids.

- *PopDataset_sum* - Total population of the subarea from the population dataset (inhabitants).
- *PARAM_AnnualGrowthRate* - Annual growth rate (%).
- *YEAR_popdensity* - Year of the population of the plan.
- *population* - Total population at the target year (inhabitants).
- *pop_density* - Population density for the target year of the plan. (inhab./ha).

The planner can choose which of the above layers to generate, with the AOI Layer being the only mandatory layer as it defines the project area for subsequent steps. While the other layers are optional, providing the mandatory information associated with them, such as water availability and average water consumption, is crucial before proceeding to the third process, ensuring a consistent result. After running the second process (Union of Areas and Attributes), these inputs can be added, as the resulting subarea polygons will include all the attributes presented above.

All steps for delineating polygons and inputting the required data are performed using QGIS native tools, such as "Add Polygon Feature" and the attribute forms and tables. If available, the planner can adopt existing layers containing the required information. To do this, the features (polygons) from the existing layer can be selected, copied, and pasted into the desired base layer. For example, consider that the planner has a layer that identifies areas with a sanitation system. In that case, they can copy and paste these areas into the newly created layer for existing sanitation systems. After copying the features, the planner should verify the attributes of the pasted features to ensure that all required information is correctly filled in.

2. Union of Areas and Attributes (Delimitation of Subareas)

This process generates a layer with subareas, the smallest polygons resulting from the intersection of the layers created in the previous step. Each subarea inherits the attributes from its parent layers defined earlier. As mentioned, the only mandatory layer is the AOI Layer, while the other three are optional. Figure 7 illustrates the subarea polygons generated from two different combinations of input layers.

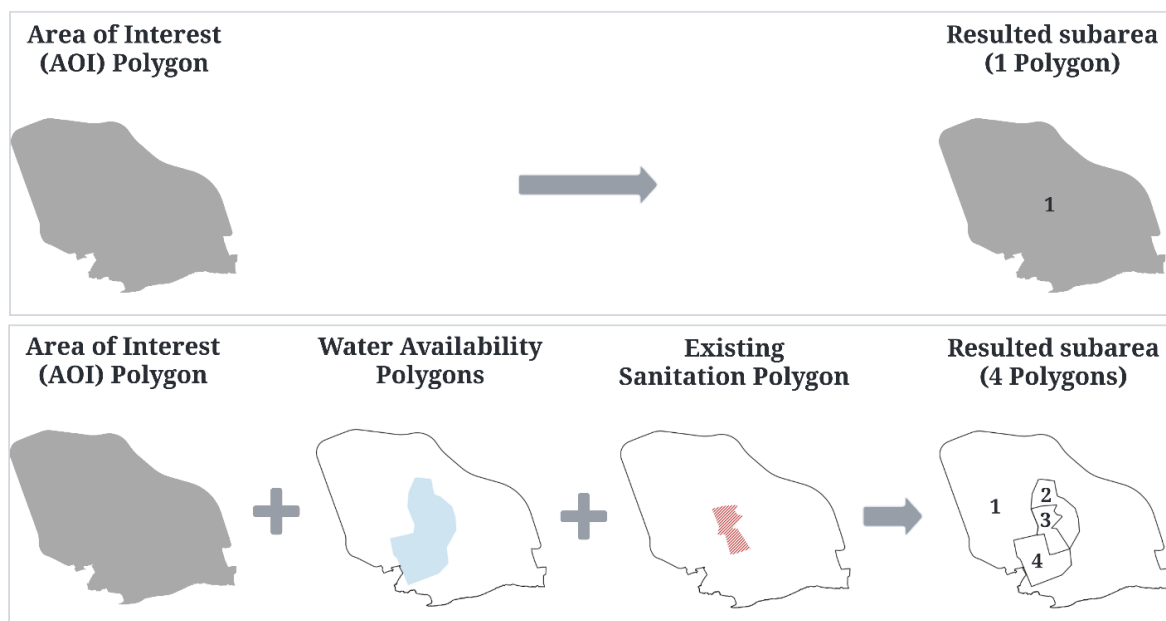


Figure 7 Examples of subarea polygons generated from different combinations of input layers

Independent of the set of input layers adopted, the resulting subarea layer always has all the following attributes fields:

- **uuid** – A string attribute containing the unique identification of the polygons.
- **area_ha** – A numerical attribute with the area in hectares of the polygons.
- **ExistingSanitation** – A Boolean attribute indicating whether a sanitation system exists in the area.
- **SanitationType** – A string attribute specifying the type of sanitation system (onsite or offsite).
- **WaterAvailable** – A Boolean attribute indicating whether water is available in the area.
- **AverageWaterConsumption** – A numerical attribute to inform the average water consumption in the respective area. The average consumption must be informed in liters per inhabitant per day.
- **InformalSettlement** – A Boolean attribute indicating areas where the occupation is informal.
- **LowIncomeArea** – A Boolean attribute indicating low-income areas.

When no parent layer is selected, the subareas layers will populate their respective attributes with no value. For example, in Figure 7, the subarea polygons identified with the number 2 have values for the ‘ExistingSanitation,’ ‘SanitationType,’ ‘WaterAvailable,’ and ‘AverageWaterConsumption’ attributes, but the ‘InformalSettlement’ and ‘LowIncomeArea’ attributes remain empty.

The subareas can be manually edited after generation. This includes geometric adjustments, such as subdividing or merging subareas, as well as modifying attributes. For instance, the average water consumption for a specific area can be updated if new information becomes available or if no Water Availability Layer was provided in the first step.

3. Consolidate data, calculate indicators, suggest sanitation type and recommendations (Sanitation System Evaluation)

This process consolidates all mandatory and additional data from the input layers into the subarea polygons, using the provided parameters to suggest the most suitable sanitation system for each subarea, along with additional information for decision-makers. At this step, the subareas are already defined and used as an input layer. The population dataset is a mandatory input, which can be in either raster or vector format. Streets, paths, and building footprint vector layers can be included as secondary data if desired.

As recommended, sanitation planning considers the population at the start of the plan and projects the population at the end of the plan to ensure the proposed solution’s long-term sustainability. Population projections for the plan's start and end (if the dataset year precedes the plan start year) are calculated using a simple linear projection based on an annual growth rate, which the planner provides, and the years of the dataset, plan start, and plan end. The population density is calculated for the start and end of the plan based on its respective projected population and the area of the subarea. Subsequently, the subareas are classified into low, middle, or high-density

areas considering the “Upper limit for low population density classification” and “Upper limit for medium population density classification” parameters.

Below are the input parameters that the user must provide before running the process, along with their attribute names in the output layer and corresponding explanations.

- *PARAM_PopDatasetYear* - Year of population dataset
- *PARAM_LowDensityPop* - Upper limit for low population density classification (inhab./ha).
- *PARAM_MediumDensityPop* - Upper limit for medium population density classification (inhab./ha).
- *PARAM_AnnualGrowthRate* - Annual growth rate (%).
- *PARAM_StartPlanYear* - Year of start of plan.
- *PARAM_PeopleHHStart* - People per household - start of plan.
- *PARAM_EndPlanYear* - Year of end of plan.
- *PARAM_PeopleHHEnd* - People per household - end of plan.
- *PARAM_WaterConsumReq* - Minimum water consumption required for sewerage systems (L/inhab./day).
- *PARAM_streetWidth* - Average street width [optional].

First, the process clips the dataset layers to the project area, as defined by the subarea polygons generated previously based on the Area of Interest (AOI) layer. This step is crucial for optimizing processing by avoiding running over a dataset that is more extensive than needed. As a result, additional output layers are created, consisting of the clipped dataset, as illustrated in Figure 8.

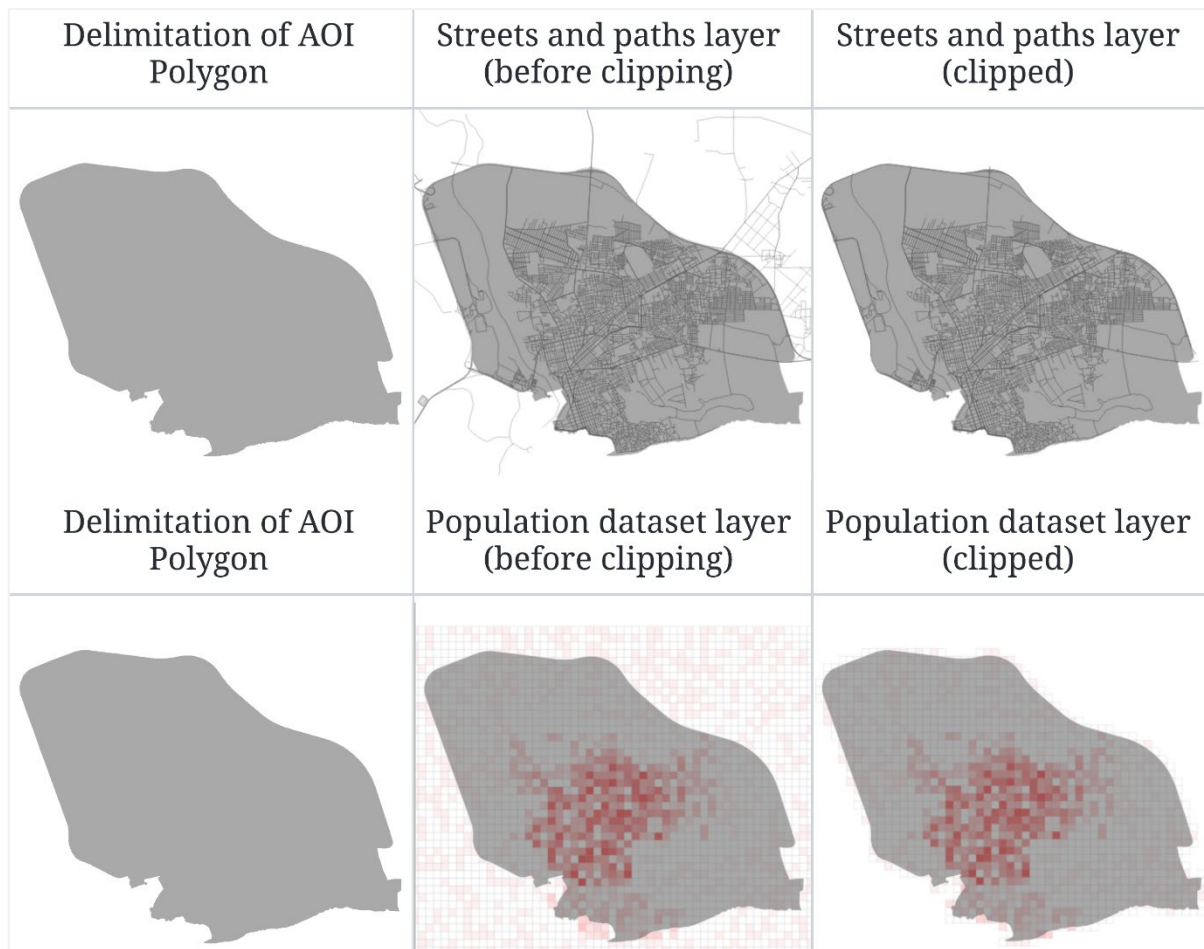


Figure 8 Illustrative examples of clipping primary data to the AOI Polygon

Secondly, the process consolidates the data within each subarea, such as calculating the total population and number of buildings for each subarea, followed by calculating indicators based on the clipped layers and provided parameters. The consolidated data and the calculated indicators include the following:

- *PopDataset_sum* - Total population of the subarea from the population dataset (inhabitants).
- *street_length_sum* - Sum of the total length of streets and paths in the subarea. Based on the register street layer (meters).
- *count_buildings* - Sum of the building in the subarea. Based on the buildings footprint layer.
- *water_consump_requirem* - Minimum water consumption for adoption of sewage systems (L/inhab./day).
- *pop_startPlan* - Total of the start of plan population for the subarea (inhabitants).
- *pop_density_startPlan* - Population density for the start of the plan. (inhab./ha).
- *classif_pop_density_startPlan* – The classification of the population density based on the start of the plan population and the parameters (low, medium, or high density).

- *households_startPlan* – Calculated using the start of plan population and parameter of average inhabitants per household (people/household).
- *pop_endPlan* - Total of the end of plan population for the subarea (inhabitants).
- *pop_density_endPlan* - Population density for the end of the plan. (inhab./ha)
- *classif_pop_density_endPlan* - The classification of the population density based on the end of plan population and the parameters (low, medium, or high density).
- *households_endPlan* - Calculated using the end of plan population and parameter of average inhabitants per household (people/household).
- *building_per_ha* – Calculated using the buildings footprint layer (if available) and the area of the subarea.

Both parameters and indicators are added to the resulting "Suggested Sanitation System by Subarea Layer." Figure 9 provides a schematic example of the subarea outcomes, showing part of the consolidated data and indicators described above.

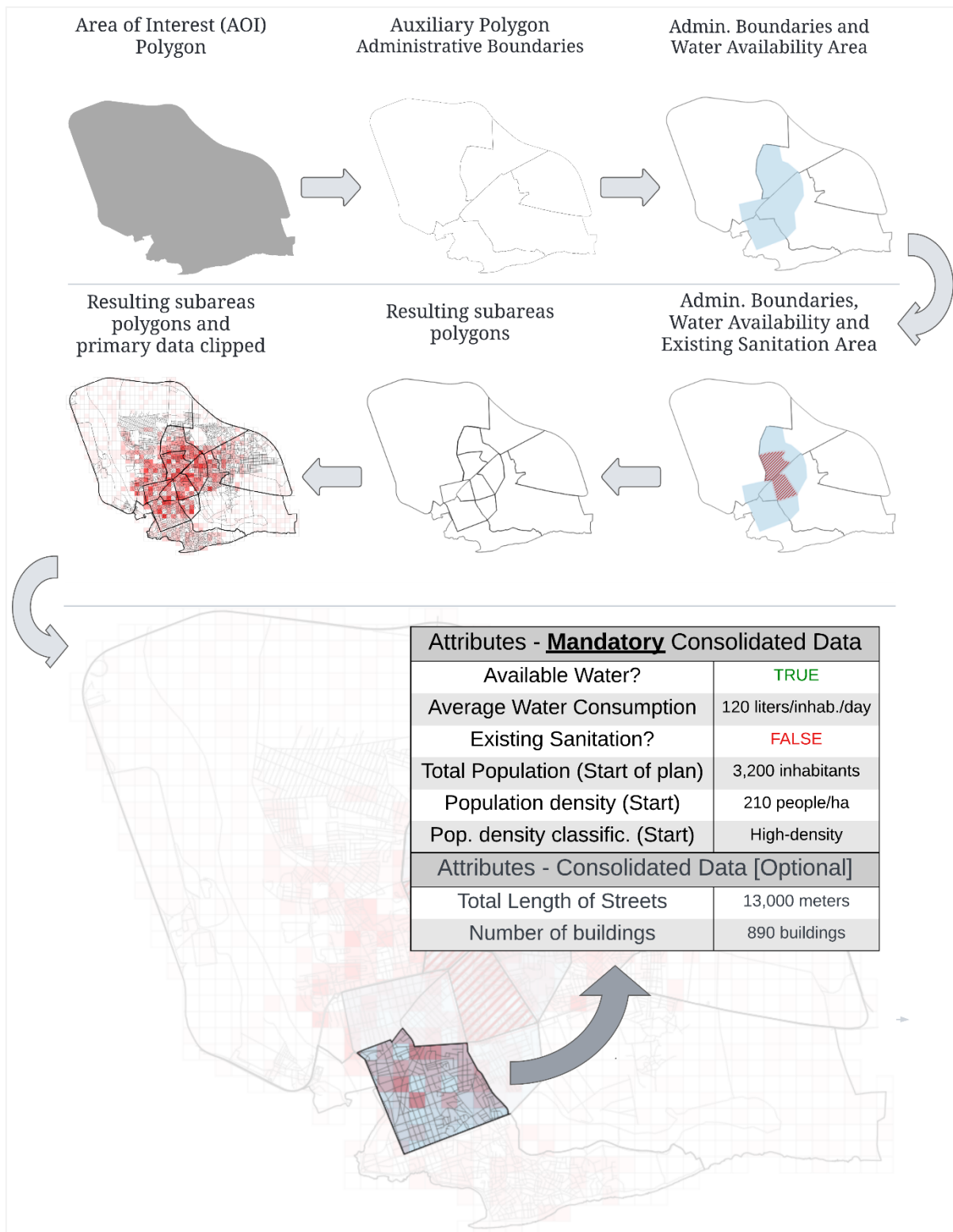


Figure 9 Illustrative examples of step 6 - Creating and consolidating subarea polygons with primary data

Finally, with all the consolidated data and indicators for each subarea, the most suitable sanitation solution is calculated by following the criteria. Suggestions are provided for both the start and end of the plan, and these may differ in the cases where the population density classification (low, medium, or high) for the start of the planning is different than at the end of planning.

Determining the most suitable sanitation system is a technical decision based on four key factors: population density, the presence and type of existing sanitation systems, water availability, and average consumption. These are the primary elements to be considered in early-stage sanitation planning, as discussed in Chapters 2.11 and 2.1.4. Figure 10 illustrates the decision tree used during this process for both the start and end of the plan.

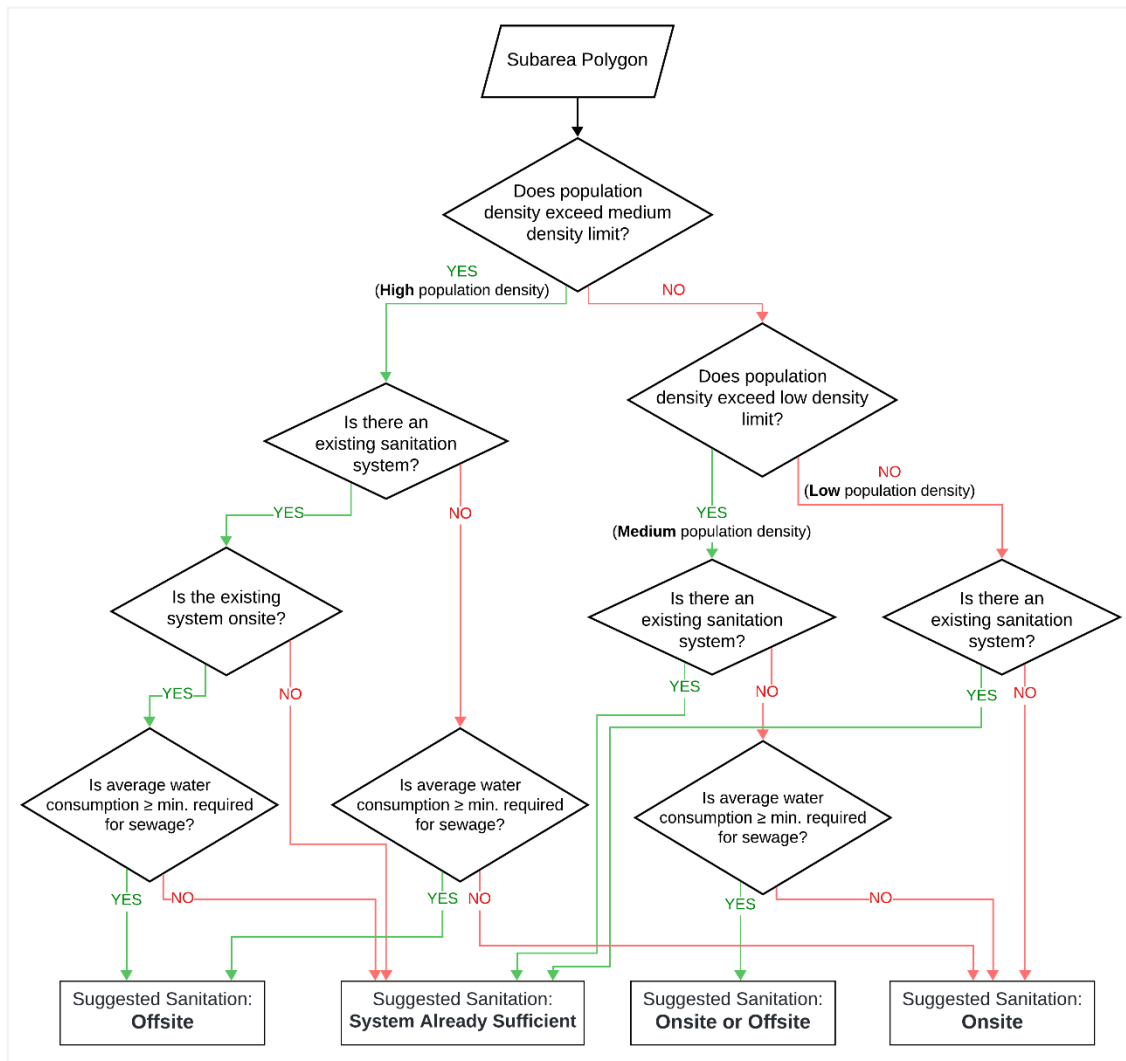


Figure 10 Suggested sanitation system decision tree

In summary, for areas classified as low population density, onsite sanitation systems are recommended regardless of water consumption, as these areas typically do not encounter issues with space or effluent infiltration. In medium-density areas, onsite systems are suggested if water consumption is below the recommended level for sewage systems; if consumption exceeds the minimum, both onsite and offsite systems are technically feasible. For high-density areas, offsite sanitation systems are recommended if water consumption meets the required minimum, while onsite systems are suggested if it falls below. The only scenario where a change to the existing sanitation system is suggested is in high-density areas currently using onsite sanitation, where a shift to offsite systems is recommended.

In addition to the technical recommendation for the sanitation system, further recommendations are also evaluated based on the criteria outlined in the following decision tree. Informal and low-income areas are considered only if identified, as they are based on secondary data.

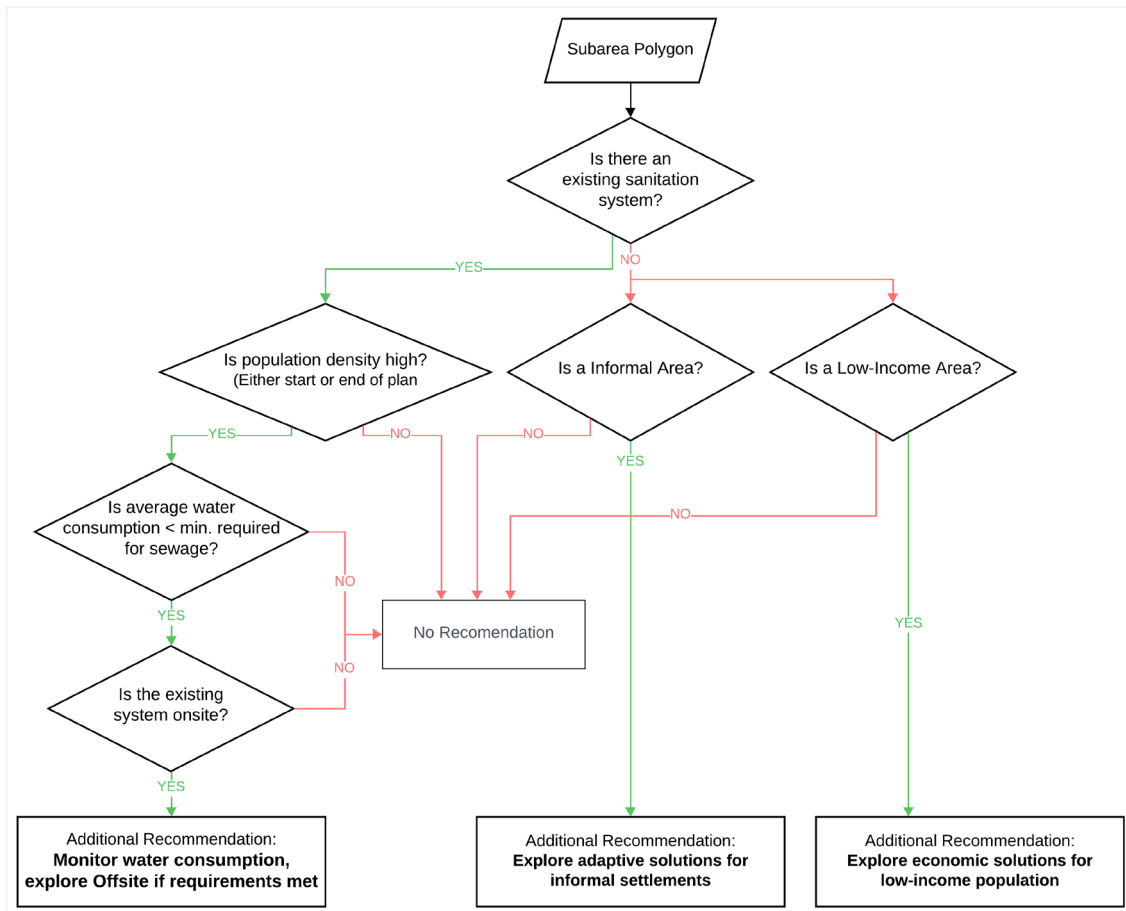


Figure 11 Additional recommendations decision-tree

The suggested sanitation systems and additional recommendations are stored on the following attributes on the "Suggested Sanitation System by Subarea Layer."

- *SuggestedSanitSystemStart* – The most suitable sanitation system for the subarea, based on the start of the plan indicators.
- *SuggestedSanitSystemEnd* - The most suitable sanitation system for the subarea, based on the end of the plan indicators.
- *AdditionalRecommendations* – Additional recommendations based on the local context.

A key aspect that enhances the flexibility and practicality of the process is the automatic recalculation of all consolidated data, indicators, suggested sanitation systems, and additional recommendations whenever changes are made to the output subareas layer. For example, the planner can adjust parameters for one or more subareas after the initial run, modifying the information and parameters for a desired subarea according to their specific requirements without needing to start from the first step of delimitating the areas and informing the attributes.

Figure 12 provides a schematic example of manually dividing a subarea, demonstrating how the primary data is recalculated.

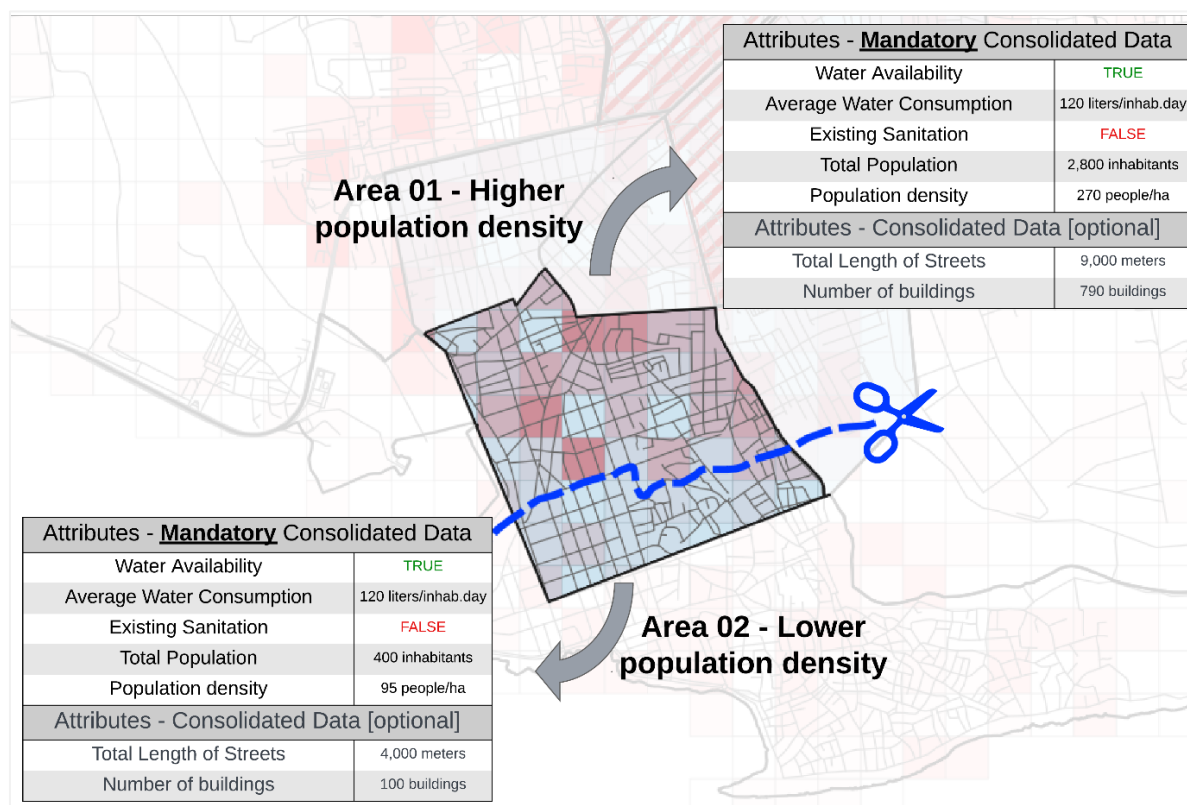


Figure 12 Illustrative examples of manually dividing a subarea polygon and the automatically recalculated primary data

The developed scripts presented in this chapter and the instructions on installing and using them are available in the thesis repository².

² <https://github.com/leonazareth/OpenSpatialSanitation>

Chapter 4 Results

This chapter explains the methodology for early-stage sanitation planning using open-access spatial data available for the case study region. The workflow follows the flowchart in Figure 6 and uses the developed QGIS scripts. The focus, however, is on explaining the practical early-stage planning steps rather than how to use the scripts. This approach allows planners to understand the concepts and apply the techniques using their preferred GIS tools.

4.1 Case study, Boca Chica – Dominican Republic

The proposed methodology was implemented in Boca Chica, Dominican Republic, to assess the benefits and challenges of using open-access spatial data and compare it with official data from the last population census.

4.1.1 Introduction

The municipality of Boca Chica is located in the province of Santo Domingo in the southeastern Dominican Republic. It is located 30 km away from the capital of the Dominican Republic, Santo Domingo, and one of the principal seaports in the Dominican Republic and the entire Caribbean, called Caucedo Port, besides one of the country's main airports, Las Américas.

The Dominican Republic is an island nation that is environmentally vulnerable to the effects and consequences of climate change. Its economy relies heavily on beach tourism, providing reliable water supply and sanitation services crucial for sustainable social and economic development. This dependency is particularly pronounced in high-intensity tourist areas such as Boca Chica, which features small plazas, restaurants, hotels, and sandy beaches that draw thousands of tourists yearly. The municipality has two districts: Boca Chica and La Caleta, as shown in Figure 13.



Figure 13 Location of the municipality of Boca Chica and its two districts (La Caleta and Boca Chica)

One of the challenges Boca Chica faces is the accelerated growth of tourism and population over the past decade. This rapid expansion has placed significant pressure on the existing infrastructure, creating challenges in expanding access to ensure water supply and safely managed sanitation for the entire city. As the demand for it continues to rise, the municipality must address these issues to sustain its economic growth and maintain the appeal of its tourist areas.

The water and sanitation services in Boca Chica are managed by the regional water and sewer corporation, CORAABO.

4.1.2 Climate and Geological Overview

Boca Chica has a typical tropical climate, with average annual temperatures between 25 and 26°C, maximum averages of 30-31°C, and minimum averages of 20-21°C, with the ocean moderating effect on temperature fluctuations. The area receives average annual rainfall between 1,000 and 1,500 mm, with peak values reaching 1,600-1,700 mm and lower values around 700-800 mm. Additionally, Boca Chica is susceptible to tropical storms and hurricanes (Ministry of Tourism of the Dominican Republic, n.d.).

According to the National Geologic Services, the study area consists predominantly of highly karstified limestones from the Los Haitises and La Isabela Formations, which allow for very high water flow and productivity. Most of the area's soil has either high or moderate permeability, with some of it found in beach deposits, which also allow good water flow but have limited capacity due to their smaller size. A small portion of the area consists of less permeable soil, primarily in the loamy deposits and sediment-filled lagoons, which do not hold significant water (National Geological Service (SGN), 2010).

4.1.3 Population – Dataset Selection

The National Office of Statistics of the Dominican Republic (ONE) recently published a new population census, with a total population (urban and rural) of 167,040 inhabitants for Boca Chica in 2022. The official census data was released after the previously presented datasets, and its aggregations are at the municipal level, spatially meaning that the population can be merged with the municipality's polygons, as illustrated in Figure 3. As discussed in Chapter 3.1.2, the existing open-access spatial population dataset estimates the population for the year 2020, which is a projection of the population based on the official datasets available for each country at the time of the development of the dataset; in the case of the Dominican Republic, it was the 2010 census. The steps to acquire these datasets are also detailed in Chapter 3.1.2.

Determining the most accurate population dataset requires a localized assessment, as each dataset's methodology can produce different results depending on the region. For instance, Figure 14 compares the aggregated population data from the four cataloged datasets with the official 2022 census data, showing the differences by province. The provinces are arranged in descending order of population size, and the percentage difference between each dataset and the 2022 census data is outlined. Positive percentages indicate that the population dataset assigned a larger population than the census, while negative percentages indicate a lower population. This comparison underlines the variability in population estimates, emphasizing the need for careful evaluation when selecting datasets for specific applications.

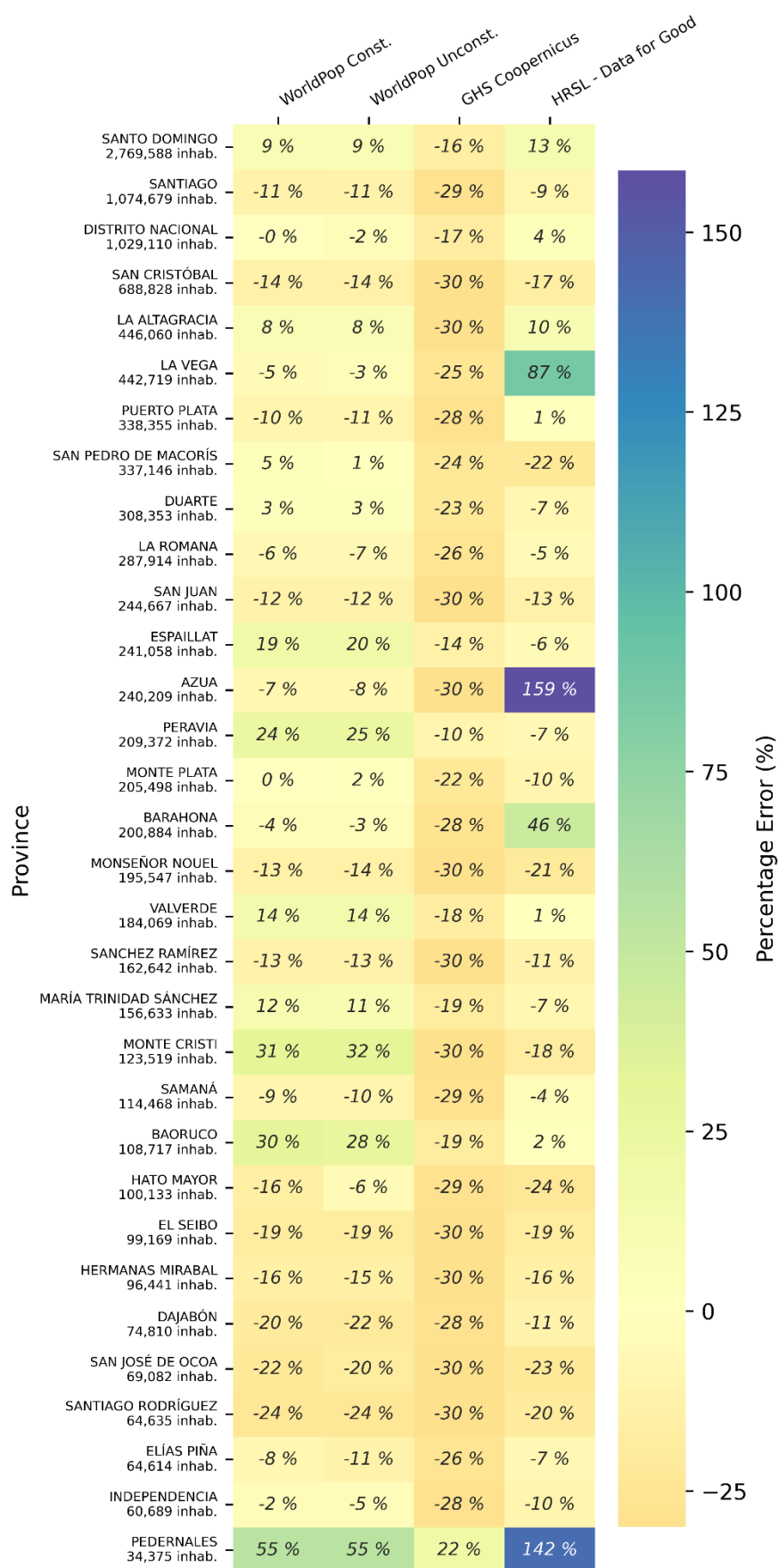


Figure 14 Comparison of 2020 Population datasets with 2022 Census by province, Dominican Republic

The dataset's discrepancy range varies significantly for the Dominican Republic. The *HRSL* dataset (Facebook Connectivity Lab and Center for International Earth Science Information Network - CIESIN - Columbia University, 2016) exhibits the most extreme positive errors, with differences reaching as high as 159%. At the same time, *GHS Copernicus* (European Commission. Joint Research Centre., 2023) shows the most substantial underestimations, with errors up to 30%. This variation highlights that each dataset has strengths and weaknesses depending on the region. The two *WorldPop* datasets (Stevens et al., 2015) offer the most accurate estimates for the province of Santo Domingo, where Boca Chica is located, with populations only 9% higher than the census data. In contrast, *GHS Copernicus* underestimates the population by 16%, while *HRSL* overestimates it by 13%. It is important to remark that these datasets represent populations from two years before the 2022 census, which may explain some of the observed discrepancies.

The same analysis was conducted, focusing on the municipalities within the Santo Domingo Province to determine the most suitable dataset for Boca Chica, as illustrated in Figure 15.

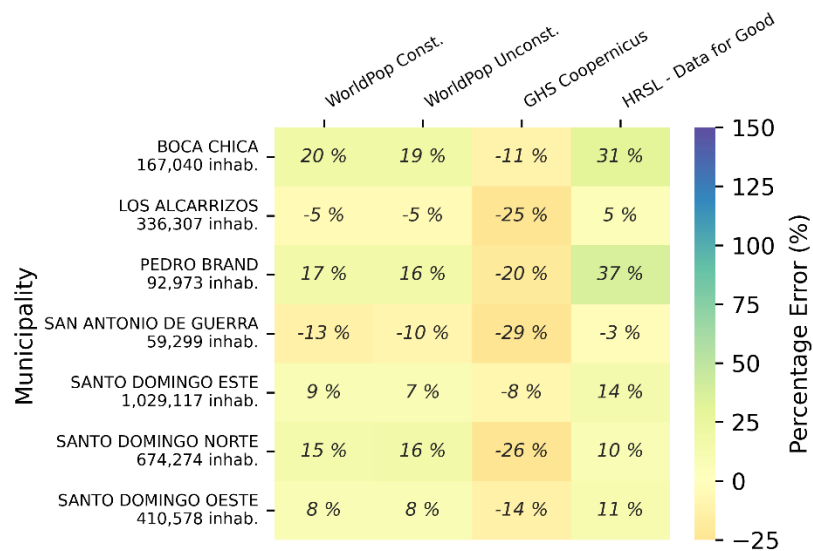


Figure 15 Comparison of 2020 population datasets with the 2022 Census by municipalities, Santo Domingo Province

Although the *GHS Copernicus* dataset provides the closest population estimates for Boca Chica, its figures are still lower than those recorded in the 2022 census. This underestimation could hamper the analysis of sanitation systems, as it may lead to results that do not fully account for the actual population needs. In this context, even with a discrepancy of around 20%, the *WorldPop* datasets remain the most reliable choice for the study area. In addition to the population values, another crucial aspect to assess is the spatial distribution of the population. Two key factors should be considered to identify potential inaccuracies. The first is when the dataset assigns population to areas without buildings, and the second is when areas with buildings are left unassigned. To accurately evaluate these discrepancies, satellite imagery from the same year as the dataset should be used.

Figure 16 illustrates the four different datasets for the area of study, represented by the colored pixels, with the higher the population, the lighter the pixel.

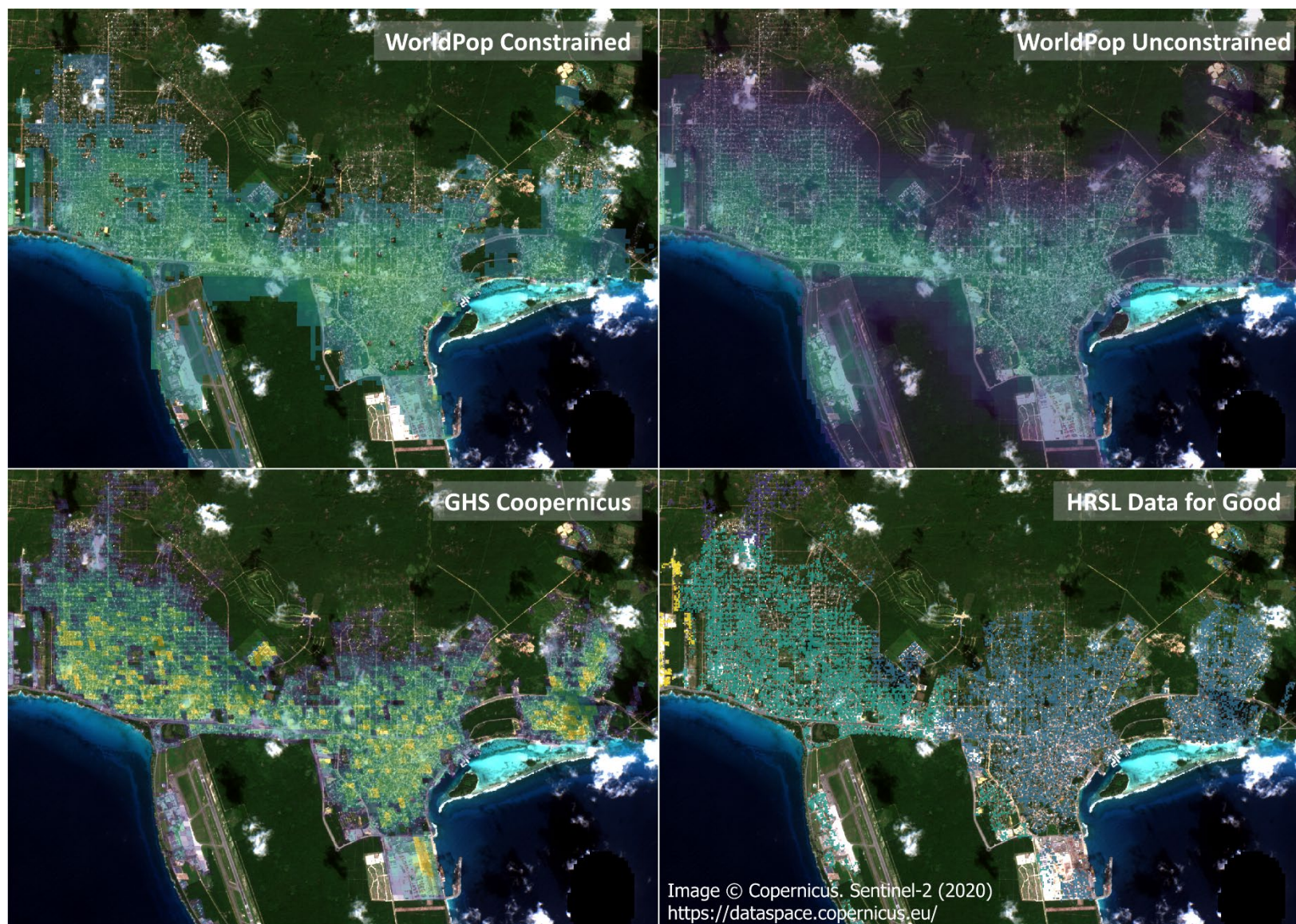


Figure 16 Spatial distribution of different population datasets in the study area

While the *WorldPop* unconstrained dataset has a slight advantage over the constrained version when considering the overall population value for Boca Chica, the constrained dataset demonstrates a more accurate spatial distribution. The unconstrained dataset incorrectly assigns population to areas with roads but without buildings in 2020, as indicated by the dark purple areas in the top-right map of the previous image. Since the total population of Boca Chica is similar for both datasets, assigning the population to areas without occupations leads to a lower population value in heavily occupied areas, which can hinder the analysis. Due to this discrepancy, the *WorldPop* constrained population dataset was opted for the current sanitation planning.

4.1.4 Existing information and data

Developing a reliable early-stage sanitation plan relies heavily on local knowledge, as much of the essential information is not publicly available and is often held by local water and sanitation utilities. Obtaining such data typically requires an official request for planners not affiliated with these utilities. In this research, the information used was obtained within the context of the project "Optimization of the Sanitary Barrier Project in Boca Chica and La Caleta Bay," conducted under the auspices of the Inter-American Development Bank (IDB). It is important to note that no sensitive information or data concerning individual users was utilized; only aggregated and non-identifiable data were adopted for this study.

The essential information obtained and employed in this research relates to identifying areas with existing sanitation and georeferenced historical data on water consumption by users. Furthermore, additional information, such as delineating informal and low-income areas, was obtained on fieldwork in the project context. This information was employed to generate "Additional Recommendations," as described in Item 3 of Chapter 3.2, highlighting areas that may require special attention in subsequent steps of the sanitation planning process.

Optional or secondary datasets included in the case study are the *Overture Map Foundation* building footprints and the *OpenStreetMap* records of roads, streets, and paths. As previously explained, these datasets, like the informal settlement and low-income area data, are not essential for early-stage planning. However, incorporating them from the initial steps of the sanitation plan can provide additional insights and are tightly connected to the subsequent steps of the sanitation planning process.

A common situation in developing countries is the lack of georeferenced information within water and sanitation utilities. Most existing information and plans are often available only in hard formats, such as PDF files or printed documents. In such cases, the data must be georeferenced using GIS software previously to adopt then the framework described in this research.

4.1.5 Delimitation of the areas – First step

The first step in early-stage sanitation planning is to delineate the area of interest and identify specific zones based on their unique characteristics. The area of interest defines the outer boundaries of the project, but within these boundaries, there are internal divisions delimitating areas with their own characteristics. Planners must identify areas with existing sanitation and water supply systems and assign or calculate the average water consumption for each area. As previously discussed, this step is crucial for evaluating and selecting the most appropriate sanitation system according to the local context. Additionally, if the information is available, it is beneficial to delimitate areas with characteristics that may require special consideration in

later stages of sanitation planning (beyond the scope of this research), such as informal settlements and low-income areas. These areas often demand adaptive and cost-effective technologies to enhance implementation success.

Furthermore, an auxiliary grid layer representing population density across various regions within the area of interest can also be created to assist planners in delineating internal boundaries more precisely. It may help the planners subdivide the project's areas based on the population concentration and distribution, which, in further steps, may lead to distinct sanitation systems according to the criteria explained in Item 3 of Chapter 3.2.

The adopted base layers for the current research were:

- Area of Interest
- Existing Sanitation Area
- Water Availability Area
- Gridded Population density (auxiliary and optional)
- Administrative Boundaries

A QGIS script, called Create Base Layers, was developed to facilitate the abovementioned process of creating the required layers to delineate the areas of the project and assign both essential and supplementary attributes, presented in Item 1 of Chapter 3.2.

4.1.5.1 Area of Interest (AOI)

The area of interest for implementing the methodology has been defined as Boca Chica's urban footprint, excluding the port and airport areas in the southern part of the peninsula. The city's expansion areas are not included, as incorporating them would require complex population growth predictions and estimates. This research adopted a linear growth model based on the annual growth rate provided by the user. Consequently, if areas that do not have a population in the dataset are included, it will continue to result in areas without a population in future projections, independent of the annual growth rate adopted. The Area of Interest layer restricts data processing within its boundaries by clipping any data that intersects with the delimited polygon. Figure 17 illustrates the Area of Interest delimited for the study case.



Figure 17 Delimitation of the area of interest for the study case

The delimitation of the Area of Interest (AOI) was performed manually. Although several methods exist to automate the identification of urban footprints, such as using GIS software or algorithms to process remote sensing data from optical, radar, or other types of sensors. Another existing alternative is to use pre-processed datasets that already delimitate urban footprints at a global scale, such as Global Urban Footprint (GUF³) (Esch et al., 2013) or Global Human Settlement Layer (GHSL⁴) (European Commission. Joint Research Centre., 2023). However, employing these methods requires considerable expertise in remote sensing or data analysis, and they are not essential for this study. Therefore, a manual approach was chosen, demonstrating that this process is feasible even for planners without advanced technical skills.

4.1.5.2 Existing Sanitation

The delimitation of existing sanitation areas is generally mapped and available in planning documents; however, in some cases, it may not be formally recorded and instead relies on the knowledge of employees from water and sanitation utilities, particularly those responsible for the operation and maintenance of the system. For Boca Chica, existing plans identified two areas with implemented sanitation systems, which were later verified through fieldwork conducted during a prior project. The first, referred to as "Area 01" in Figure 18, consists of a sanitary sewerage system serving the area of the city's central tourist zone near the beach, which includes a mix of residential and commercial properties, such as hotels and restaurants. In this area, the collected effluent is transported via a few pumping stations and pressurized pipes to a wastewater treatment plant in the city's southern part. The second, identified as "Area 02", is a

³ Available on request at: <https://www.dlr.de/de/eoc/forschung-transfer/projekte-und-missionen/global-urban-footprint/guf-daten-und-datenzugang>

⁴ Available at : <https://human-settlement.emergency.copernicus.eu/download.php>

residential apartment complex with a sanitary sewerage system and a decentralized wastewater treatment plant for managing the collected effluent.



Figure 18 Areas with existing sanitation system and location of existing wastewater treatment plant

It is important to emphasize that the delimitation of areas with existing sanitation should only be done for areas where sanitation is safely managed. This distinction is important because areas with existing sanitation are treated differently when determining the most suitable sanitation solutions, and they are only recommended if the characteristics of population density and water consumption indicate that the existing system is not the most recommended (See Figure 10). It is also necessary to inform the type of sanitation system implemented, whether Onsite or Offsite (Offsite considers any sewerage system, which means centralized or decentralized).

4.1.5.3 Water Supply area and average consumption

Records of water supply system assets in cities of developing countries are often unavailable, and when they are, they may not be up to date. It is more common to find records of connections that represent the system's users, as this information is crucial for the financial management of the utilities. Both types of information can be utilized to determine areas with access to water supply. Moreover, another common challenge is the presence of areas with informal connections to the water supply system, often through illegal connections. In such cases, fieldwork becomes essential for planners to identify and address these issues accurately.

In addition, it is important to identify the average water consumption for each zone within the study area to delineate areas with water availability. This information may be available in planning documents, mainly where the water supply system is sectorized—meaning the system is divided into smaller zones to facilitate management. In such cases, meters are installed to

record the water consumption of each zone. Alternatively, if individual meters are installed, the data can be aggregated into clusters and consolidated to determine the average water consumption per zone. Moreover, records used for tariff charges often include the number of inhabitants per connection, as charges are commonly based on the number of people rather than individual meters. These records can be cross-referenced with local water consumption profiles to estimate the average water consumption per connection, which can then be clustered as the explained case for individual meters.

In the study area, records of connections were available, consisting of a mix of connections with individual meters and those with estimated consumption based on the number of inhabitants per connection. Additionally, information classifying connections as either residential or commercial was also available. Since it is in the context of early-stage sanitation planning, this study considered only residential connections to avoid bias in the average water consumption due to large commercial water users. Figure 19 illustrates the distribution of connections across Boca Chica's area, represented by the purple dots.

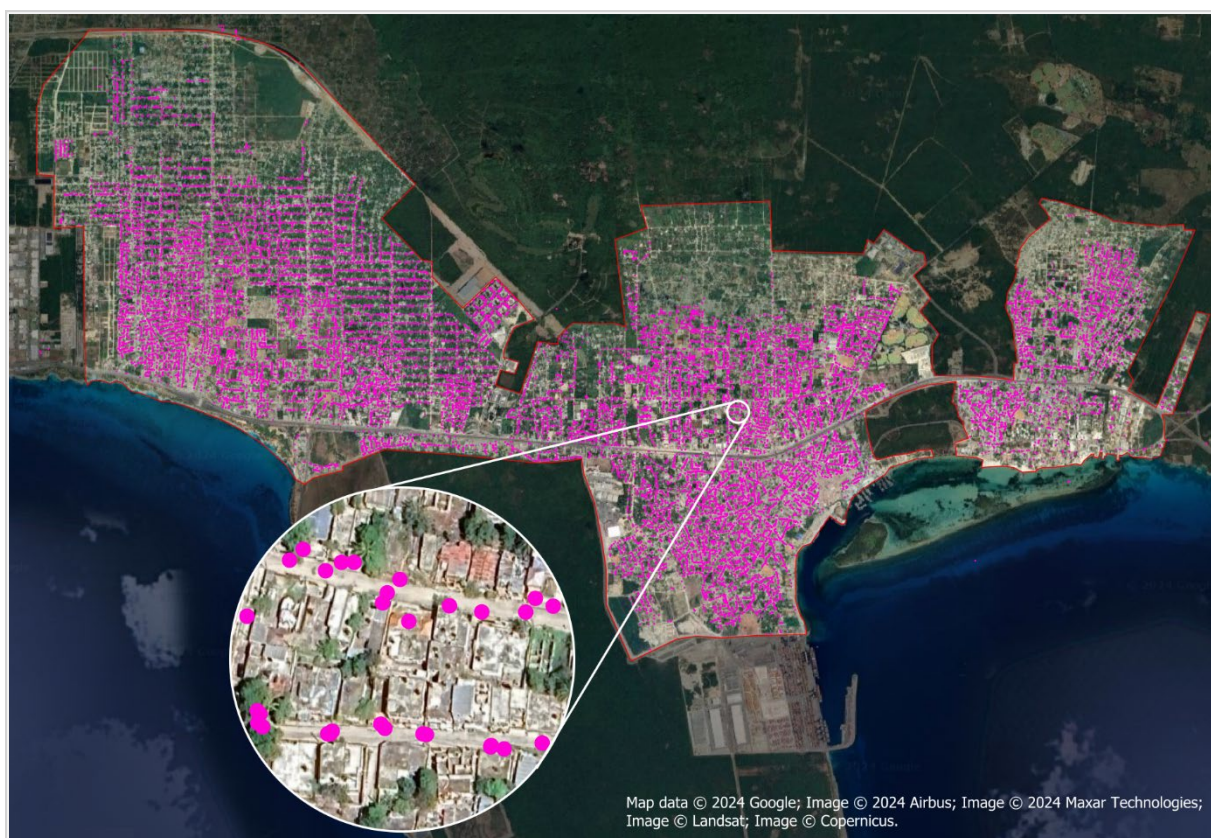


Figure 19 Record of connections of water supply users in Boca Chica

The delimitation of zones with similar average water consumption was carried out in four steps, as shown in Figure 20—the first step involved delineating the polygons that represent areas where water supply records were available. In the second step, these polygons were further subdivided into smaller areas, considering roads and streets. The third step involved calculating the average water consumption for each polygon by averaging the consumption data from all water connections within that area. In the fourth step, neighboring areas were combined if their average water consumption fell within the same range, set in 30-liter intervals per inhabitant per day, from 120 to 276. In cases where main roads divide the subareas, they remained separate, even if two polygons had the same water consumption range.

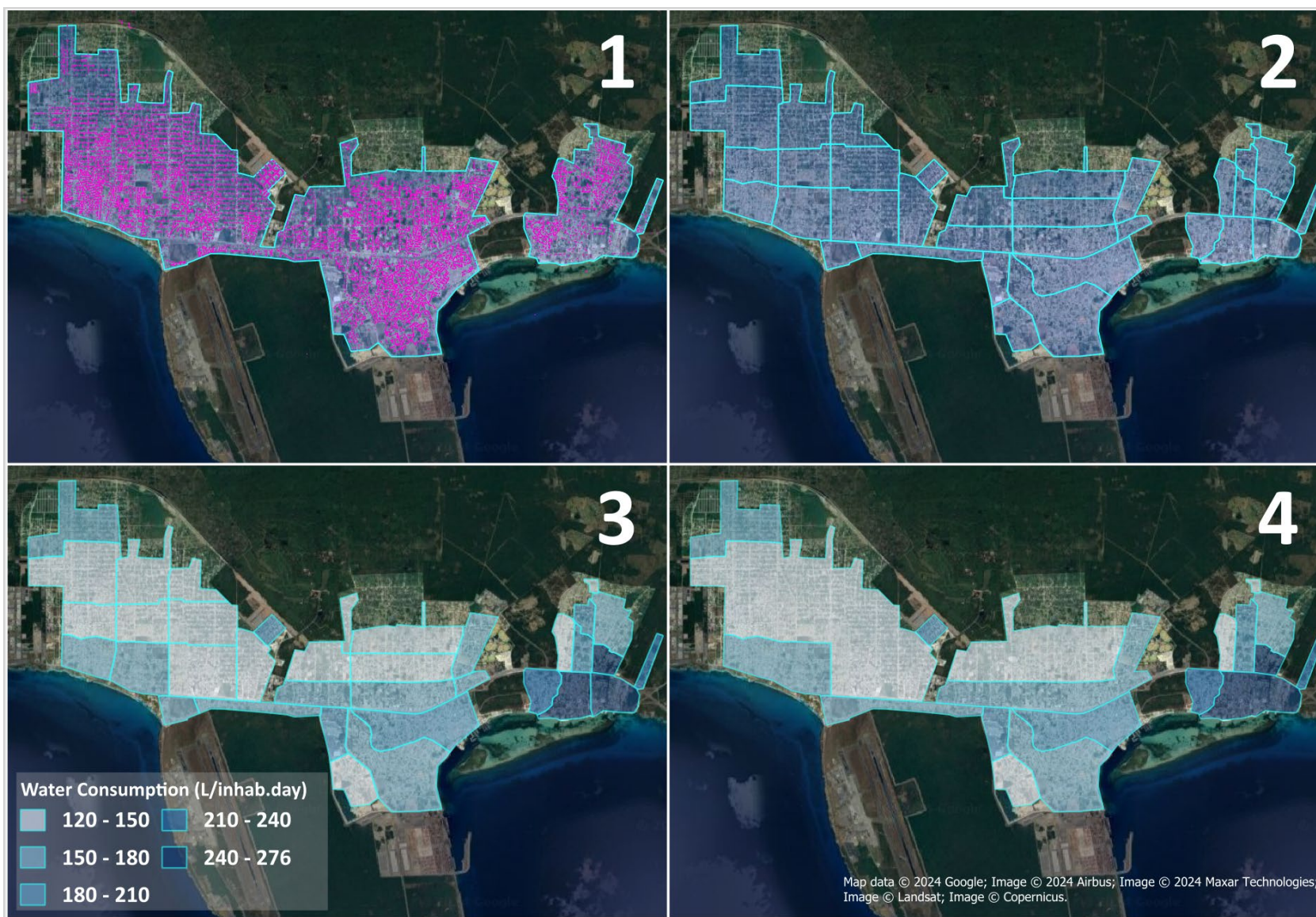


Figure 20 Workflow for delimitation of zones with different average water consumption

The average water consumption in Boca Chica is relatively high, as in many Latin American countries. The lowest average consumption identified was 120 liters per inhabitant per day in locations far from the city center, such as expansion and peri-urban areas. It is important to note that the zones identified previously with existing Offsite sanitation systems have some of the highest water consumption in the city, ranging from around 220 to 240 liters per inhabitant per day in "Area 01," near the beach, and approximately 190 liters per inhabitant per day for the apartment building complex, identified as "Area 02."

4.1.5.4 Gridded population density (auxiliary and optional layer)

A gridded population density layer may be helpful for planners to visualize how population density varies across the project area. Population density is a key factor influencing the most suitable sanitation system. As shown in Figure 10, it is adopted to calculate the suggested system and understand this distribution auxiliary delineating areas with similar population densities. Figure 21 illustrates the standardized gridded population density layer based on the selected *WorldPop* Constrained dataset for the study area.

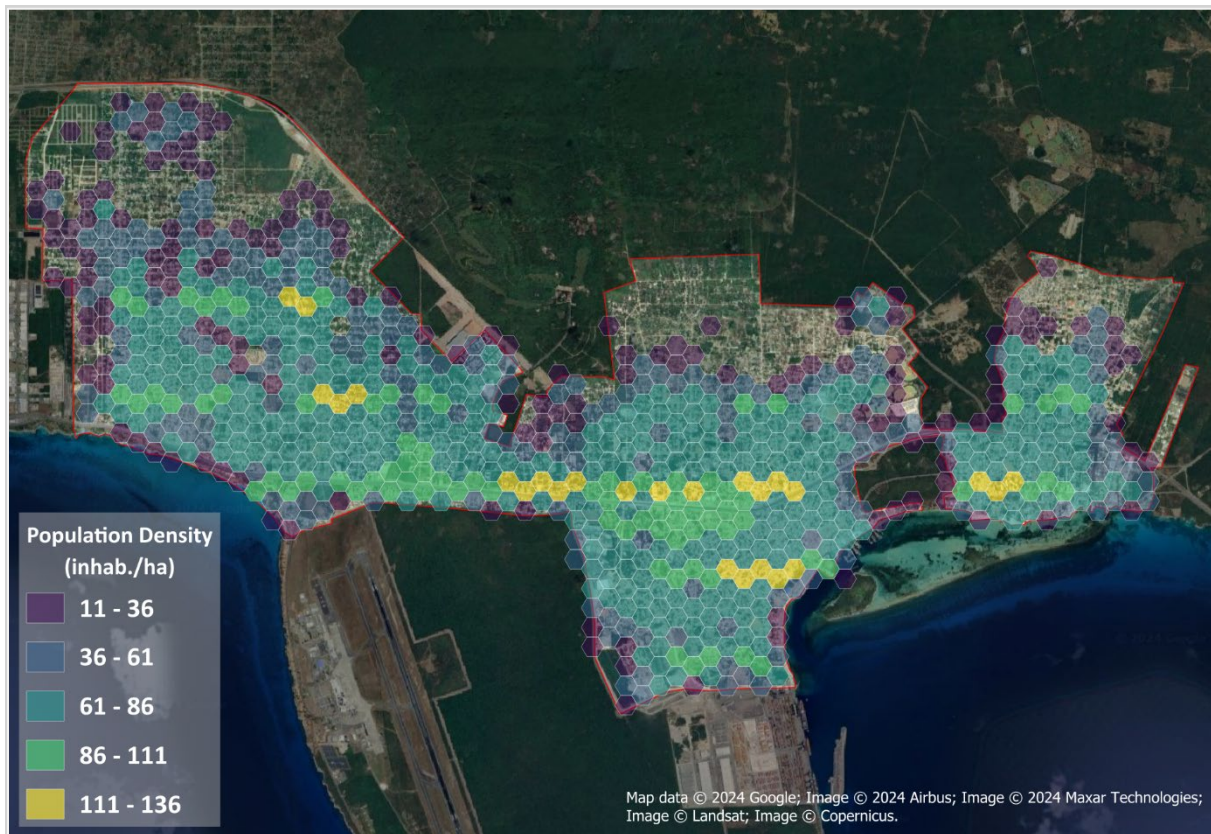


Figure 21 Gridded population density for the study area

Several steps should be followed to create a gridded population density layer. The first step is to determine the shape and size of the grid; rectangular and hexagonal grids are the most commonly used. The grid size must be selected, ensuring it is bigger than the resolution or unit of the population dataset to avoid generating many empty grids without population data. Once the grid is created, the planner must intersect the population from the dataset with the grids to calculate the population density for each grid cell. If the planner is more interested in the population density for the end of the planning period, the projection year should be adjusted accordingly. This requires considering several input parameters, such as the dataset's year, the population projection's target year, and the annual growth rate. Population projection is a critical

component of any sanitation plan, and the planner must decide on the most suitable type of projection and target year for their specific context.

While these steps provide a general guide, allowing it to be reproduced using various GIS tools, the QGIS scripts to "Create Gridded Population," developed in the present research, facilitate this process by allowing the generation of the gridded population layer, as shown in Figure 21, using a linear population projection based on an annual growth rate, provided as a percentage, calculated using the following expression.

Equation 4.1—1 Population growth – Linear projection

$$\text{Population}_{\text{target}} = \text{Population}_{\text{dataset}} \times (1 + \text{Annual Growth Rate})^{(\text{Year}_{\text{target}} - \text{Year}_{\text{dataset}})}$$

4.1.5.5 Administrative boundaries and other

When available, official administrative boundaries, such as neighborhood or city region divisions, can help subdivide the study area. However, in areas experiencing rapid, unplanned urban growth, local governments often face the challenge of keeping these boundaries up-to-date. In the case of Boca Chica, the National Office of Statistics of the Dominican Republic (ONE⁵) provides a neighborhood layer from 2021, represented by the dotted white lines in Figure 22.



Figure 22 Official division of neighborhood of the study area

⁵ <https://portalgeoestadistico.one.gob.do/>

For the proposed methodology, the administrative boundaries layer should also be adopted to further subdivide the study area based on factors like population density and to delineate areas identified as informal settlements or low-income areas—characteristics that heavily influence the choice of sanitation systems. While official administrative boundaries can serve as an initial guide for internal divisions, planners should adapt them as necessary, as sanitation systems are influenced more by these specific characteristics than by the artificial boundaries themselves.

The workflow used in the case study to define the subdivisions is presented in Figure 23. First, population density grids with at least 61 inhabitants per hectare for the year 2024 were filtered and compared with the official neighborhood divisions. Next, new internal divisions were created to separate areas with population densities above this threshold, using streets and roads as boundaries by adopting satellite imagery as an essential auxiliary during this delimitation. Following this, two areas identified during fieldwork as vulnerable were delineated—one classified as a low-income area and the other as an informal settlement. These characteristics were incorporated as attributes in the standardized Administrative Boundaries layer. The final consolidated result of these delimitations is presented in Step 4 of the figure.

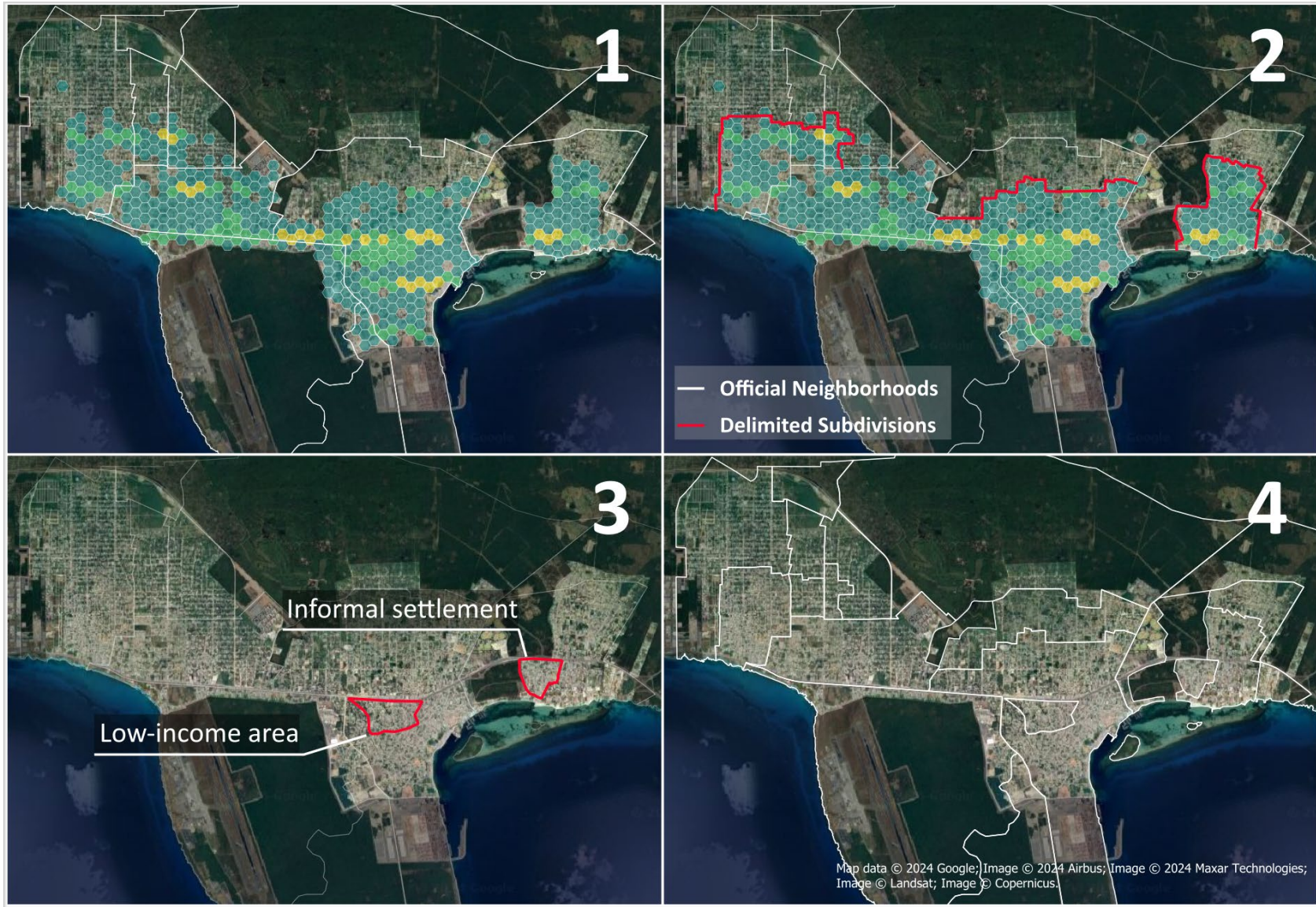


Figure 23 Workflow for delimitation of administrative boundaries layer subdivisions

4.1.6 Creation of subareas – Second step

The second step of the proposed methodology is to merge all the information from the different base layers into a single layer. This consolidation makes it easier to visualize and adjust multiple aspects as needed and facilitates the calculation of indicators and results in the third step of the process.

In this research, the consolidated layer with all the relevant information is called the "Subarea Layer." To generate the subareas, the layers presented before—existing sanitation systems, water availability, and administrative boundaries—are intersected, merging the information and the delimited areas and clipped by the “Area of Interest Layer,” resulting in all the information being grouped into one layer within the defined study area. While these steps can be replicated using different tools, the QGIS script "Delimitation of Subareas," developed in this research, simplifies the process. It lowers the GIS knowledge required for the planner by simplifying all the steps into one: selecting the previously created layers and running the model. Figure 24 shows the separately created base layers and the final consolidated "Subarea Layer."

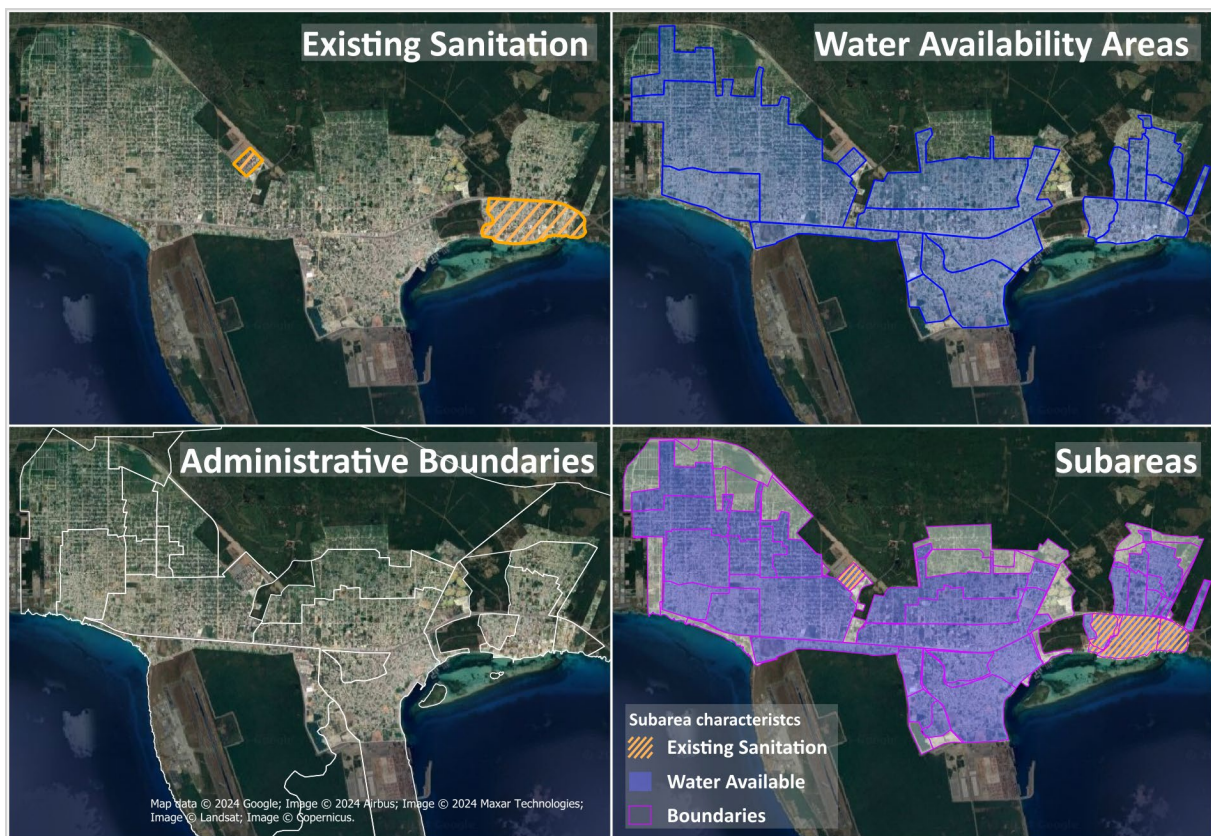


Figure 24 Consolidation of Base layers into a single Subarea layer

4.1.7 Calculate sanitation system and indicators – Third step

The calculation of the suggested sanitation system characterizes the third step. To do that, the planner must inform essential parameters, calculate the indicators, and determine the suggested sanitation system and additional recommendations based on Figure 10 and Figure 11 (as explained in Item 3 of Chapter 3.2). However, the sanitation planning process is complex and involves multiple factors. It is characterized as an iterative process that requires several adjustments along the way. Simplifying this process is essential to allow planners to easily

modify initial parameters and adjust the previously defined areas, allowing the comparison of different possibilities and alternatives, which results in an optimized plan.

The proposed methodology has the advantage of easing adjustments at any stage of the planification. In practice, it means that even after calculating the indicators and results of this third step, changes in the parameters and boundaries of the subareas can be made without breaching the sanitation planning process's workflow. Therefore, starting with smaller subareas and, if necessary, grouping them later is preferable. This approach allows for a better distinction of the subareas and their characteristics, such as population density and average water consumption, and enables planners to test different parameters and scenarios before deciding whether to keep areas separate or merge them with neighboring subareas.

4.1.7.1 Assumptions and parameters

The National Institute of Drinking Water and Sewerage (INAPA) of the Dominican Republic is the government agency responsible for setting water and sanitation standards⁶. INAPA indicates that the project horizon for a sanitation plan should range from 20 to 50 years, depending on the importance of the infrastructure. For this research, 30 years was adopted, the typical period for most sanitation infrastructures. Therefore, the year for the start of planning considered was 2024, and the end of planning was 2054.

The annual growth rate for the province of Santo Domingo, where Boca Chica is located, was based on reports from the National Statistics Bureau (ONE). These reports⁷ indicate an annual growth rate of 1.3 percent per year between the previous census in 2010 and the most recent one in 2022. The average household occupancy rate in Boca Chica, according to the 2022 census data, is 2.94 persons per household, which was used for both the start and end of the planning period.

According to Tilley et al. (2014), a minimum water consumption of 60 liters per inhabitant per day is necessary to consider the implementation of sanitary sewerage systems (offsite systems) as a feasible solution. Lower consumption can reduce the operational efficiency of the system, leading to frequent clogging and the need for maintenance.

A key parameter for the calculations is defining the ranges for classifying population density as low, medium, or high. As explained in Chapter 2.1.1, the proposed methodology assumes that onsite sanitation is always the most suitable solution for low-density populations. For medium-density areas, onsite and offsite options are feasible, requiring further investigation. In high-density areas, offsite sanitation is generally recommended unless the average water consumption is below 60 liters per inhabitant per day, which would compromise system efficiency.

Other criteria must be considered in cases with no specific guidelines for these density ranges, as occurs in the Dominican Republic. To address this, a study was conducted to compare capital and operational costs for containment, emptying transportation, and treatment of a low-cost sewerage system and a septic tank connected to a soak pit. The evaluation selected four areas of the study case, each with different street topology characteristics that directly impact sewerage system costs: the length of streets per hectare and the average street widths. For each area, eight different population densities, ranging from 25 to 200 inhabitants per hectare, were

⁶ Available at:

https://www.inapa.gob.do/phocadownload/Proyectos/Reglamentos_y_requerimientos_de_proyectos/Reglamento_TecnicoParaDisenoDeObrasEIstalacionesHidroSanitarias.pdf

⁷ Available at: <https://www.one.gob.do/datos-y-estadisticas/temas/censos/poblacion-y-vivienda/2022/>

adopted to simulate and compare the sewerage costs versus the septic tank and soak pit alternatives to identify at which population density the costs would be the same. However, a definitive conclusion could not be reached mainly due to the challenges of evaluating treatment costs for isolated subareas. The existing treatment cost curves were generated using real-world plants designed to serve entire cities or large city regions. These curves reflect the economies of scale, where the cost per inhabitant tends to decrease as the population served by the treatment plant increases (Damania et al., 2019). Applying these cost curves to isolated subareas resulted in values significantly higher than expected, as the population of the individual subareas being evaluated would represent only a portion of the total population treated by those plants. Additionally, a reliable cost curve for fecal sludge treatment plants was unavailable, preventing a comprehensive comparison that accounts for all the components involved in each technology.

Due to the lack of guidelines and the inability to reach a definitive comparison, the criteria used in Indonesia were adopted, with 100 inhabitants per hectare set as the upper limit for low density, as presented in Table 3. The upper limit for medium density—when offsite sanitation becomes the most suitable solution—is set at 160 inhabitants per hectare, as Mara (2005) first suggested and later confirmed by Manga et al. (2020) and Kabange and Graham (2017).

Lastly, an optional parameter that can be provided when the planner chooses to include a dataset of streets and paths is the average street width. As previously explained, datasets for building footprints and streets are optional and do not affect the results of the suggested sanitation system. However, they can be helpful during the evaluation of the result and in the subsequent phases of sanitation planning. They have been incorporated into the developed process scripts to ease the transition to those next steps.

4.1.7.2 Calculations and evaluation of subareas indicators

The assumptions, parameters, and calculations explained in Chapter 3.2 were applied to each of the 70 previously defined subareas. The developed QGIS script, "Sanitation System Evaluation," accomplishes this by using the "Subarea Layer" and the population dataset layer—the WorldPop constrained dataset—as input. Optional layers for streets (*OpenStreetMap* dataset) and building footprints (*Overture Maps* dataset) were also included. Additionally, the assumptions and parameters previously outlined for the study case were adopted.

The calculation output is a new layer called "Subareas - Suggested Sanitation," which includes the same 70 subareas with all parameters, calculated indicators, and aggregated results. This allows the planner to create different visualization styles for the subareas based on specific parameters or indicators of interest. Figure 25 illustrates two possible visualizations employed in the study case to explore the possibility of merging neighboring subareas with similar characteristics. In this case, the average water consumption and population density were evaluated at the end of the planning.

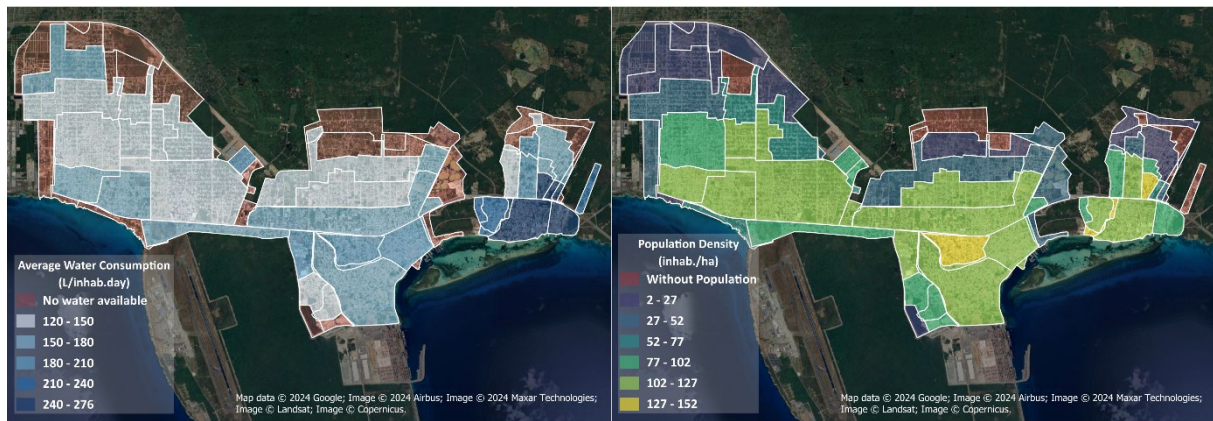


Figure 25 Visualizations of the “Subareas - Suggested Sanitation” Layer. Average water consumption (left) and population density (right) at the end of the planning period.

Four small subareas with water consumption but no attributed population were identified during the analysis, indicating a problem in the population dataset for these specific regions. Figure 26 highlights these subareas in red. They were merged with the neighboring subareas with similar characteristics to avoid being dismissed, except for the subarea more to the east of the city. It was kept separate and treated individually due to its distinct characteristics as a new residential complex, ensuring it would not be removed in the subsequent step.

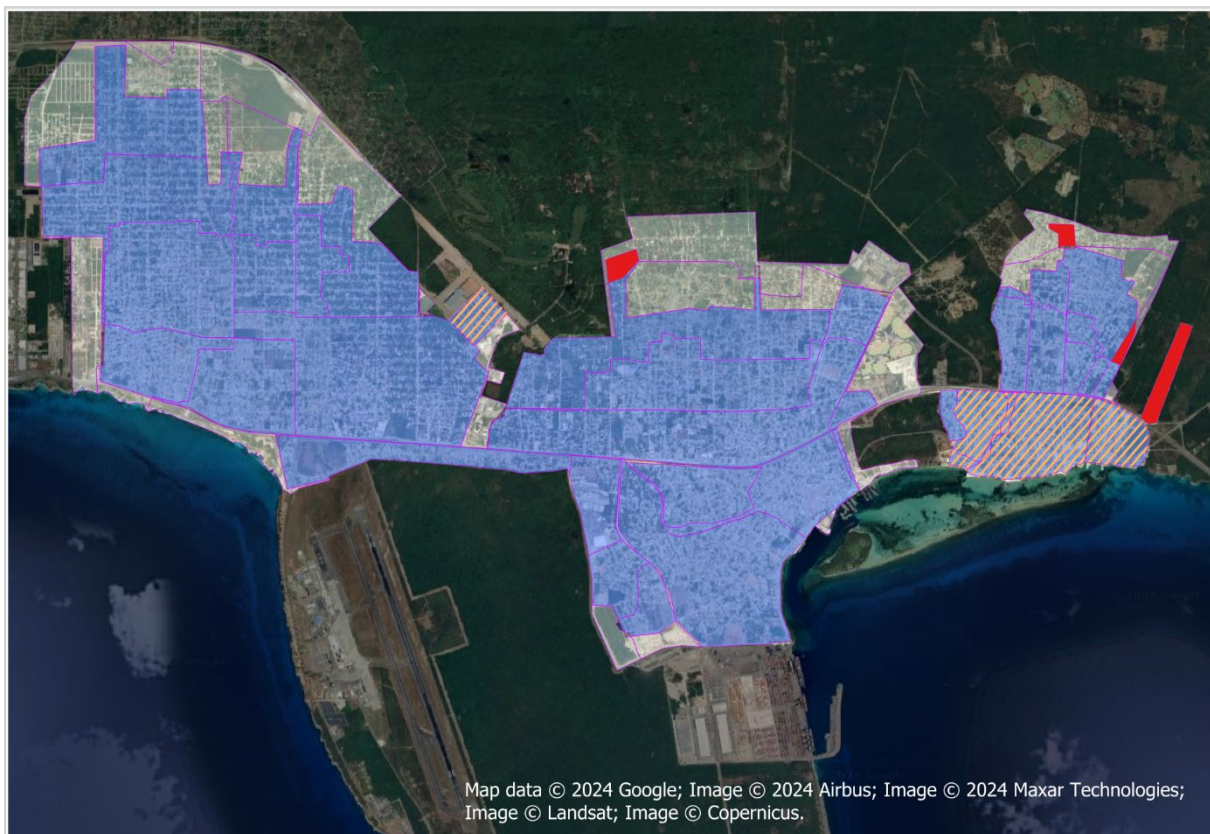


Figure 26 Identification of subareas with data discrepancies

4.1.7.3 Adjustments and suggested sanitation system

Two steps were taken to reduce the number of subareas. First, subareas with no population data were deleted if confirmed by evaluating the non-existence of apparently residential buildings

by using satellite imagery and the building footprint layer as auxiliaries. Second, neighboring subareas with similar characteristics were merged. Subareas were considered similar if they had average water consumption within the same interval of 30 liters per inhabitant per day and 25 inhabitants per hectare for population density, as previously defined. The subareas were edited manually using native QGIS tools for feature manipulation. Afterward, the developed script "Recalculate Subareas" was run to recompute each subarea's total population, street length, and buildings, as well as recalculate indicators and results that are the suggested sanitation system and additional recommendations.

With the steps outlined above, reducing the number of subareas from 70 to 32 was possible, simplifying the assessment of the suggested sanitation systems for the study case subareas. Figure 27 illustrates the proposed sanitation system for each subarea at the start and end of the planning period.



Figure 27 Suggested sanitation system for the start of planning (left) and end of planning (right)

The figure shows that based on the parameters considered and primary data for the subareas, such as population and water consumption, an onsite sanitation system is the most suitable for all subareas at the start of planning. By the end of the planning period, over a 30-year project horizon, the population density in some areas increases to medium, making both onsite and offsite sanitation systems feasible. This highlights the need for further evaluation of factors such as groundwater level, soil type, land availability, and the socio-cultural and economic aspects of the population to select the best alternative.

The most densely occupied subarea is the zone designated as a low-income area (see Figure 23), with a density of around 92 inhabitants per hectare at the start of planning, increasing to around 136 inhabitants per hectare by the end of planning. Nevertheless, this area does not reach the threshold of 150 inhabitants per hectare required to be classified as highly densely occupied. Figure 28 shows the QGIS form configured to display relevant information for the selected subarea, including additional recommendations, in this case, considering the low-income area.

Subareas - SuggestedSanitation_03 - Feature Attributes

Information of the area

Existing Sanitation? ☐ Type of existing sanitation (NULL)

Available water? ☒ Average water consumption (liters/inhab./day) 184

Informal Settlement? ☐ Low-Income Area? ☒

Total Population (dataset) 5236.23 Population dataset year 2020

Area (ha) 59.58 Total street length 14891.02

Number of buildings 2083 Building density (building/ha) 34.96

Parameters

Low-density pop. limit (inhab.ha) 100 Medium-density pop. limit (inhab.ha) 160

Water consumpt. requirement (liters/inhab.day) 50 Annual growth rate (%) 1.3

Start of planning year 2024 End of planning year 2054

People/household - Start 2.94 People/household - End 2.94

Average street width (m) 5

Minimum water consumption for sewage systems? ☒

Indicators

Population - Start 5514 Population - End 8123

Population density - Start (inhab./ha) 92.55 Population density - End (inhab./ha) 136.34

Pop. density classification - Start Low density Pop. density classification - End Medium density

Number of households - Start 1876 Number of households - End 2763

Suggested Sanitation System - Start Onsite **SuggestedSanitSystemEnd** Onsite or Offsite

identifier SA-03

Additional Recommendations

Explore economic solutions to ensure affordability for the low-income population

Recalculate the subarea

Figure 28 QGIS form displaying details of the selected subarea

As highlighted previously, adjustments to both the geometry and parameters of the subareas are allowed. As shown in the image above, all parameters the planner adopts when executing the "Sanitation System Evaluation" script are stored individually for each layer feature. This gives the flexibility to apply different parameters to specific project zones, for example, adopting different annual growth and occupancy rates (people per household) to areas with distinct demographic characteristics. To make adjustments, the planner must change the desired parameters in the feature form and click "Recalculate the Subarea" at the bottom. This will recalculate the indicators and results only for the selected subarea.

A detailed visualization of the low-income area is presented in Figure 29. It shows the two optional layers: the *OpenStreetMap* dataset for streets and the *Overture Map* dataset for building footprints. As indicated by the population data, the area is densely occupied, corroborated by the building footprint layer, showing more than two thousand buildings in less than 60 hectares, as shown in Figure 28. Additionally, satellite imagery helps assess the area's characteristics and confirms the calculated indicators discussed earlier.

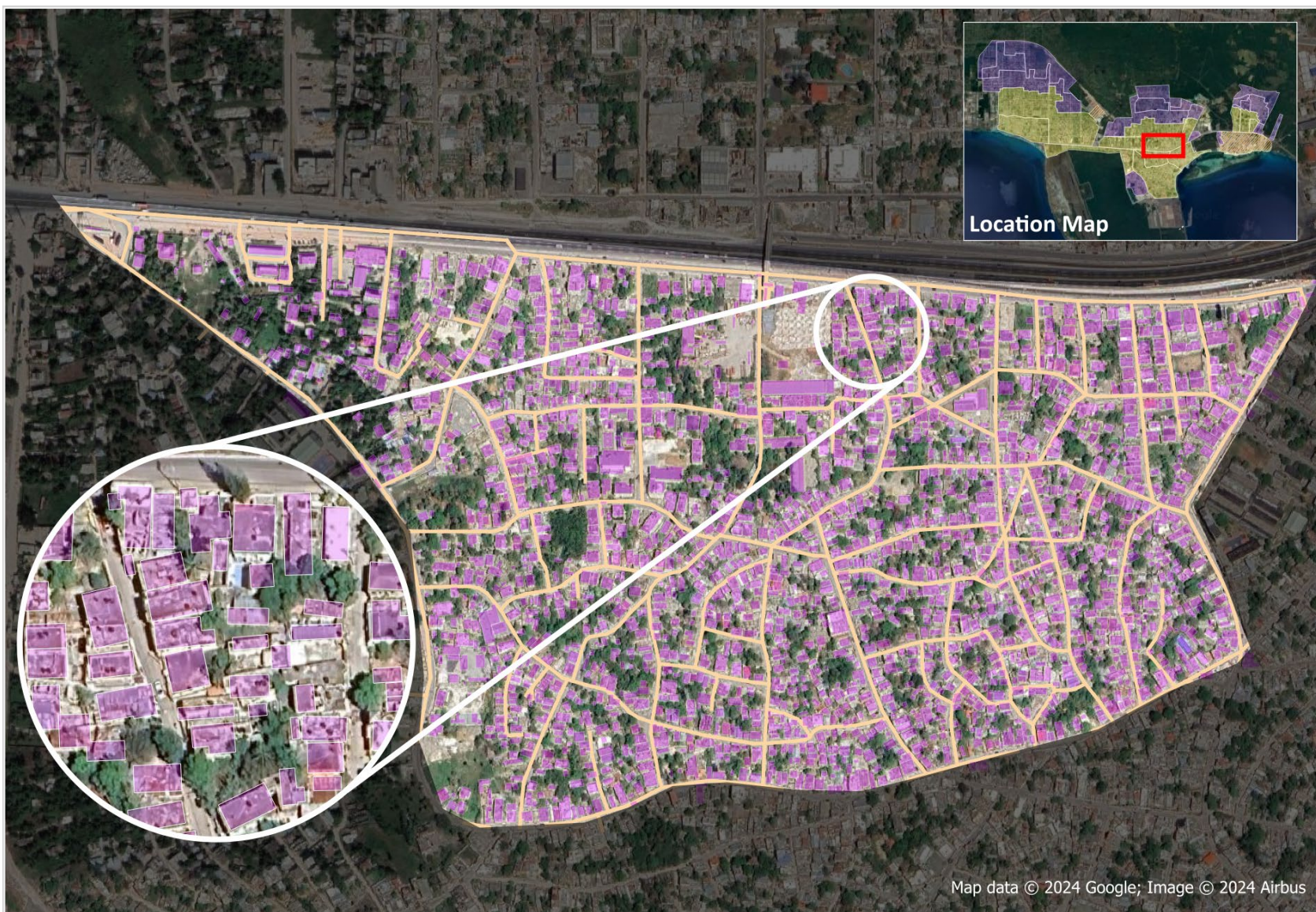


Figure 29 Detailed visualization of a densely populated low-income area, highlighting optional datasets of streets (OpenStreetMap) and building footprints (Overture Map).

After verifying data consistency and evaluating the resulting suggested sanitation systems, a further step was taken to group neighboring features with the same recommended system for the end of the planning period. This process resulted in 19 distinct subareas for the study case. It is important to note that the zones in the northern part of the study area did not merge with the surrounding areas, as these zones lacked an available water system. Figure 30 illustrates the final configuration of the study area with the delimited subareas. Furthermore, the figure highlights areas where the boundaries of the existing water system should be investigated further to guarantee whether there was an error in the geolocation of the water users or if it only represents recent areas of expansion of the system. The high-resolution version of the layouts of the suggested sanitation system is available in Appendix B.

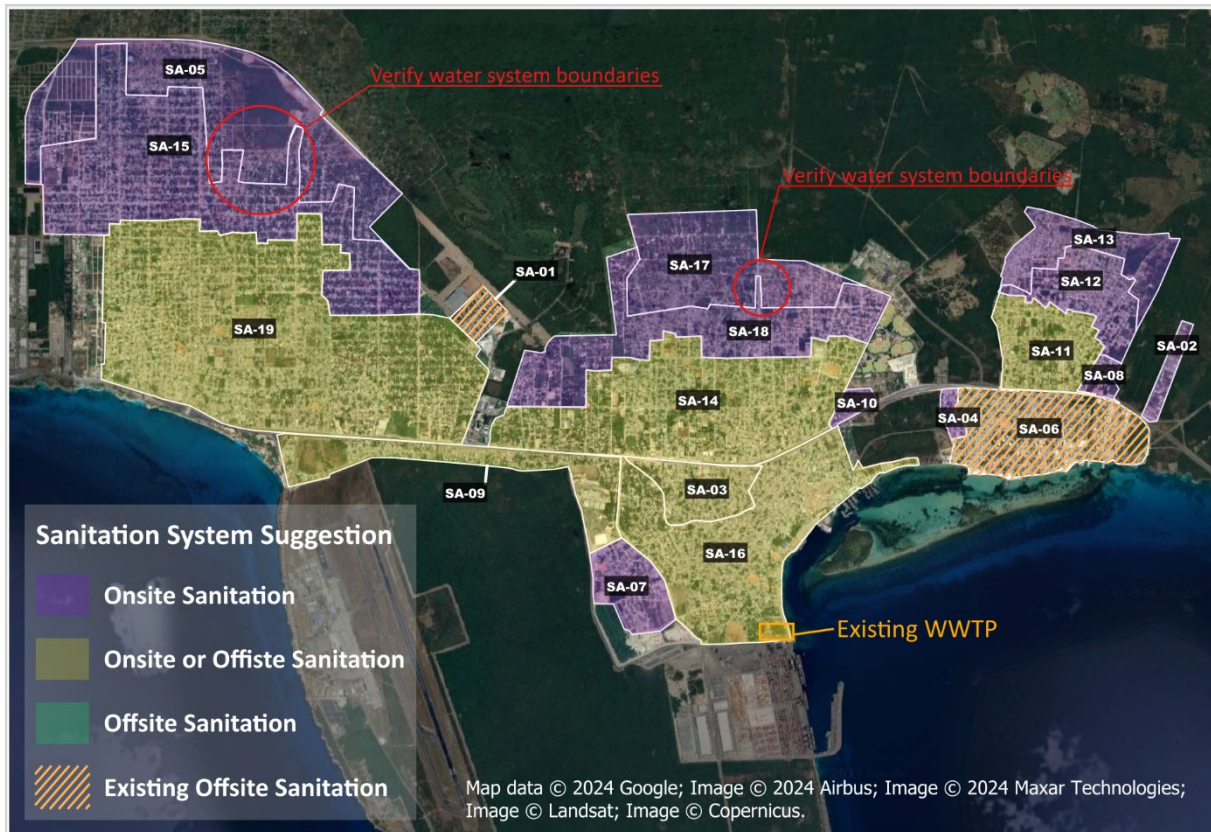


Figure 30 Final delimitation of the Subareas with the respective suggested sanitation system for the end of planning

Based on the parameters and datasets used in this study, the total population in the study area is estimated to be around 151,000 people at the start of the plan and about 222,000 by the end of the planning period. In areas with existing sanitation systems, the population is about 12,000 at the start and 17,000 at the end of the planning period, as Table 9 exhibit. Considering the datasets of *OpenStreetMap* and *Overture Maps*, the study area also includes over 565 kilometers of streets, roads, paths, and nearly 62,000 buildings. Based on the water user registry, the average water consumption is 173 liters per inhabitant per day, which is high compared to other Latin American countries. The tables with all the characteristic indicators and results of the subareas are in Appendix C.

Table 9 Population by type of sanitation suggested for the case study

Suggested Sanitation System	Population Start of the plan	Population End of the plan
Onsite	24,646	36,310
Onsite or Offsite	114,665	168,933
Existing sanitation	11,958	17,618
Total	151,269	222,861

The area identified as "SA-04" in Figure 30 illustrates one of the advantages of incorporating spatial data and applying the proposed methodology. When considering this subarea's characteristics isolated, an onsite sanitation system would appear to be the most suitable solution. However, given its proximity to an area with an existing offsite system, it may be more advantageous to expand the existing system to serve the population in this area. A less obvious case includes subareas "SA-03," "SA-16," and "SA-07," which are located near the existing wastewater treatment plant (WWTP). In the subsequent phases of the sanitation planning process, a further investigation of these subareas is recommended to determine whether onsite or offsite sanitation is the better option, particularly concerning the long-term costs of implementation and operation of the system. Other subareas, such as those in the northern part of Boca Chica, are clearly better suited for onsite sanitation due to their low population density and lack of access to a regular water supply system.

Finally, as outlined in Chapters 2.1.1 and 2.1.3, while the presented study case focuses on early-stage planning, the subsequent phases of sanitation planning must consider the local context, including technical, socio-cultural, and financial factors, to select the appropriate technology. The optimal sanitation system is the one that best balances these criteria, ensuring feasibility, cost-effectiveness, and community acceptance. A well-chosen system not only meets the population's immediate needs but also adapts to future growth and changes, enhancing the sustainability of the sanitation infrastructure over the project horizon.

Chapter 5 Discussion

Based on the findings, population density and water consumption are two essential criteria to consider when selecting the most suitable sanitation system during early-stage sanitation planning, in addition to delimitation of areas with existing sanitation systems. Population density determines the feasibility of adopting specific types of sanitation due to its direct relationship with the cost of implementation. In low-density areas, onsite systems, such as septic tanks or pit latrines, are generally more suitable as they have the space available for infiltration into the ground and safe storage of fecal sludge necessary for such solutions. On the other hand, offsite systems would be expensive in these areas, as they require extensive sewage networks to connect dispersed households. However, as population density increases, the relative costs of onsite and offsite systems balance out. At high densities, sewage systems become the most viable option due to economies of scale when the implementation costs per connection are reduced. In addition, the technical viability of offsite solutions requires a minimum per capita water consumption, thus ensuring proper operation since they rely on sufficient hydraulic flow to transport the wastewater.

Regarding the primary data mentioned above, existing water consumption and sanitation data is usually unavailable as open-access spatial data, as it is information held by the water and sanitation utilities and, therefore, relies on local knowledge. However, existing open-access gridded population datasets are excellent alternatives for the population density data required. These datasets offer more granular information, providing population distribution in smaller units, such as a 100-meter grid, rather than aggregated at the municipal level, which is common in official censuses, especially in developing countries. As highlighted in Chapter 2, the granularity of population distribution is crucial for sanitation planning. It allows each region within the study area, called subareas in this study, to be more precisely characterized. This level of detail better reflects the real needs of the different subareas, resulting in a more consistent sanitation plan.

The selection of open-access spatial datasets in this research adopted a series of criteria: global data coverage, open licensing, adherence to OGC standards, and sufficient resolution. This ensures that the dataset quality meets sanitation planning requirements and facilitates the reproduction of the proposed methodology in any region worldwide. Four primary gridded population datasets from three different research groups were cataloged.

1. *WorldPop* with two distinct datasets (constrained and unconstrained), developed by the *WorldPop* Research Group at the University of Southampton.
2. High-Resolution Settlement Layer (*HRSL*) from a partnership between Meta and the Center for International Earth Science Information Network (CIESIN) at Columbia University.
3. Global Human Settlement Layer (*GHSL*) from the European Commission's Copernicus Program.

The above datasets are generated using a top-down methodology, which disaggregates census data into smaller grid cells. This process integrates multiple datasets, including satellite imagery, land cover, and building footprints, to estimate the likelihood and number of people

living in each grid cell. The top-down approach can be further classified into two types: constrained datasets, which restrict population distribution to areas with identified buildings, and unconstrained datasets, which spread the population across the entire area. All of the cataloged datasets are constrained except for the unconstrained variant of *WorldPop*. Among the selected datasets, most offer a resolution of 100 meters, with *HRS* providing a finer resolution of 30 meters.

The accuracy of these datasets varies significantly depending on the specific context. Gridded population datasets are known to underestimate the population in high-density residential areas and overestimate in non-residential areas with numerous buildings and constructions, as explored by Kuffer et al. (2022). As demonstrated in the case study, there is no generally superior dataset; each model's performance varies across different areas according to the methodologies adopted in its development. Two factors were evaluated for the proposed case study in Boca Chica, the Dominican Republic: the total population compared to the 2022 census and the spatial distribution within the study area. The dataset selected for the case study was the *WorldPop* constrained dataset, which showed a population 20% higher than the most recent census available - a considerable difference. However, it compensates for this discrepancy by offering a more detailed spatial distribution than the aggregated population data provided at the municipal level by the census. In addition, minor errors were identified in the spatial distribution, such as the absence of populations in areas with existing water connections, according to data from the local water and sanitation utility.

In addition, the research included optional data to enhance the characterization of the area, such as the identification of low-income areas and informal settlements, as well as open-access secondary spatial datasets that provide information on existing streets, roads, paths, and building footprints. Defining informal settlements and low-income areas requires local knowledge, as these characteristics are difficult to identify using remote sensing and satellite images alone (Thomson et al., 2021). Concerning transportation infrastructure data and building footprints, there are several options for open-access spatial data. This research adopted the *OpenStreetMap* dataset for streets and the *Overture Map* dataset for building footprints. Both provided satisfactory accuracy regarding the completeness and accuracy of the infrastructure mapped in Boca Chica, even in densely occupied areas, as set out in Chapter 4.1.7.3.

These findings emphasize the need for planners to carefully evaluate and select the dataset that best aligns with the planning area while understanding the limitations of the chosen dataset. Complemented by other spatial data such as streets and building plans, an accurate population dataset is essential for developing sanitation plans consistent with local characteristics. The datasets used in this research are constantly evolving and updated. Future versions are expected to improve accuracy by incorporating corrections from recently released censuses and more accurate auxiliary spatial data—such as the streets and building footprints datasets presented in this research— which will likely lead to more accurate constrained models.

The methodology proposed in this thesis uses scripts developed for QGIS, an open-source GIS software, to reduce barriers to adopting advanced spatial analysis methods in developing countries. These scripts automate essential steps in sanitation planning, allowing planners to input the necessary parameters quickly and analyze the results. The flexibility of QGIS scripts is a significant advantage, as it allows planners to readjust parameters and subarea boundaries without returning to the initial steps to recalculate the results, simplifying the planning process. In addition, allowing users to change the parameters of each subarea individually makes it possible to explore various scenarios according to their unique characteristics, such as adopting different population growth rates for new developing areas compared to centralized and already saturated zones. A challenge in developing the scripts was balancing ease of use with the need

for flexibility to accommodate various types of data (raster or vector), data formats, and the different ways information is stored in the datasets.

Although all the datasets used in this research can be acquired through the links provided in the data catalog, acquiring some requires basic GIS knowledge. Only the *WorldPop* dataset offers an API and a QGIS-specific plugin, simplifying data acquisition. If other dataset providers offered similar web services or APIs, developing scripts for data acquisition would be possible, further facilitating access to the data. The detailed documentation of the proposed methodology, presented in Chapter 3, ensures that the steps are reproducible in other GIS software if desired. However, capacity building remains crucial, as some GIS skills are still required to adopt the methodology, highlighting the ongoing need for training in this field.

Applying the proposed methodology in Boca Chica, Dominican Republic, shows the advantages of incorporating cataloged spatial data into early-stage sanitation planning. Although population data from the recent 2022 census is available for the Dominican Republic, it is aggregated at the municipal level. As discussed in Chapter 2, this level of aggregation does not allow for consistent sanitation planning. Despite some discrepancies, the 100-meter spatial resolution of the selected dataset allows a better understanding of the distribution of the population in the planning area, which is crucial for delimiting subareas and determining the appropriate sanitation systems based on their characteristics. In addition, using spatial data allows planners to evaluate not only the isolated subareas but also the surrounding areas' characteristics. This broader perspective helps to identify better alternatives, such as those presented in Chapter 4.1.7.3, where it was found that extending existing offsite sanitation infrastructure to nearby areas could be a more efficient solution when considering the surrounding context. Initially, these areas seemed more suitable for onsite solutions based on their individual characteristics, but the analysis of nearby infrastructure suggested this other possibility.

A significant challenge in this research, especially in the case study, was to find references to determine the limits for classifying population density as low, medium, or high. These classifications are crucial in sanitation planning because they are strongly related to the most appropriate sanitation system to be implemented. The lack of specific guidelines for the Dominican Republic led to adopting parameters from Indonesia, which may not be the most appropriate for the local demographic and socio-cultural characteristics. Attempts were made to simulate scenarios for selected subareas, varying their population densities and comparing the costs of sewage systems to those of septic tanks for each population density. The purpose was to identify the population density at which the costs of the two systems would be equivalent, indicating when onsite solutions are no longer the most suitable, and both solutions are viable. However, it was not possible to reach a definitive conclusion due to the challenges of accurately estimating treatment costs for isolated sub-areas. Existing treatment cost curves have been developed based on economies of scale for entire cities or regions of large cities, where the cost per inhabitant decreases as the population served increases. Applying these cost curves to smaller areas, and consequently small populations, results in overestimated values, as these subareas would only represent a portion of the effluent treated by these plants. In addition, there are no reliable cost curves for fecal sludge treatment plants, which made it difficult to compare all the components of a sanitation system.

This research only considered domestic users, excluding commercial and industrial contributions, which can influence sanitation planning. This simplification is justified for early-stage planning; however, later stages of sanitation planning should assess the impacts of non-domestic users, as their sewage contributions and specific needs may suggest changes to the systems initially considered. In addition, a linear population projection was adopted in this

study, assuming a constant annual growth rate over time, which may not be suitable for long-term planning. The planner must assess the ideal projection method for each study area; for example, in Boca Chica, official government statistics indicate a downward population growth trend over the years. In such cases, a projection method incorporating variable growth rates would result in more reliable population projections, thus improving planning results.

Another limitation of the proposed methodology is that it does not allow the planner to define areas of potential expansion where, according to the population data set adopted, there is no population at the moment, but could become populated areas in the future. This restriction prevents the consideration of possible expansions of the urban area currently occupied. Therefore, further research and development are suggested to allow these considerations.

Chapter 6 Conclusion and Recommendations

This research proposed a new methodology to address the gap in adopting open-access spatial data for early-stage sanitation planning. While the sanitation sector uses spatial data mainly to operate and maintain existing systems, with limited application in the planning phase, other sectors have already adopted open-access spatial data at this stage. The proposed methodology, supported by the QGIS scripts developed and the cataloged open-access spatial datasets provides an accessible and replicable framework for planners. The primary datasets in this study included gridded population data, while additional datasets included streets, roads, and building footprints.

The case study in Boca Chica, Dominican Republic, highlighted the potential of incorporating spatial data during early-stage sanitation planning, especially in developing countries, where data is often outdated or unavailable and lacks the necessary granularity. The flexibility of the developed QGIS scripts allows planners to modify parameters, aggregate or disaggregate subareas, and recalculate indicators easily, enabling analysis of various scenarios, which results in a more dynamic and optimized sanitation planning process. The study's findings highlight the advantages of using the gridded population dataset and the additional street and building footprints datasets to characterize the location better, contributing to more informed decision-making. Although some discrepancies were observed between the official census data and the gridded population datasets for the study area, this is compensated for as the spatial resolution of the selected datasets allowed for a more accurate assessment of population distribution, something that is not possible with census data aggregated at the municipal level. This finer granularity is essential for assessing the most appropriate sanitation systems based on the unique characteristics of each subarea.

Considering the limitations identified in this research, such as the use of linear growth projections and the impossibility of defining potential areas of population expansion, future work should incorporate more sophisticated population projection models. Including other datasets, such as socioeconomic indicators, soil type, groundwater levels, and flood risk areas, would also benefit an even more comprehensive characterization of the subareas, thus providing more information for sanitation planning.

Integrating APIs and web services to automate the acquisition and processing of cataloged datasets would significantly benefit planners with limited GIS knowledge, further facilitating the adoption of spatial data in sanitation planning phases. However, this mainly depends on the dataset providers deliver such services - a feature currently only available for WorldPop datasets. In addition, ongoing capacity-building efforts in GIS and spatial data analysis are essential to empower local planners and stakeholders.

The evidence presented confirms that open-access spatial data integration can enhance early-stage sanitation planning by offering a flexible and replicable methodology that democratizes access to advanced planning tools and can be applied in various contexts, including developing countries. Integrating high-quality spatial data and automating analysis and calculation processes provide a stronger foundation for effective sanitation planning, ultimately contributing to more sustainable sanitation solutions suited to the local context.

References

- Angel, S., Lamson-Hall, P., & Blanco, Z. G. (2021). Anatomy of density: Measurable factors that constitute urban density. *Buildings and Cities*, 2(1), 264–282. <https://doi.org/10.5334/bc.91>
- Blatt, A. J. (2016). Open-Access Geospatial Data: Promise and Potential. *Journal of Map & Geography Libraries*, 12(2), 216–222. <https://doi.org/10.1080/15420353.2015.1125405>
- Cader, C., Pelz, S., Radu, A., & Blechinger, P. (2018). Overcoming Data Scarcity for Energy Access Planning with Open Data – The Example of Tanzania. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLII-4/W8, 23–26. <https://doi.org/10.5194/isprs-archives-XLII-4-W8-23-2018>
- Damania, R., Desbureaux, S., Rodella, A.-S., Russ, J., & Zaveri, E. (2019). *Quality Unknown: The Invisible Water Crisis*. World Bank Publications.
- Data, S., & Center (SEDAC), A. (n.d.). *Gridded Population of the World (GPW)*, v4. <https://sedac.ciesin.columbia.edu/data/collection/gpw-v4>
- De Moura, E. N., & Procopiuck, M. (2020). GIS-based spatial analysis: Basic sanitation services in Parana State, Southern Brazil. *Environmental Monitoring and Assessment*, 192(2), 96. <https://doi.org/10.1007/s10661-020-8063-2>
- Diaw, M. T., Cissé-Faye, S., Gaye, C. B., Niang, S., Pouye, A., Campos, L. C., & Taylor, R. G. (2020). On-site sanitation density and groundwater quality: Evidence from remote sensing and *in situ* observations in the Thiaroye aquifer, Senegal. *Journal of Water*,

- Sanitation and Hygiene for Development*, 10(4), 927–939.
<https://doi.org/10.2166/washdev.2020.162>
- Esch, T., Marconcini, M., Felbier, A., Roth, A., Heldens, W., Huber, M., Schwinger, M., Taubenbock, H., Muller, A., & Dech, S. (2013). Urban Footprint Processor—Fully Automated Processing Chain Generating Settlement Masks From Global Data of the TanDEM-X Mission. *IEEE Geoscience and Remote Sensing Letters*, 10(6), 1617–1621.
<https://doi.org/10.1109/LGRS.2013.2272953>
- European Commission. Joint Research Centre. (2023). *GHSL data package 2023*. Publications Office. <https://data.europa.eu/doi/10.2760/098587>
- Facebook Connectivity Lab and Center for International Earth Science Information Network - CIESIN - Columbia University. (2016). *High Resolution Settlement Layer (HRSL)* [Dataset].
- Furlong, C., Singh, S., Shrestha, N., Sherpa, M. G., Lüthi, C., Zakaria, F., & Brdjanovic, D. (2020). Evaluating financial sustainability along the sanitation value chain using a financial flow simulator (eSOSView™). *Water Science and Technology*, 82(11), 2220–2233. <https://doi.org/10.2166/wst.2020.456>
- Gambrill, M., Gilsdorf, R. J., & Kotwal, N. (2020). Citywide Inclusive Sanitation—Business as Unusual: Shifting the Paradigm by Shifting Minds. *Frontiers in Environmental Science*, 7, 201. <https://doi.org/10.3389/fenvs.2019.00201>
- Gao, Q., Blum, K. M., Gago-Ferrero, P., Wiberg, K., Ahrens, L., & Andersson, P. L. (2019). Impact of on-site wastewater infiltration systems on organic contaminants in groundwater and recipient waters. *Science of The Total Environment*, 651, 1670–1679.
<https://doi.org/10.1016/j.scitotenv.2018.10.016>
- Geoportal*. (n.d.). <https://portalgeoestadistico.one.gob.do/>

- Ghangrekar, M. M. (2022). *Wastewater to Water: Principles, Technologies and Engineering Design*. Springer Nature Singapore. <https://doi.org/10.1007/978-981-19-4048-4>
- Gondhalekar, D., Nussbaum, S., Akhtar, A., Kebschull, J., Keilmann, P., Dawa, S., Namgyal, P., Tsultim, L., Phuntsog, T., Dorje, S., Namgail, P., & Mutup, T. (2013). Water-related health risks in rapidly developing towns: The potential of integrated GIS-based urban planning. *Water International*, 38(7), 902–920. <https://doi.org/10.1080/02508060.2013.855447>
- GRASS GIS. (n.d.). *Bringing advanced geospatial technologies to the world*. Retrieved June 18, 2024, from <https://grass.osgeo.org/about/history/>
- Haller, L., Hutton, G., & Bartram, J. (2007). Estimating the costs and health benefits of water and sanitation improvements at global level. *Journal of Water and Health*, 5(4), 467–480. <https://doi.org/10.2166/wh.2007.008>
- Hammouri, N., Talafha, M., Hamarneh, Q., Annab, Z., Al-Ruzouq, R., & Shanableh, A. (2022). Vulnerability Hotspots Mapping for Enhancing Sanitation Services Provision: A Case Study of Jordan. *Water*, 14(11), 1689. <https://doi.org/10.3390/w14111689>
- Hierink, F., Boo, G., Macharia, P. M., Ouma, P. O., Timoner, P., Levy, M., Tschirhart, K., Leyk, S., Oliphant, N., Tatem, A. J., & Ray, N. (2022). Differences between gridded population data impact measures of geographic access to healthcare in sub-Saharan Africa. *Communications Medicine*, 2(1), 117. <https://doi.org/10.1038/s43856-022-00179-4>
- High Resolution Settlement Layer (HRSL). (2016). *Center for International Earth Science Information Network*. <https://www.ciesin.columbia.edu/data/hrsl/>
- Kabange, R. S., & Graham, J. P. (2017). Community Population Density Variation Implications on Sanitation System Cost – The Case of Kotoko Community in Suame (Kumasi), Ghana. *Civil and Environmental Research*.

- Kerstens, S. M., Leusbrock, I., & Zeeman, G. (2015). Feasibility analysis of wastewater and solid waste systems for application in Indonesia. *Science of The Total Environment*, 530–531, 53–65. <https://doi.org/10.1016/j.scitotenv.2015.05.077>
- Kerstens, S. M., Spiller, M., Leusbrock, I., & Zeeman, G. (2016). A new approach to nationwide sanitation planning for developing countries: Case study of Indonesia. *Science of The Total Environment*, 550, 676–689. <https://doi.org/10.1016/j.scitotenv.2016.01.104>
- Ketema, A. A., Dalecha, T., Assefa, E., & Dar, E. (2014). Water supply and sanitation system preplanning and CLARA Simplified Planning Tool Application for Arba Minch, Ethiopia. *Sustainable Sanitation Practice*, 19, 44–52.
- Kuffer, M., Owusu, M., Oliveira, L., Sliuzas, R., & Van Rijn, F. (2022). The Missing Millions in Maps: Exploring Causes of Uncertainties in Global Gridded Population Datasets. *ISPRS International Journal of Geo-Information*, 11(7), 403. <https://doi.org/10.3390/ijgi11070403>
- Kumar, M., Singh, R. B., Singh, A., Pravesh, R., Majid, S. I., & Tiwari, A. (2023). *Geographic Information Systems in Urban Planning and Management*. Springer Nature Singapore. <https://doi.org/10.1007/978-981-19-7855-5>
- Kvarnström, E., & McConville, J. (2007). *Sanitation Planning—A Tool to Achieve Sustainable Sanitation: Water Supply and Sanitation for All. Berching, Germany, Sept 26–28, 2007.* 12-. http://www.huber.de/hp126032/Download-presentations-_International-Symposium_.htm
- Loetscher, T., & Keller, J. (2002). A decision support system for selecting sanitation systems in developing countries. *Socio-Economic Planning Sciences*, 36(4), 267–290. [https://doi.org/10.1016/S0038-0121\(02\)00007-1](https://doi.org/10.1016/S0038-0121(02)00007-1)

- Maguire, D. J. (1991). An overview and definition of GIS. In *Geographical information systems: Principles and applications* (Vol. 1, pp. 9–20). Longman Scientific & Technical Harlow.
- Manga, M., Bartram, J., & Evans, B. E. (2020). Economic cost analysis of low-cost sanitation technology options in informal settlement areas (case study: Soweto, Johannesburg). *International Journal of Hygiene and Environmental Health*, 223(1), 289–298. <https://doi.org/10.1016/j.ijheh.2019.06.012>
- Mara, D. (1996). *Low-cost sewerage*. John Wiley London.
- Mara, D. (2005). Sanitation for all in periurban areas? Only if we use simplified sewerage. *Water Supply*, 5(6), 57–65. <https://doi.org/10.2166/ws.2005.0050>
- Mara, D. (2008). Sanitation Now: What is Good Practice and What is Poor Practice? *Proceedings of the IWA International Conference ‘Sanitation Challenge: New Sanitation and Models of Governance’*. <http://eprints.whiterose.ac.uk/9218/>
- Mara, D., Lane, J., Scott, B., & Trouba, D. (2010). Sanitation and Health. *PLoS Medicine*, 7(11), e1000363. <https://doi.org/10.1371/journal.pmed.1000363>
- Minghini, M., Mobasher, A., Rautenbach, V., & Brovelli, M. A. (2020). Geospatial openness: From software to standards & data. *Open Geospatial Data, Software and Standards*, 5(1), 1, s40965-020-0074-y. <https://doi.org/10.1186/s40965-020-0074-y>
- Minh, H. V., & Hung, N. V. (2011). Economic Aspects of Sanitation in Developing Countries. *Environmental Health Insights*, 5, EHI.S8199. <https://doi.org/10.4137/EHI.S8199>
- Ministry of Tourism of the Dominican Republic. (n.d.). *Go Dominican Republic: Official Tourism Website of the Dominican Republic*. Retrieved August 25, 2024, from <https://www.godominicanrepublic.com/>
- Ministry of Urban Development, Government of India. (2008). *Technology options for urban sanitation in India*. https://mohua.gov.in/upload/uploadfiles/files/Urban_Sanitation.pdf

- Narayan, A., & Lüthi, C. (2020, August). Solving urban sanitation sustainably and equitably. *Water Environment Federation*, 46(4), 18–21.
- National Geological Service (SGN). (2010). *Geological Map of the Dominican Republic Scale 1:50,000, Boca Chica (6271-II) [Map]*. National Geological Service. https://www.sgn.gob.do/images/mapas/cartog_geologica_sgn/cgeo_rd/
- Nelson, K. L., & Murray, A. (2008). Sanitation for Unserved Populations: Technologies, Implementation Challenges, and Opportunities. *Annual Review of Environment and Resources*, 33(1), 119–151. <https://doi.org/10.1146/annurev.environ.33.022007.145142>
- Openshaw, S. (1984). The modifiable areal unit problem. *Concepts and Techniques in Modern Geography*.
- Overture Maps Foundation. (2024, March 2). *Who we Are—Overture Maps Foundation*. <https://overturemaps.org/about/who-we-are/>
- Pajares, E., Muñoz Nieto, R., Meng, L., & Wulforst, G. (2021). Population Disaggregation on the Building Level Based on Outdated Census Data. *ISPRS International Journal of Geo-Information*, 10(10), 662. <https://doi.org/10.3390/ijgi10100662>
- Parkinson, J., Lüthi, C., & Walther, D. (2014). *Sanitation21—A Planning Framework for Improving City-wide Sanitation Services*. IWA, Eawag-Sandec, GIZ.
- QGIS Development Team. (2024). *QGIS Geographic Information System* [Computer software]. QGIS Association. <https://www.qgis.org>
- Schrecongost, A., Pedi, D., Rosenboom, J. W., Shrestha, R., & Ban, R. (2020). Citywide Inclusive Sanitation: A Public Service Approach for Reaching the Urban Sanitation SDGs. *Frontiers in Environmental Science*, 8, 19. <https://doi.org/10.3389/fenvs.2020.00019>

- Seibel, M. (2023). *How to Succeed as a GIS Rebel: A Journey to Open Source GIS* (1st ed., Vol. 1). Locate Press.
- Sinnatamby, G. (1986). The design of shallow sewer systems. *United Nations Centre for Human Settlements, Nairobi*.
- Spuhler, D., & Lüthi, C. (2020). Review of frameworks and tools for urban strategic sanitation planning: Considering technology innovations and sustainability. *Journal of Water, Sanitation and Hygiene for Development*, 10(4), 768–785. <https://doi.org/10.2166/washdev.2020.062>
- Stevens, F. R., Gaughan, A. E., Linard, C., & Tatem, A. J. (2015). Disaggregating Census Data for Population Mapping Using Random Forests with Remotely-Sensed and Ancillary Data. *PLOS ONE*, 10(2), e0107042. <https://doi.org/10.1371/journal.pone.0107042>
- Tatem, A. J. (2017). WorldPop, open data for spatial demography. *Scientific Data*, 4(1), 170004. <https://doi.org/10.1038/sdata.2017.4>
- The FAIR principles. (n.d.). *Utrecht University*. Retrieved June 15, 2024, from <https://www.uu.nl/en/research/research-data-management/guides/how-to-make-your-data-fair>
- Thomson, D. R., Gaughan, A. E., Stevens, F. R., Yetman, G., Elias, P., & Chen, R. (2021). *Evaluating the Accuracy of Gridded Population Estimates in Slums: A Case Study in Nigeria and Kenya*.
- Tiecke, T. G., Liu, X., Zhang, A., Gros, A., Li, N., Yetman, G., Kilic, T., Murray, S., Blankespoor, B., Prydz, E. B., & Dang, H.-A. H. (2017). *Mapping the world population one building at a time* (arXiv:1712.05839). arXiv. <http://arxiv.org/abs/1712.05839>
- Tilley, E., Ulrich, L., Lüthi, C., Reymond, P., & Zurbrugg, C. (2014). *Compendium of Sanitation Systems and Technologies*.

- Türker, U. (2011). Alternative sewerage solution: Condominial method and its application. *Physics and Chemistry of the Earth, Parts A/B/C*, 36(5–6), 179–186.
<https://doi.org/10.1016/j.pce.2010.03.038>
- United Nations (Ed.). (2017). *Principles and recommendations for population and housing censuses: 2020 round* (Revision 3). United Nations.
- Von Dobschütz, S., & Wafler, M. (n.d.). *Selecting Sustainable Sanitation Systems*. Retrieved June 15, 2024, from <https://sswm.info/sanitation-project-implementation/sanitation-solutions/selecting-sustainable-sanitation-systems>
- Walter, N., Stockman, J., Odeph, C., & Butler, B. (2023, April 4). Using Meta’s relative wealth index and high resolution population density data to help expand electricity access in Uganda. World Resources Institute. *World Resources Institute*.
<https://www.wri.org/update/using-metas-relative-wealth-index-and-high-resolution-population-density-data-help-expand>
- World Bank Group. (2023, November 16). Global Water Security and Sanitation Partnership: Annual Report 2023. *In World Bank*.
<https://www.worldbank.org/en/topic/water/publication/global-water-security-and-sanitation-partnership-annual-report-2023>
- World Health Organization. (2018). *Guidelines on sanitation and health*. World Health Organization. <https://iris.who.int/handle/10665/274939>
- WorldPop, & Center for International Earth Science Information Network (CIESIN), C. U. (2018). *Global High Resolution Population Denominators Project*. School of Geography and Environmental Science, University of Southampton; Department of Geography and Geosciences, University of Louisville; Departement de Geographie, Universite de Namur. <https://dx.doi.org/10.5258/SOTON/WP00660>

Zhou, Q., Zhang, Y., Chang, K., & Brovelli, M. A. (2022). Assessing OSM building completeness for almost 13,000 cities globally. *International Journal of Digital Earth*, 15(1), 2400–2421. <https://doi.org/10.1080/17538947.2022.2159550>

Appendices

Appendix A. - Research Ethics Declaration Form



Research Ethics Committee
IHE Delft Institute for
Water Education
E ResearchEthicsCommittee@un-ihe.org

Date: 18/07/2024
To: Leonardo Porto Nazareth
MSc Programme: Water and Sustainable Development
Approval Number: IHE-RECO 2024-ipo004

Subject: Research Ethics approval

Dear Leonardo,

Based on your application for Ethical Approval, the Research Ethics Committee (RECO) of IHE Delft RECO gives ethical clearance for your research topic "Using Open-Access Spatial Data for Enhanced Sanitation Planning".

This approval is valid until September 30, 2024.

The approval is based on the information submitted in the research ethics application form and endorsed by your mentor or supervisor. The approval of the Ethical Review Board concerns ethical aspects, as well as data management and privacy issues (including the GDPR). It should be noticed that any changes in the research design oblige a renewed review by the Ethical Review Board.

Keep this letter for your records and include a copy of it in the final version of your MSc thesis, together with your personal ethics reflection.

On behalf of the Research Ethics Committee, I wish you success in the completion of your research.

Yours sincerely,

Dr. Emanuele Fantini
Coordinator, Research Ethics Committee IHE Delft

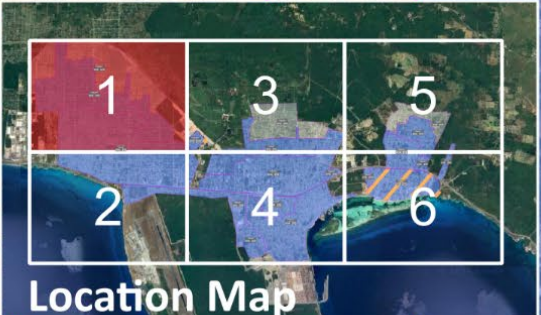
Copy to: Archive.

I declare that in conducting my research, I have adhered to the principles of research integrity and ethical considerations outlined in the “Netherlands Code of Conduct for Research Integrity” and the specific IHE guidelines. My research does not involve the collection of personal data. It focuses on analyzing and cataloging open-access spatial datasets, developing a methodology and QGIS scripts, and applying the proposed methodology to a study case, ensuring that all methodologies are transparent and publicly accessible.

Throughout the research process, I have been committed to minimizing potential ethical risks by maintaining a transparent and responsible approach to data use, anonymizing and aggregating the utilized data on the study case, as well as obtaining previous permission. The study aims to positively impact sanitation planning using globally available, non-sensitive data without causing harm or ethical concerns for anyone.

Appendix B. - Layout of the sanitation systems suggested for the subarea

Map data © 2024 Google; Image © 2024 Airbus; Image © 2024 Maxar Technologies;
Image © Landsat; Image © Copernicus.

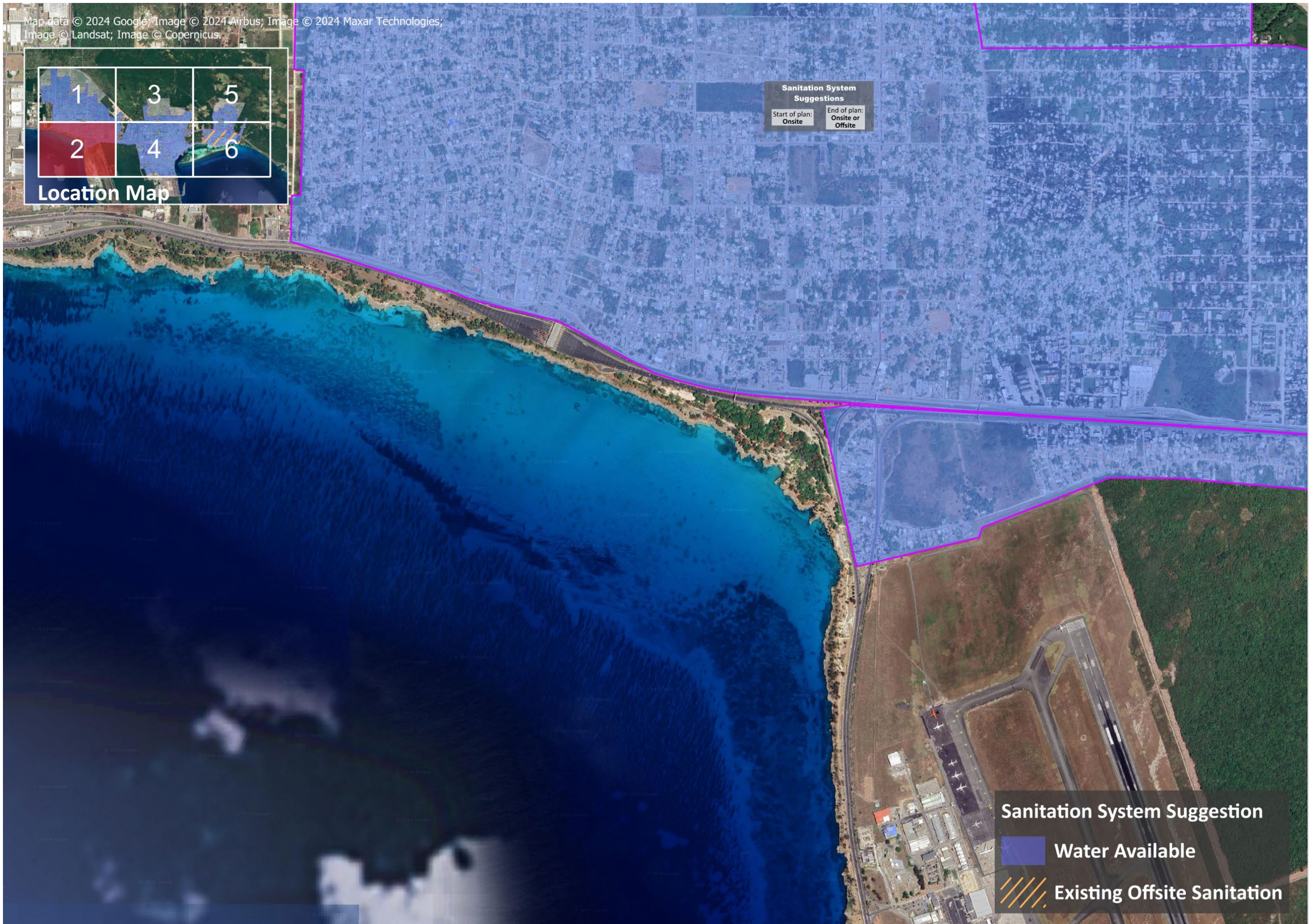


Sanitation System Suggestions
Start of plan: Onsite
End of plan: Onsite

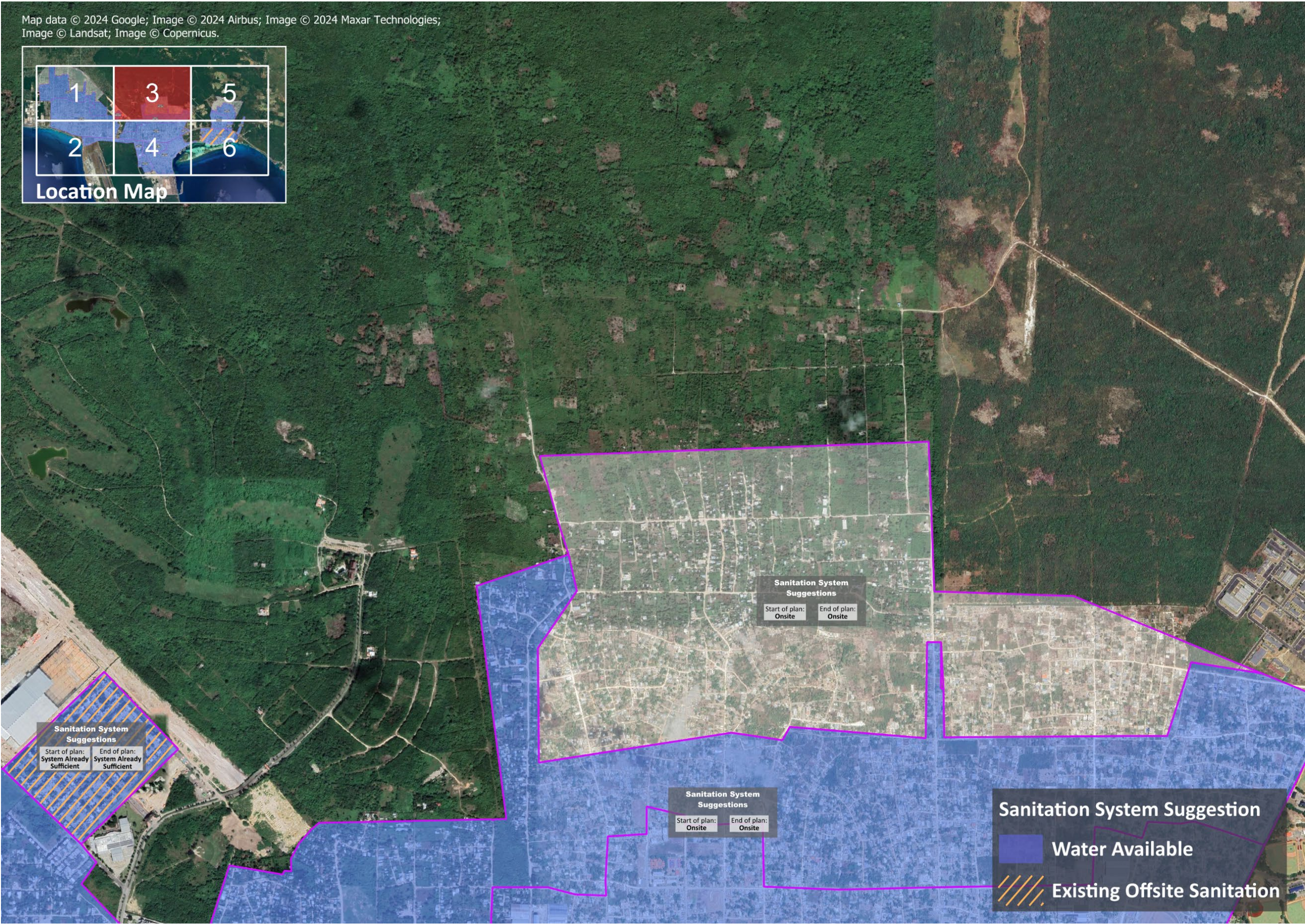
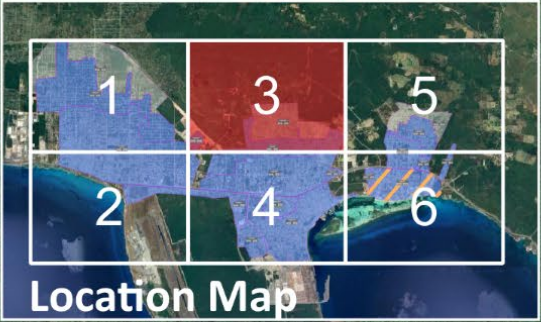
Sanitation System Suggestions
Start of plan: Onsite
End of plan: Onsite

Sanitation System Suggestions

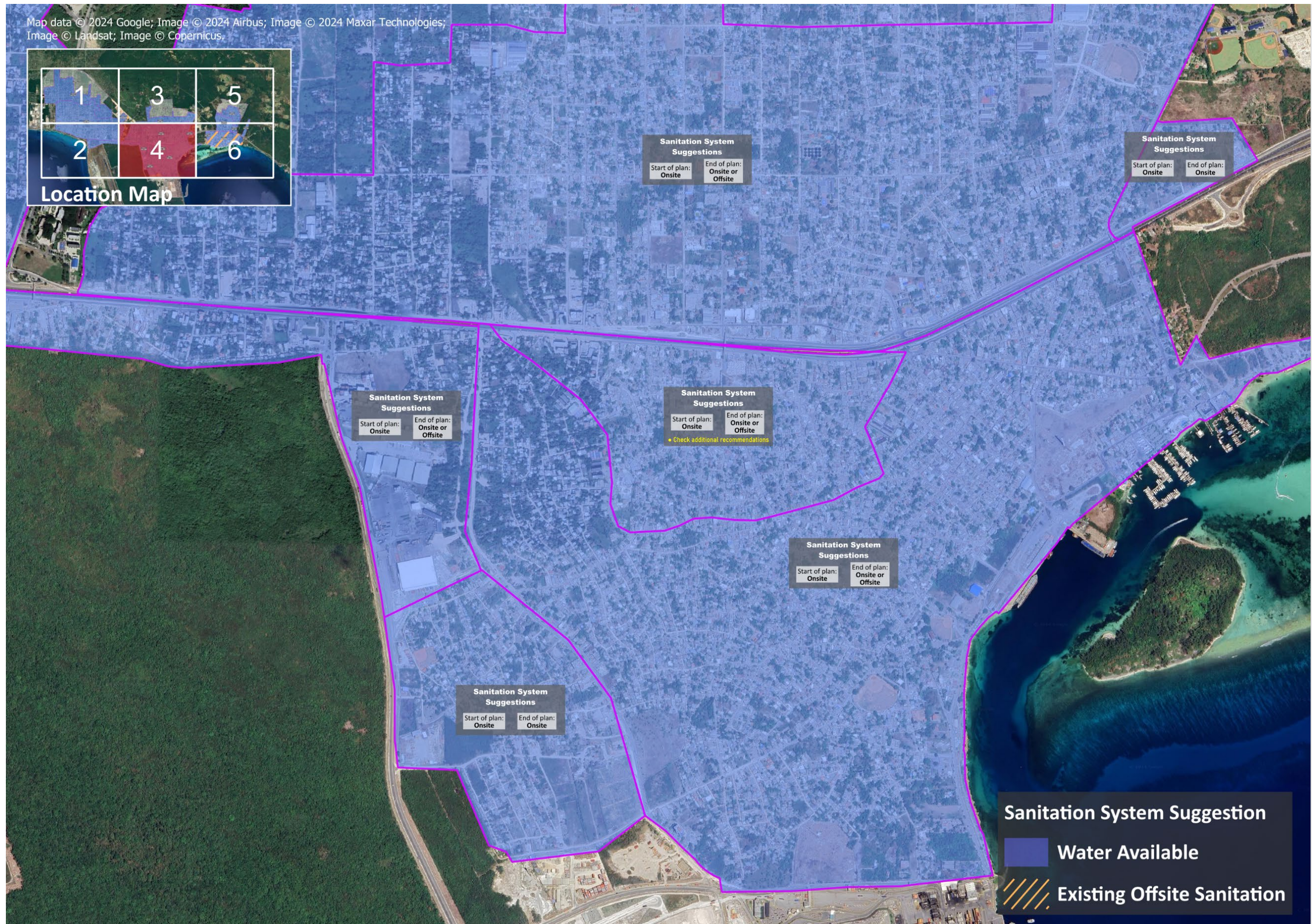
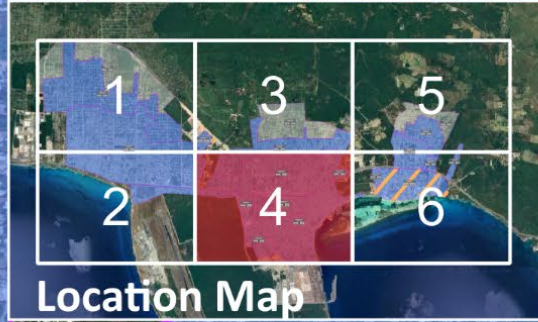
Sanitation System Suggestion
Water Available
Existing Offsite Sanitation



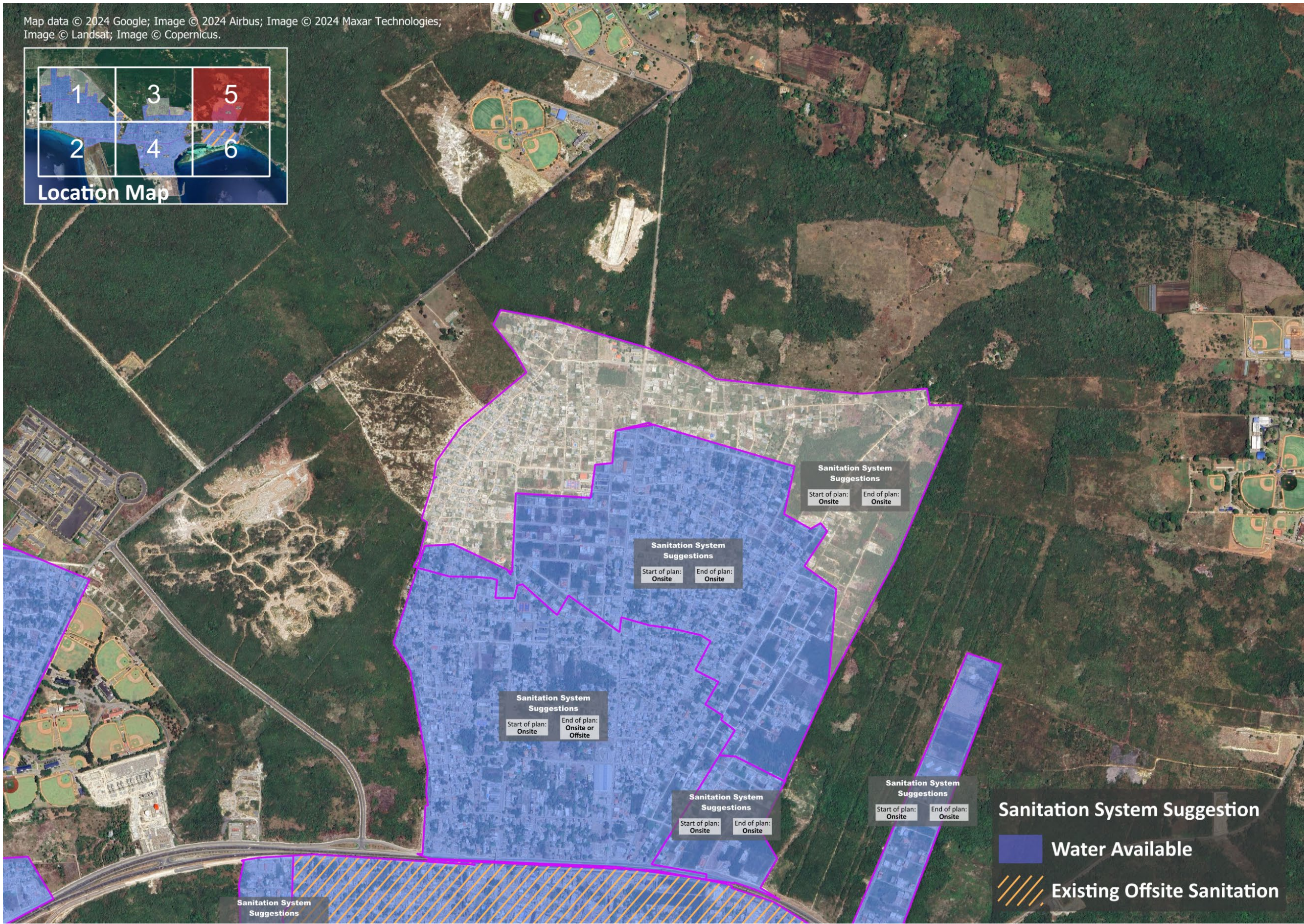
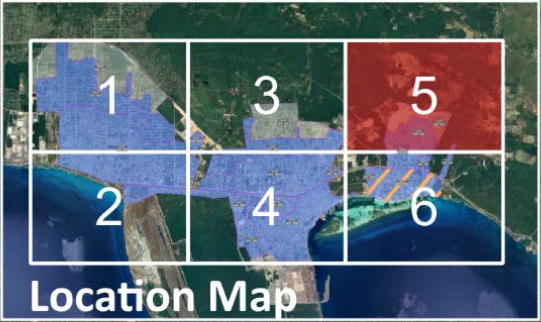
Map data © 2024 Google; Image © 2024 Airbus; Image © 2024 Maxar Technologies; Image © Landsat; Image © Copernicus.

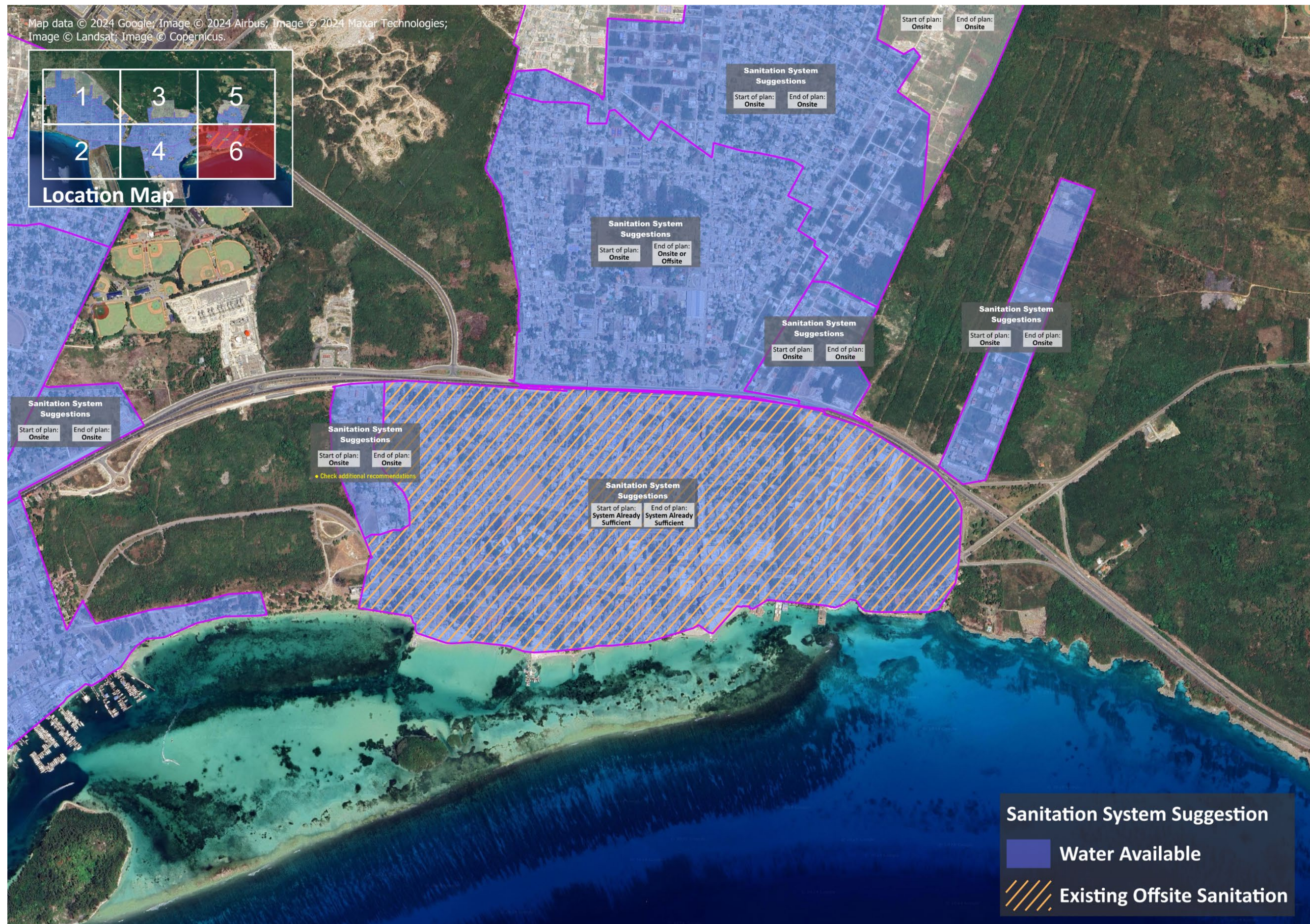


Map data © 2024 Google; Image © 2024 Airbus; Image © 2024 Maxar Technologies;
Image © Landsat; Image © Copernicus.



Map data © 2024 Google; Image © 2024 Airbus; Image © 2024 Maxar Technologies;
Image © Landsat; Image © Copernicus.





Appendix C. - Subarea Characteristics, Indicators, and Calculated Results Tables

ID	Existing sanitation?	Existing sanitation type	Informal settlement?	Low-income area?	Water system available?	Average Water Consumption (L/inhab.day)	Area (ha)	Population from the dataset	Total street length (m)	Total number of buildings
SA-01	YES	Offsite			YES	189	17	1,078	6,690.90	179
SA-02					YES	231	16	-	3,322.25	88
SA-03				YES	YES	184	60	5,236	14,891.02	2,083
SA-04			YES		YES	187	9	487	998.81	218
SA-05							297	2,049	36,874.47	2,641
SA-06	YES	Offsite	YES		YES	237	149	10,278	29,001.98	2,894
SA-07					YES	128	56	3,304	11,015.99	966
SA-08					YES	217	12	431	2,284.65	102
SA-09					YES	168	115	7,530	22,313.37	1,625
SA-10					YES	174	11	604	2,478.55	278
SA-11					YES	175	87	6,225	17,209.81	2,250
SA-12					YES	152	76	1,275	16,321.45	1,323
SA-13							80	94	11,527.96	873
SA-14					YES	154	330	25,385	73,141.57	8,466
SA-15					YES	123	368	10,183	64,162.67	7,361
SA-16					YES	166	277	20,257	58,468.27	7,740
SA-17							167	294	21,897.46	2,063
SA-18					YES	132	196	4,683	37,485.81	4,406
SA-19					YES	145	645	44,258	137,690.24	16,307

ID	Start of the plan					End of the plan					No. of buildings per hectare
	Population (inhab.)	Population density (inhab./ha)	Population density classification	No. of households	Suggested sanitation system	Population (inhab.)	Population density (inhab./ha)	Population density classification	No. of households	Suggested sanitation system	
SA-01	1,135	65.27	Low density	386	The sanitation system already exists	1,673	96.20	Low density	569	The sanitation system already exists	10
SA-02	-	-	Without population	-	Onsite	-	-	Without population	-	Onsite	6
SA-03	5,514	92.55	Low density	1,876	Onsite	8,123	136.34	Medium density	2,763	Onsite or Offsite	35
SA-04	513	58.70	Low density	174	Onsite	756	86.50	Low density	257	Onsite	25
SA-05	2,157	7.27	Low density	734	Onsite	3,178	10.71	Low density	1,081	Onsite	9
SA-06	10,823	72.58	Low density	3,681	The sanitation system already exists	15,945	106.93	Medium density	5,423	The sanitation system already exists	19
SA-07	3,480	61.61	Low density	1,184	Onsite	5,126	90.76	Low density	1,744	Onsite	17
SA-08	454	36.58	Low density	154	Onsite	669	53.91	Low density	228	Onsite	8
SA-09	7,929	68.77	Low density	2,697	Onsite	11,681	101.31	Medium density	3,973	Onsite or Offsite	14
SA-10	636	60.51	Low density	216	Onsite	937	89.15	Low density	319	Onsite	26
SA-11	6,555	74.93	Low density	2,230	Onsite	9,658	110.40	Medium density	3,285	Onsite or Offsite	26
SA-12	1,343	17.67	Low density	457	Onsite	1,978	26.02	Low density	673	Onsite	17
SA-13	99	1.25	Low density	34	Onsite	146	1.84	Low density	50	Onsite	11
SA-14	26,731	80.96	Low density	9,092	Onsite	39,382	119.28	Medium density	13,395	Onsite or Offsite	26
SA-15	10,723	29.13	Low density	3,647	Onsite	15,798	42.92	Low density	5,373	Onsite	20
SA-16	21,331	76.99	Low density	7,255	Onsite	31,427	113.43	Medium density	10,689	Onsite or Offsite	28
SA-17	309	1.85	Low density	105	Onsite	456	2.73	Low density	155	Onsite	12
SA-18	4,932	25.15	Low density	1,678	Onsite	7,266	37.05	Low density	2,471	Onsite	22
SA-19	46,605	72.28	Low density	15,852	Onsite	68,662	106.49	Medium density	23,354	Onsite or Offsite	25

Appendix D. - Authorization Letter for Use of Aggregated Data

AUTHORIZATION LETTER FOR USE OF AGGREGATED DATA IN ACADEMIC RESEARCH

Date: September 4, 2024

To Whom It May Concern,

I, Sergio Perez Monforte, Sector Specialist at the Water and Sanitation Division (INE/WSA) of the Inter-American Development Bank (IDB), hereby grant permission to Leonardo Porto Nazareth, an MSc student in Water and Sustainable Development at IHE Delft Institute for Water Education, to use aggregated data from the project "Optimization of the Sanitary Barrier Project in Boca Chica and La Caleta Bay" in his master's thesis research titled "Using Open-Access Spatial Data for Enhanced Sanitation Planning" (MSc Thesis Identifier WSD.24-1072190).

This authorization pertains specifically to the use of processed and aggregated data, which has been compiled, analyzed, and anonymized to ensure confidentiality. The use of this data is granted under the following terms and conditions:

Confidentiality and Anonymization:

All sensitive data will be anonymized to ensure that no personally identifiable information or specific details that could compromise the confidentiality of individuals, organizations, or locations will be disclosed. Only aggregated values derived from the data will be used in the thesis.

Academic Use Only:

The use of the data is strictly for academic purposes within the context of the aforementioned thesis. No part of the data will be used for commercial purposes or shared with third parties.

Compliance with Data Protection Protocols:

The data will be handled in compliance with all relevant data protection protocols to ensure its security and integrity.

By signing this authorization letter, I confirm that the use of processed and aggregated data is allowed for the specific academic purposes outlined above.



Signature

Sergio Perez Monforte
Sector Specialist
Water and Sanitation Division (INE/WSA)
Inter-American Development Bank (IDB)

Appendix E. - Declaration on Use of AI

During the preparation of this thesis, the author used Grammarly in order to improve the readability and language quality of the present research. As it was used throughout the writing process, his contributions are present throughout the document and not in specific sections.

The author declares that the tool has been applied only to improve the language and not to modify the research's ideas, structure, or content. All research, development, analysis, and conclusions are the original work of the author.